Kubernetes API

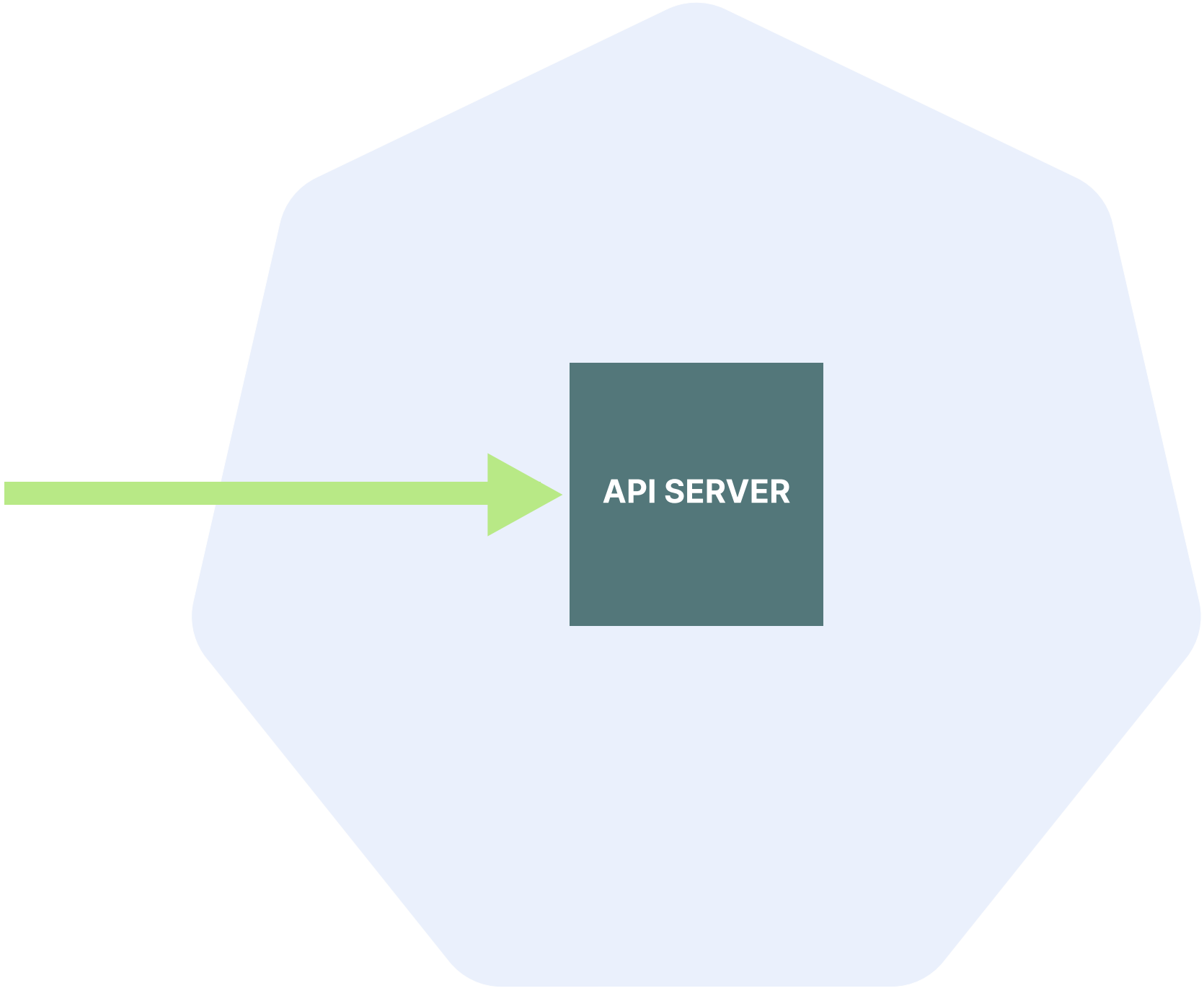
11

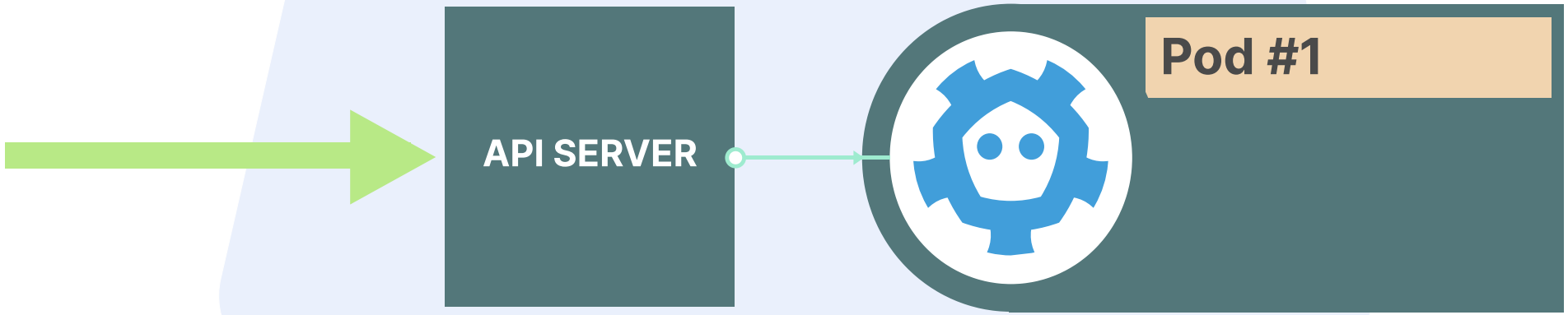
The lifecycle of a request

kubectl apply -f my-pod.yaml






API SERVER


API SERVER
Pod #1

Kubectl

Client-side validation

Generators

Runtime object

Kubectl

Client-side validation

Generators

Runtime object

Client-side validation

Generators

Runtime object

API discovery

kubectl apply -f my-pod.yaml







API negotation

```
~$ cat pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: example-pod
  labels:
    app: web
spec:
  containers:
  - name: app
    image: nginx

~$ _
```

```
~$ cat pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: example-pod
  labels:
    app: web
spec:
  containers:
  - name: app
    image: nginx

~$ _
```

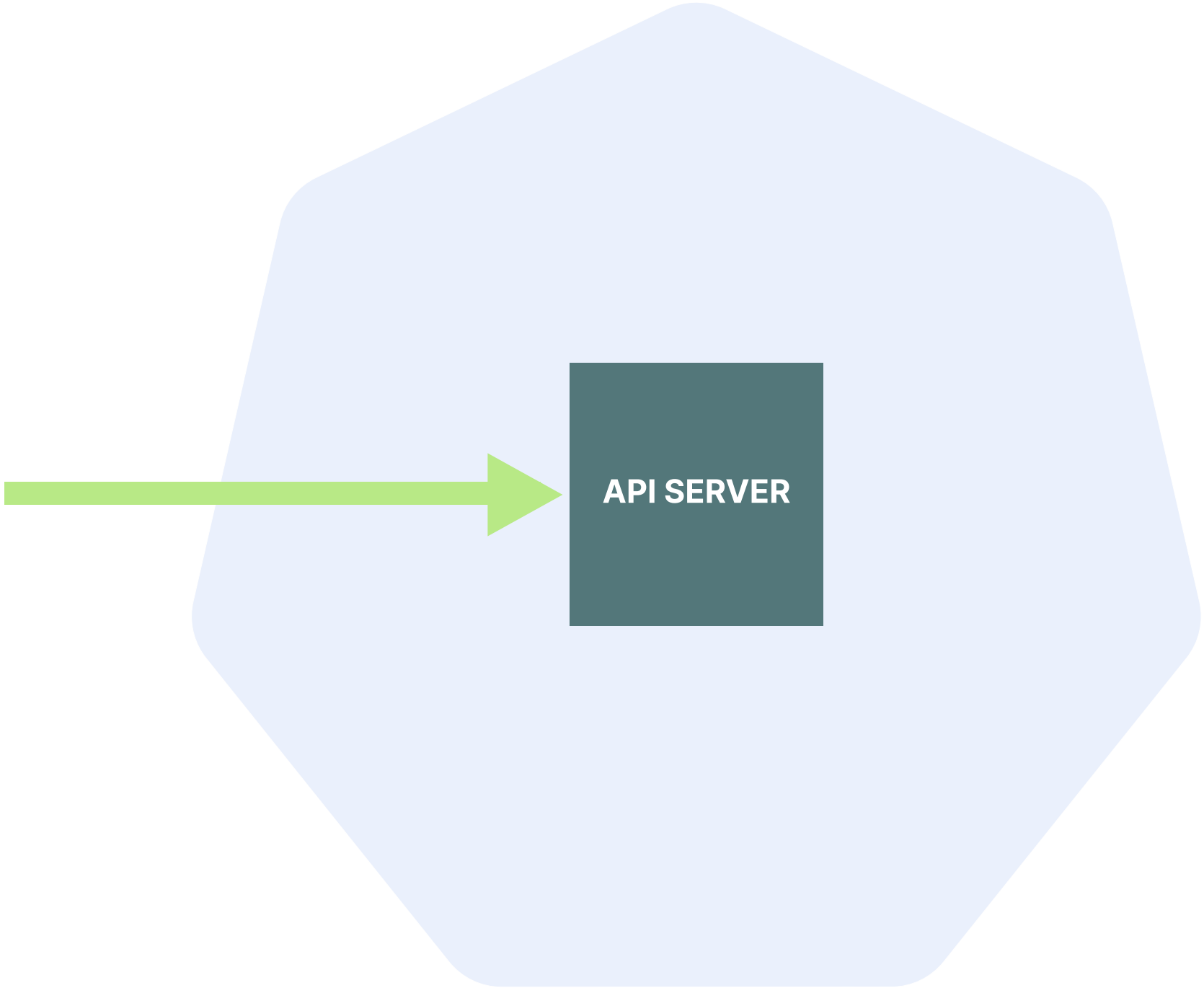
/apis/v1/pod

Versioned REST client

Send the request

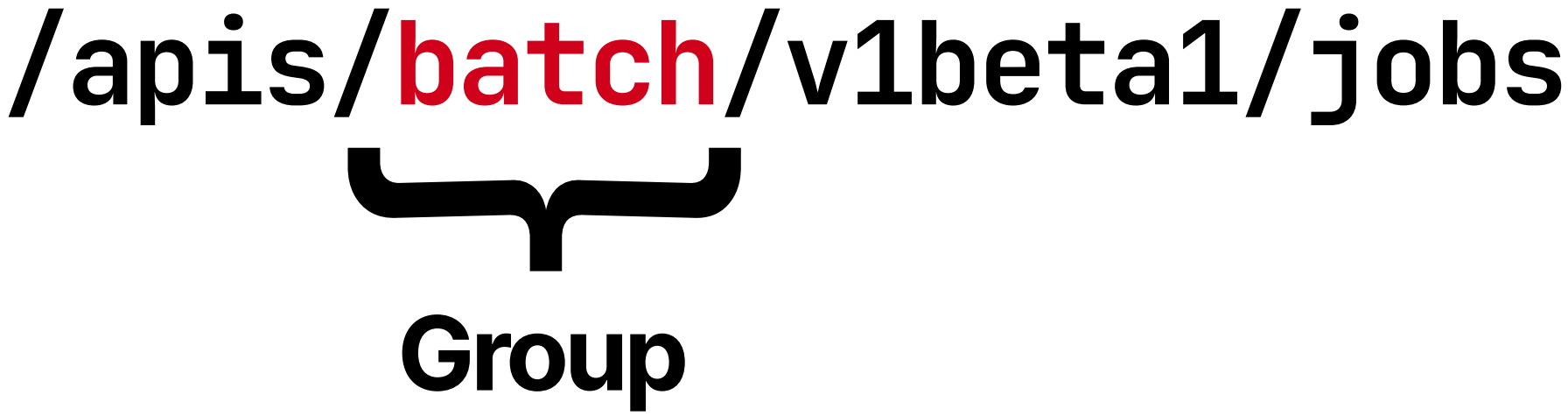
Versioned REST client

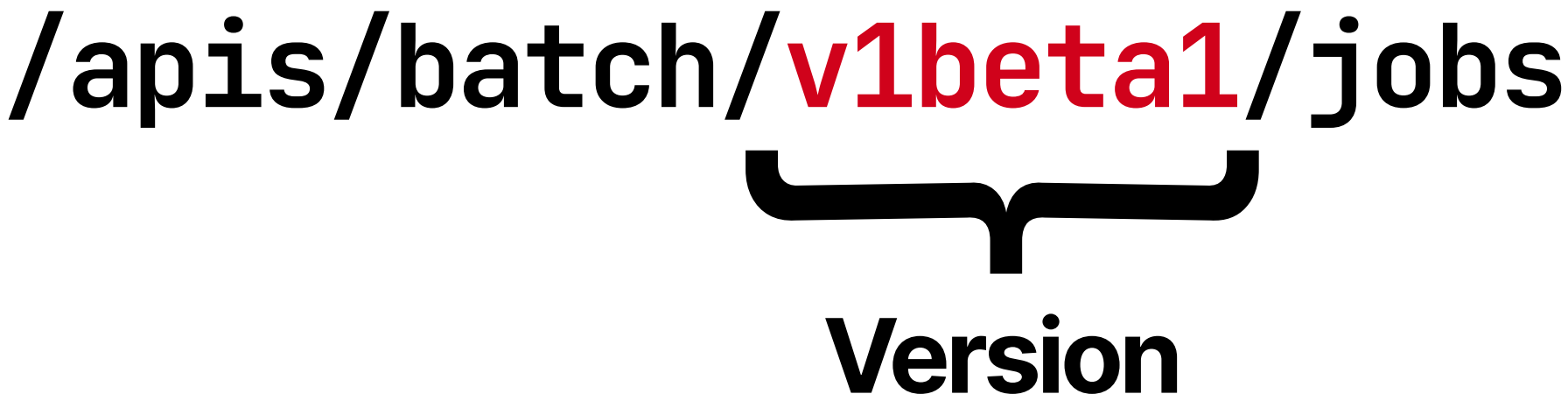
Send the request


API SERVER

API paths

/apis/batch/v1beta1/jobs


/apis/batch/v1beta1/jobs
Group


/apis/batch/v1beta1/jobs
Version

/apis/batch/v1beta1/jobs

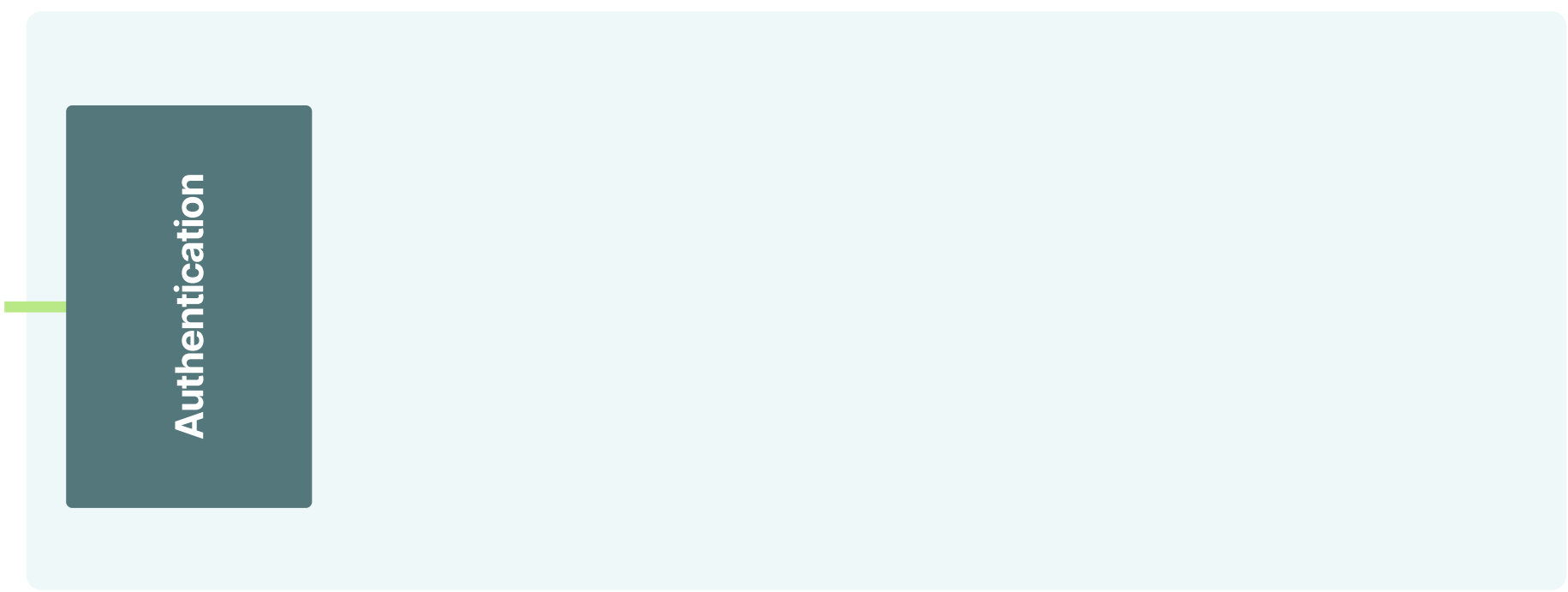


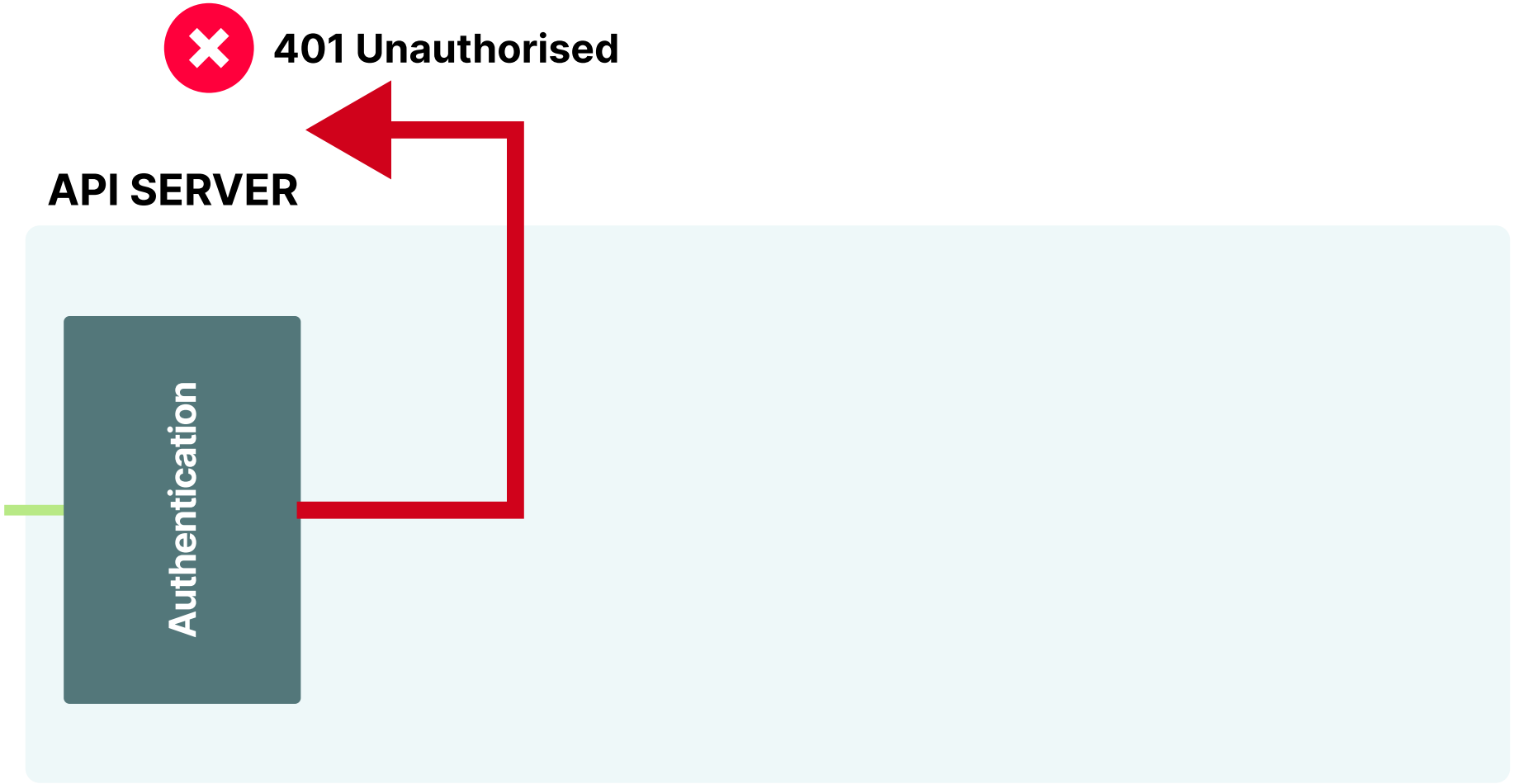
Resource

```
~$ cat job.yaml
apiVersion: batch/v1beta1
kind: Job
metadata:
  name: pi
spec:
  template:
    spec:
      containers:
      - name: pi
        image: perl
        command: ["echo",  "hello"]
      restartPolicy: Never
  backoffLimit: 4
```

Authentication


API SERVER


API SERVER
Authentication


401 Unauthorised
API SERVER
Authentication

API SERVER





User info

```
Username
UID
Groups (optional)
Extras (optional)
```

Kubernetes authentication

User types

User info

Authentication plugins

Authentication in Kubernetes

User types

User info

Authentication plugins

User types

User info

Authentication plugins

User types

Authentication

User types

Normal users

Service accounts



ADMIN

MANAGE USERS
(CREATE, UPDATE, DELETE)

USER
MANAGEMENT
SYSTEM

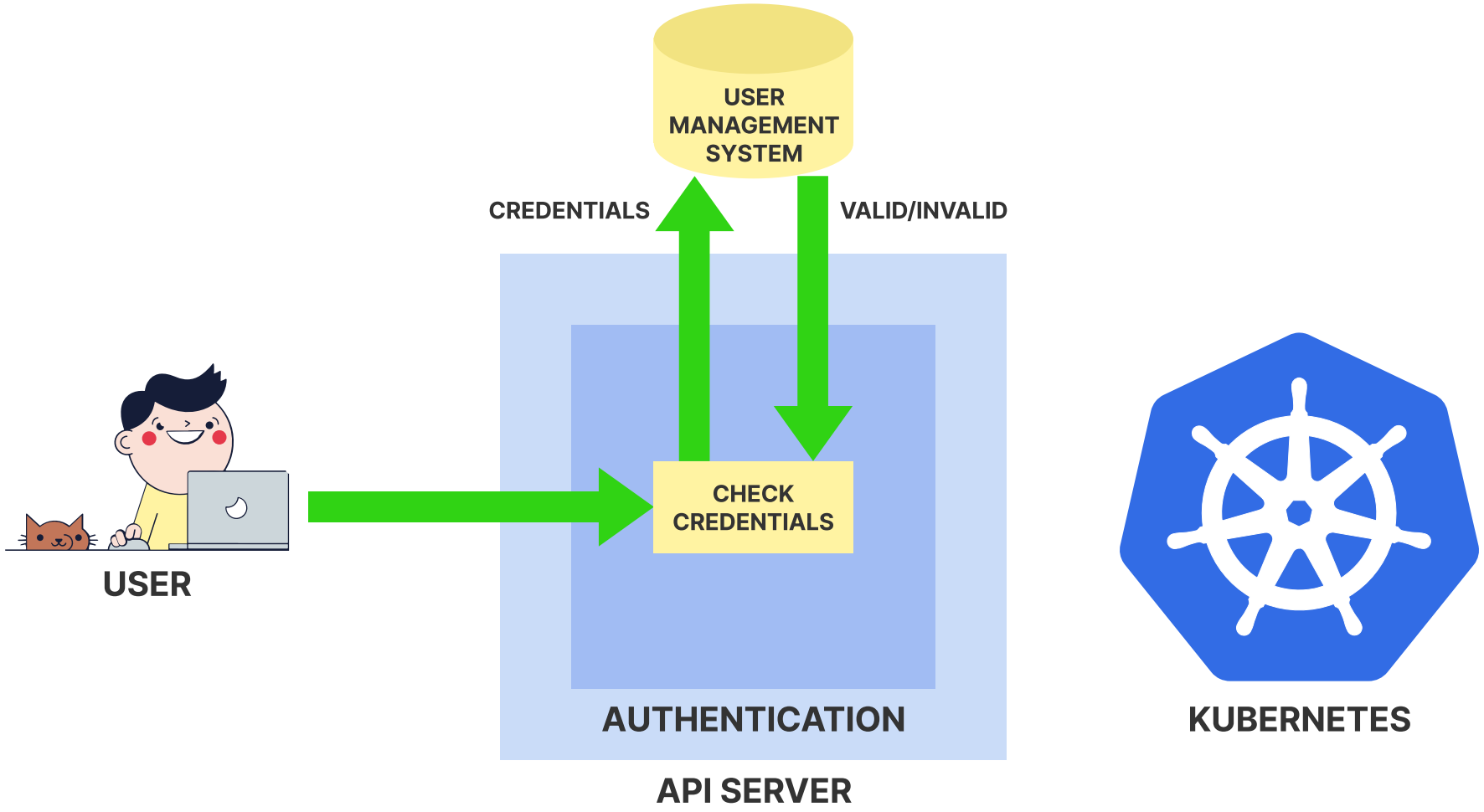
CHECK
CREDENTIALS

AUTHENTICATION

API SERVER



KUBERNETES


USER MANAGEMENT SYSTEM
CREDENTIALS
VALID/INVALID
CHECK CREDENTIALS
USER
AUTHENTICATION
API SERVER
KUBERNETES

User types

Normal users

Service accounts

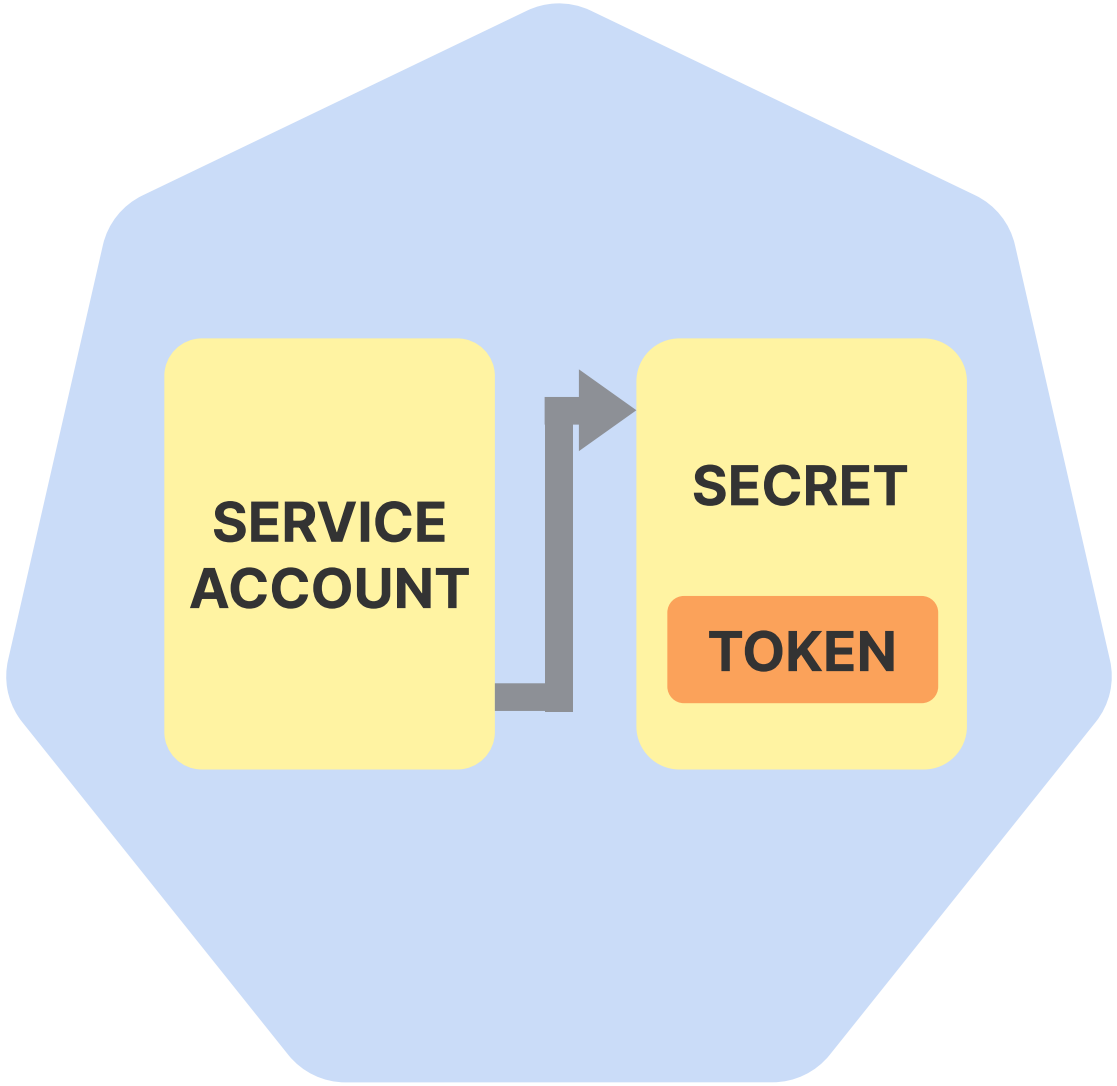


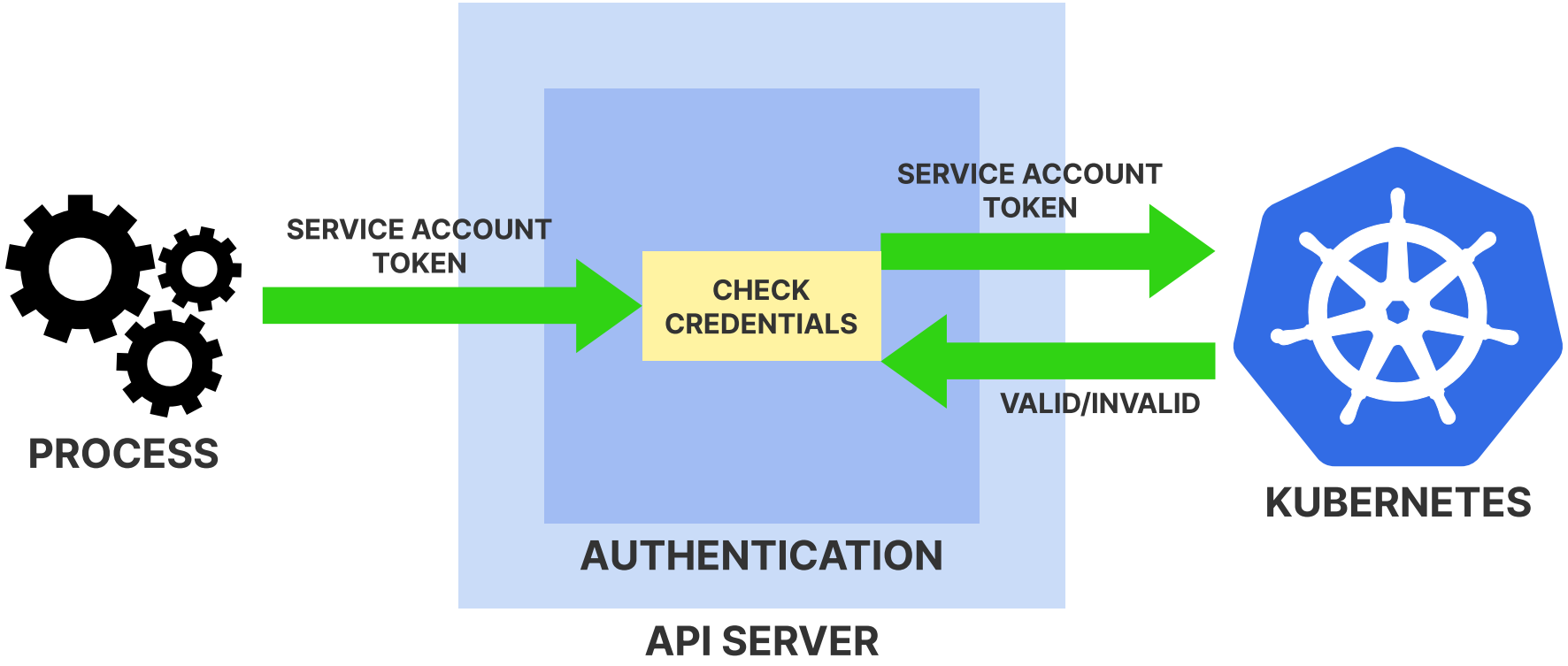
CREATE
SERVICEACCOUNT



KUBERNETES

```
apiVersion: v1
kind: ServiceAccount
metadata:
 name: foo
```


SERVICE ACCOUNT
SECRET
TOKEN
KUBERNETES


SERVICE ACCOUNT
TOKEN
CHECK
CREDENTIALS
SERVICE ACCOUNT
TOKEN
VALID/INVALID
PROCESS
KUBERNETES
AUTHENTICATION
API SERVER

ServiceAccount

User types

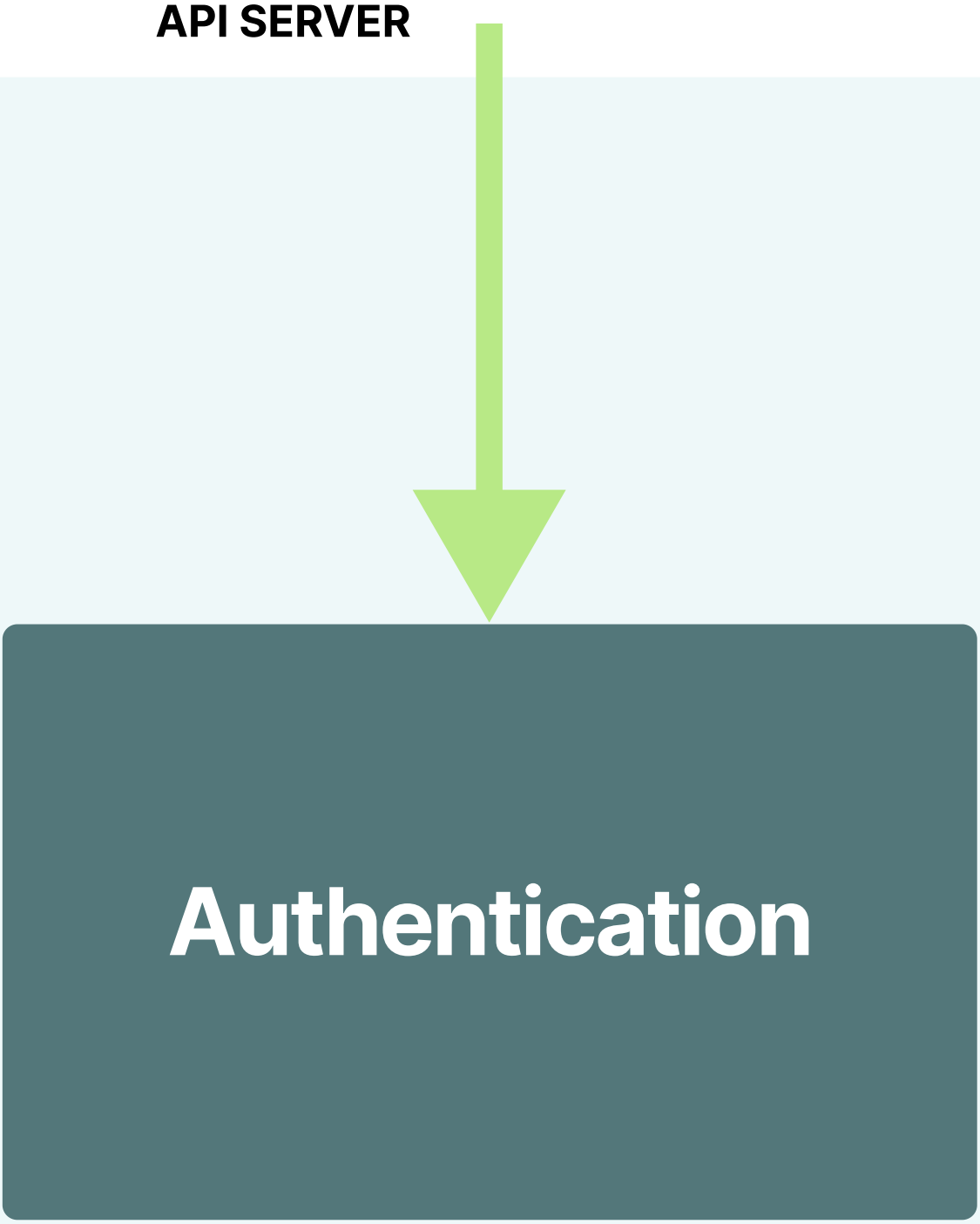
```
~$ cat service-account.yaml
apiVersion: v1
kind: ServiceAccount
metadata:
  name: build-robot

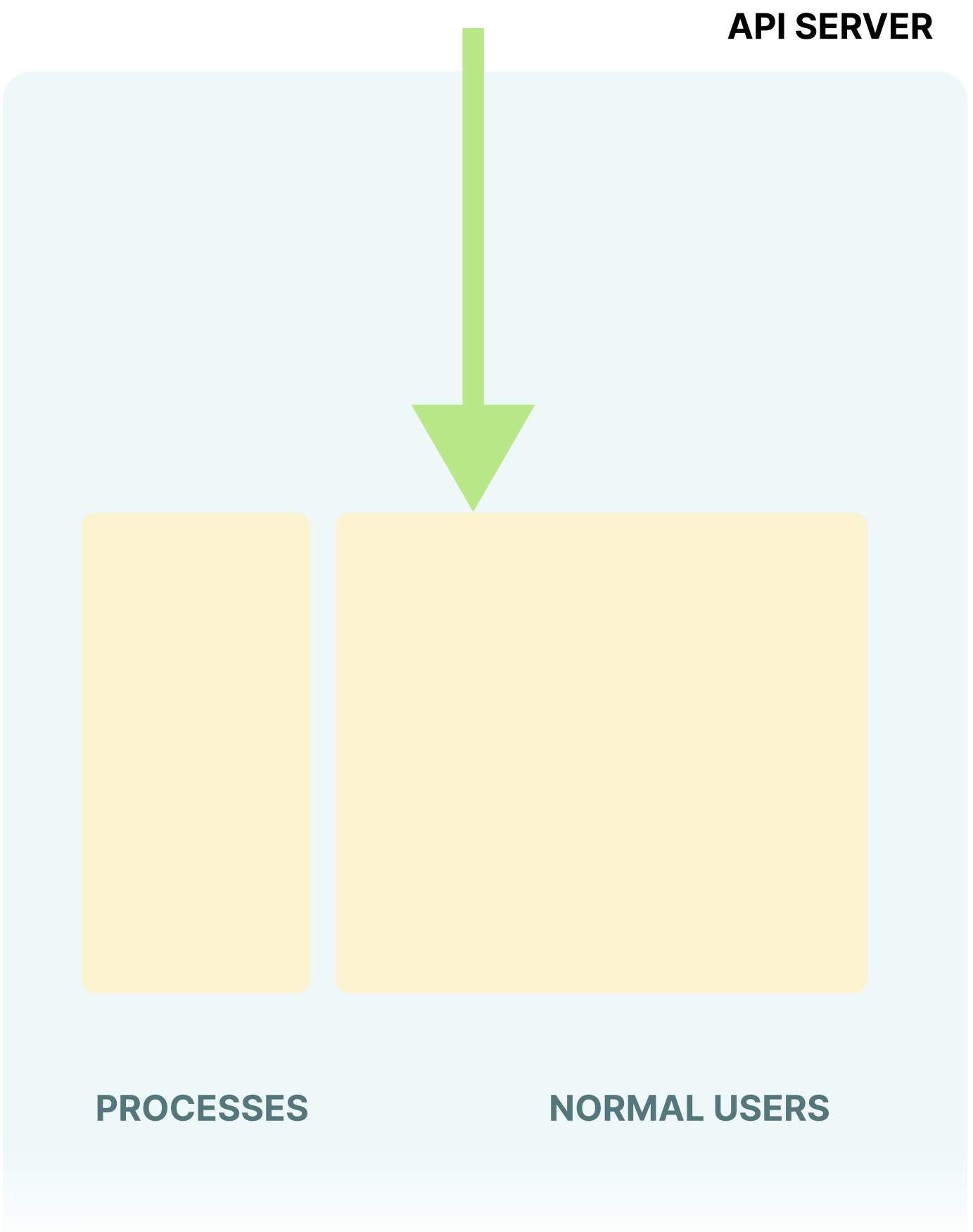
~$ _
```

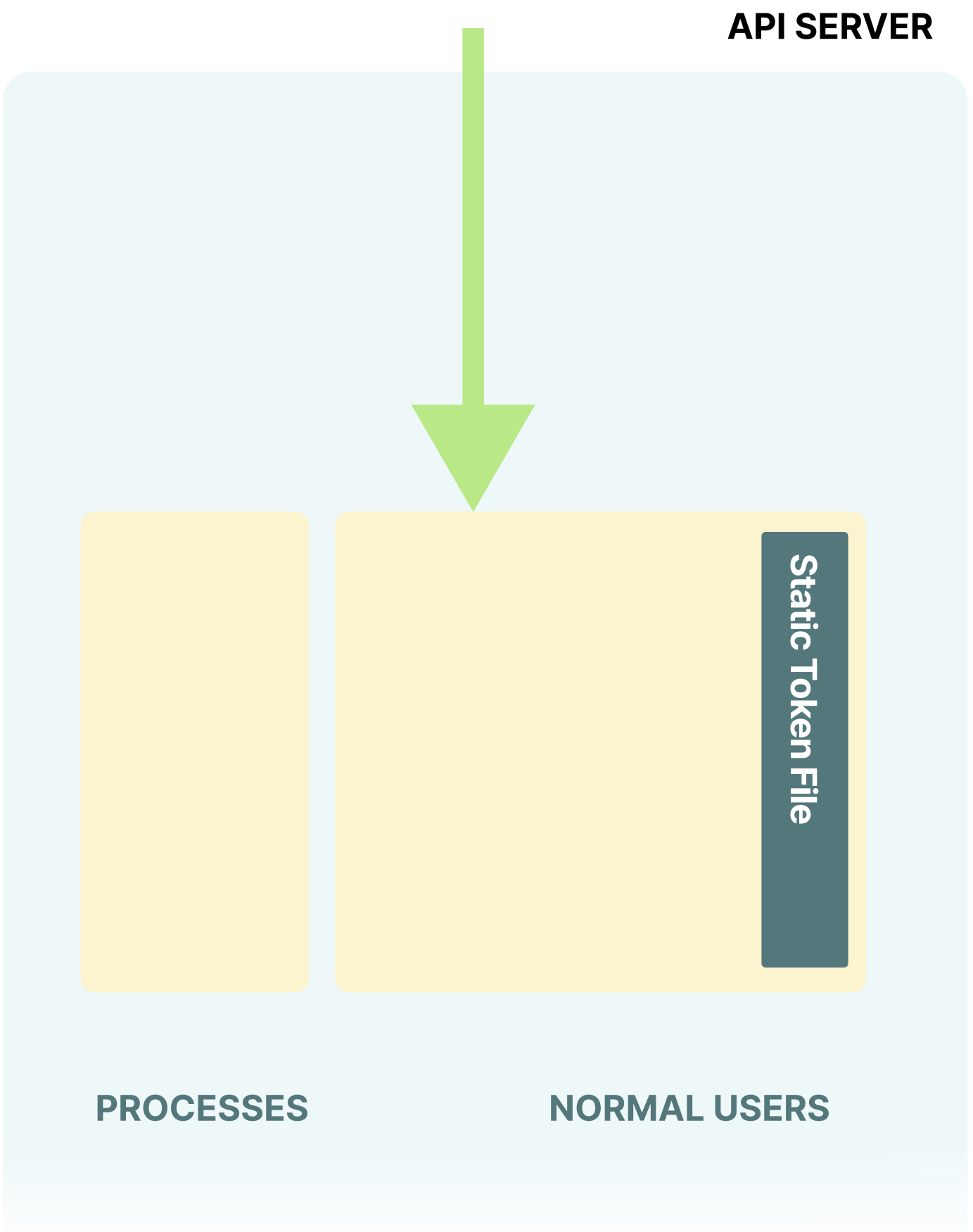
Authentication plugins

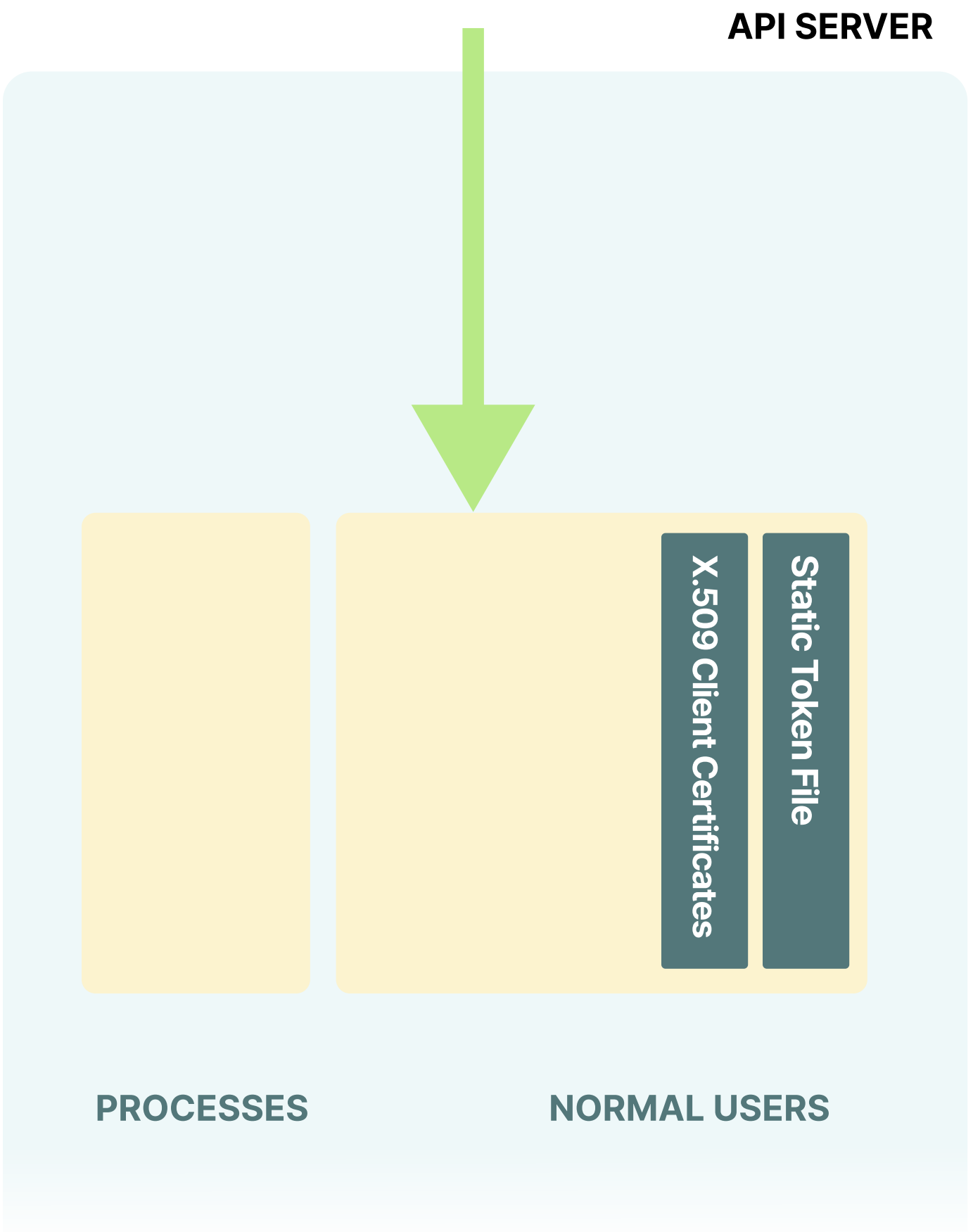
Authentication

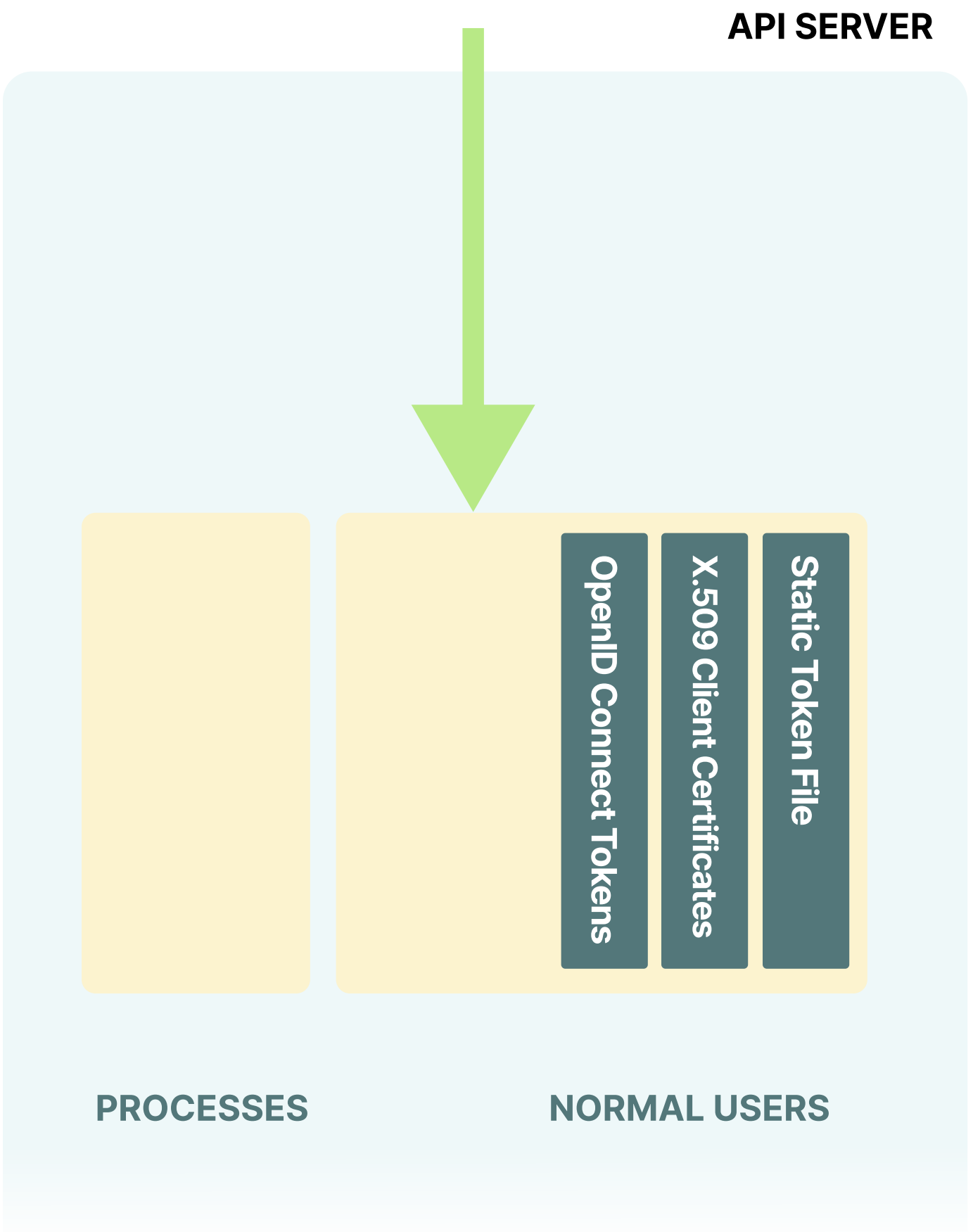
API SERVER

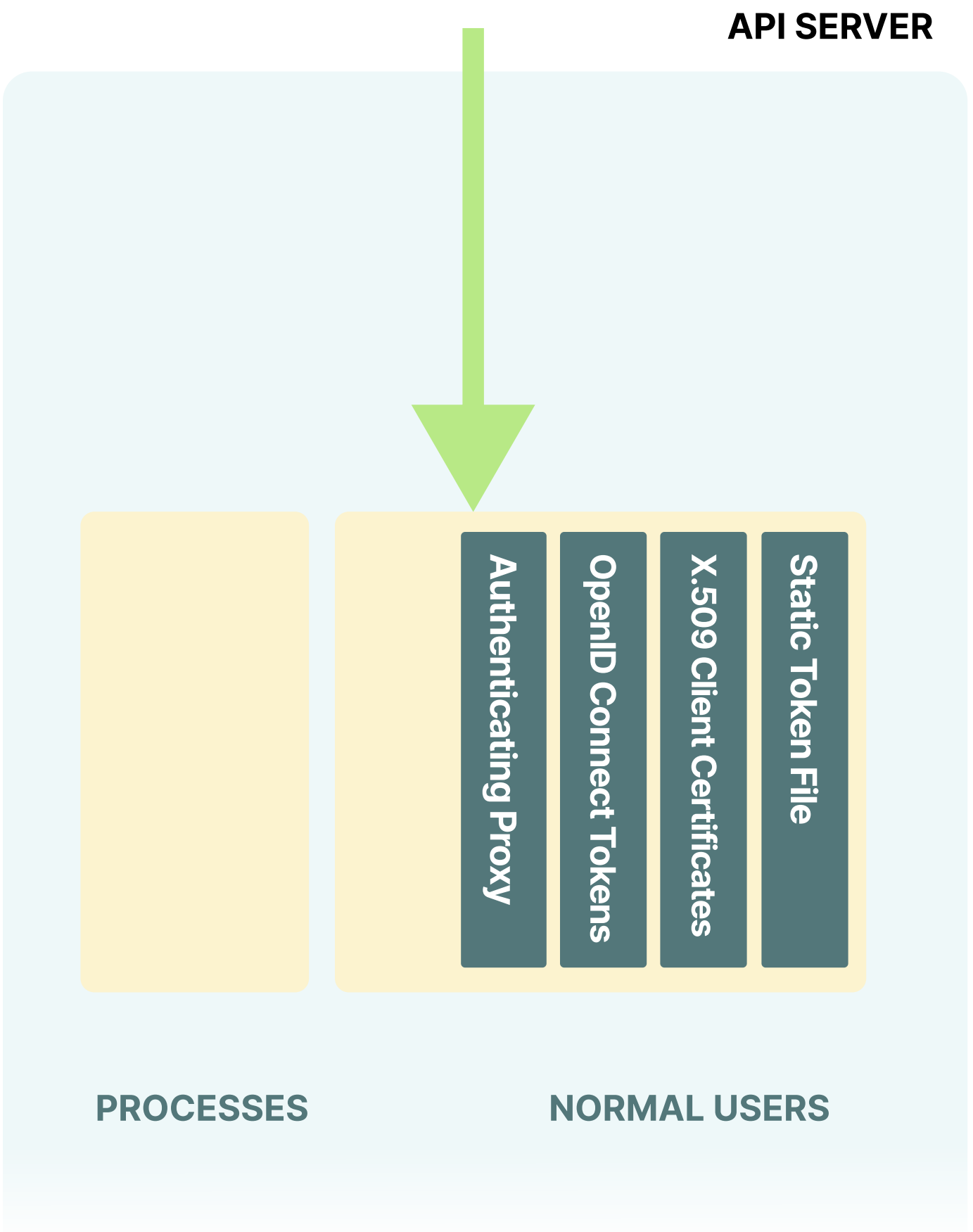


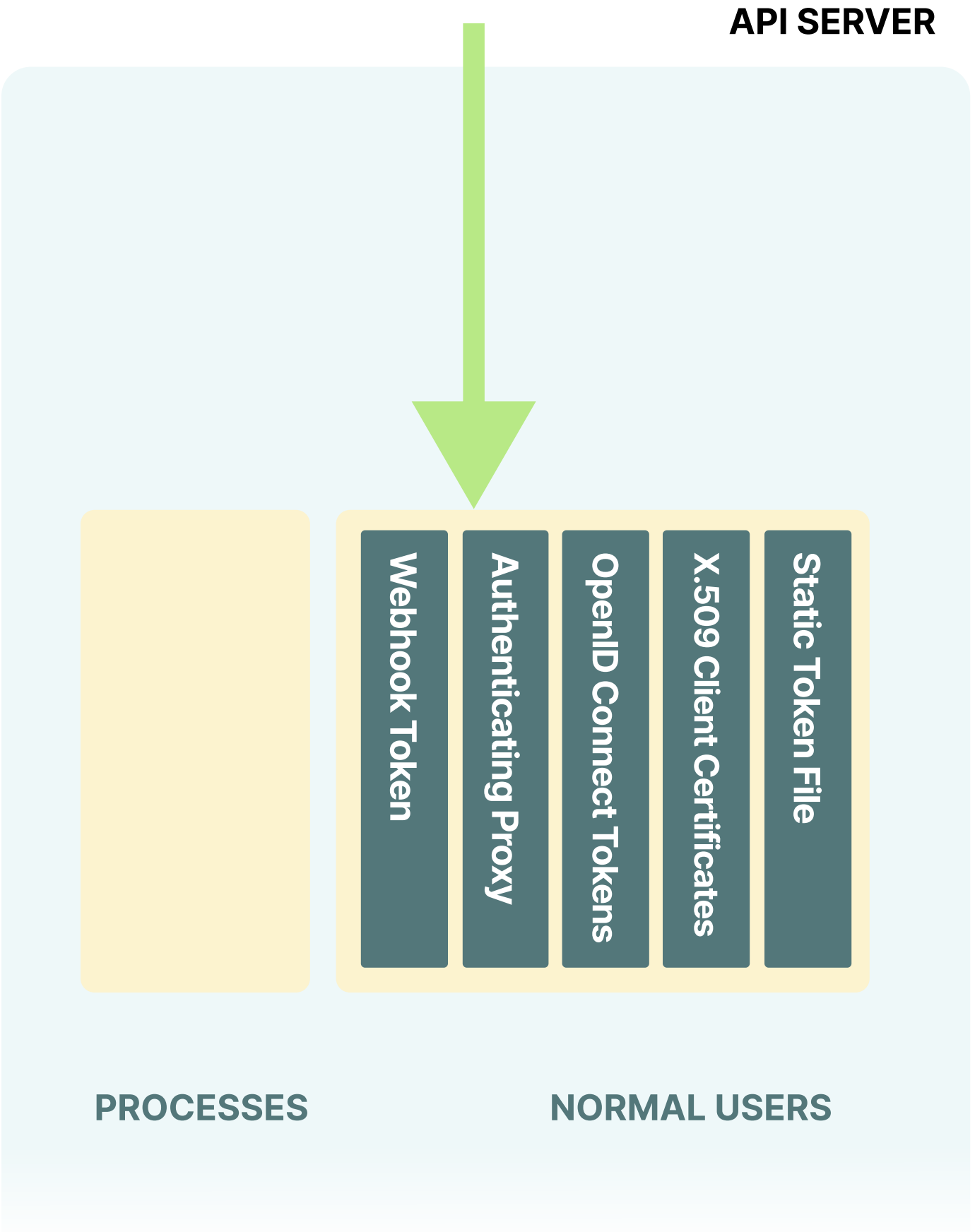

API SERVER
PROCESSES
NORMAL USERS

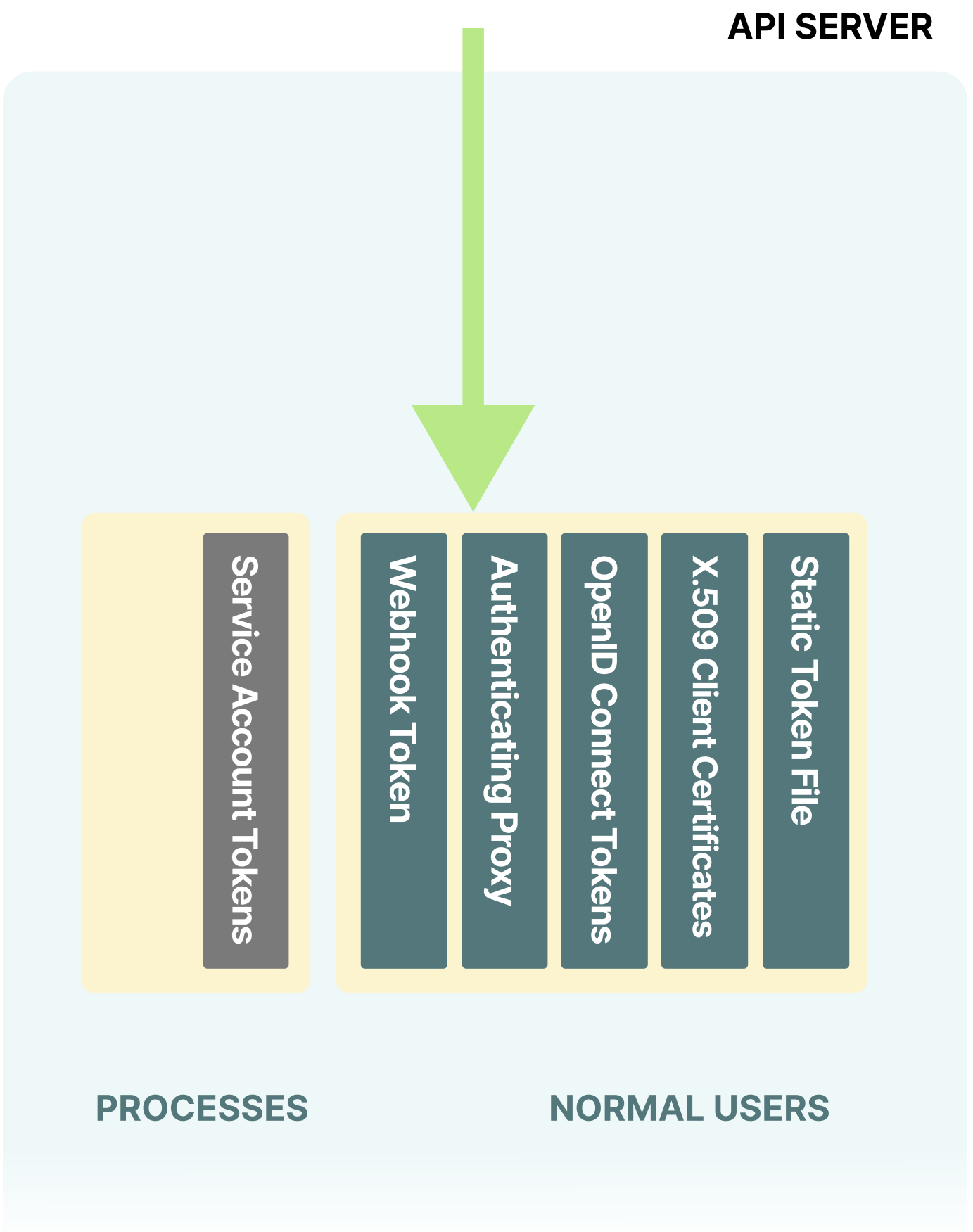

API SERVER
Static Token File
PROCESSES
NORMAL USERS

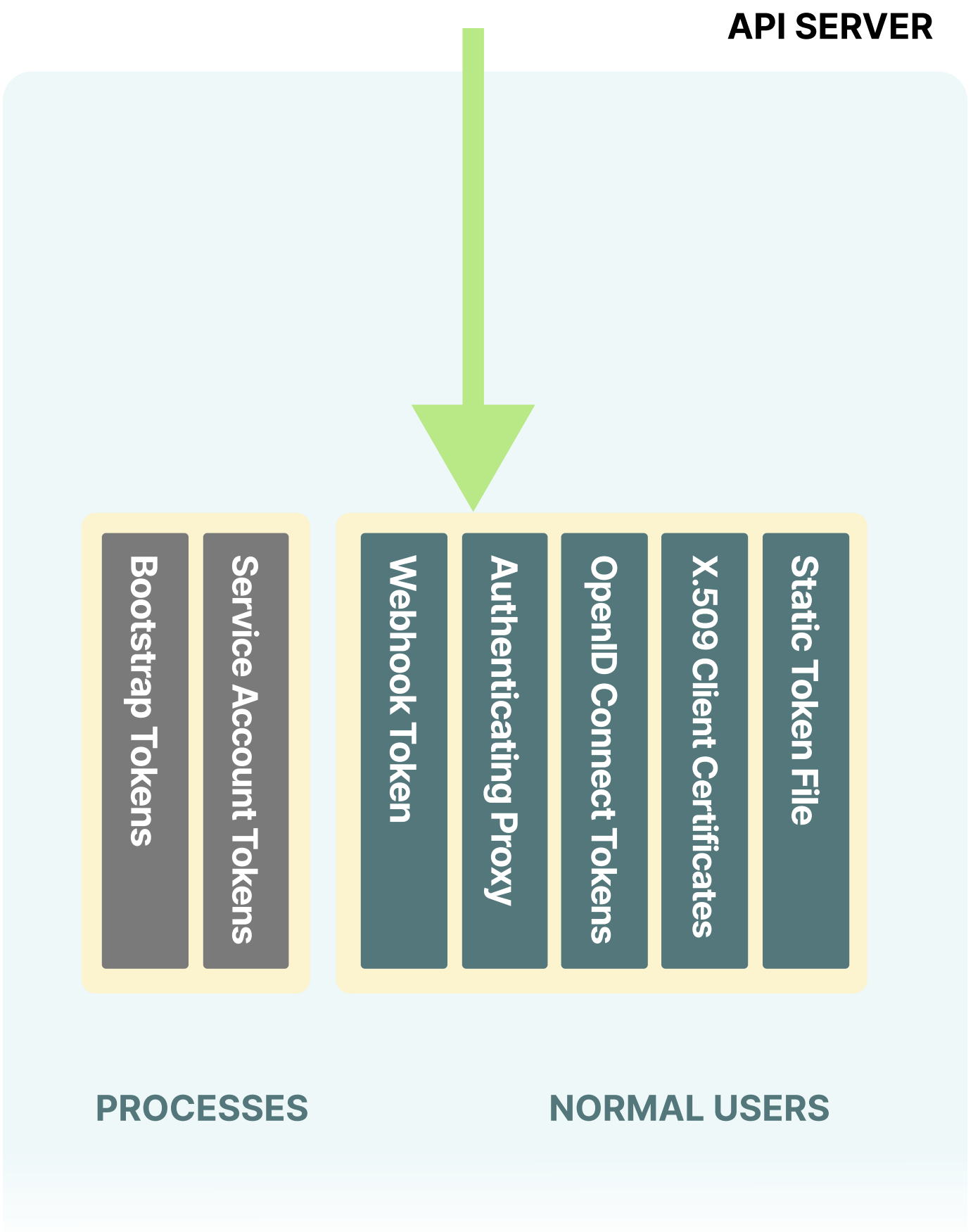

API SERVER
X.509 Client Certificates
Static Token File
PROCESSES
NORMAL USERS

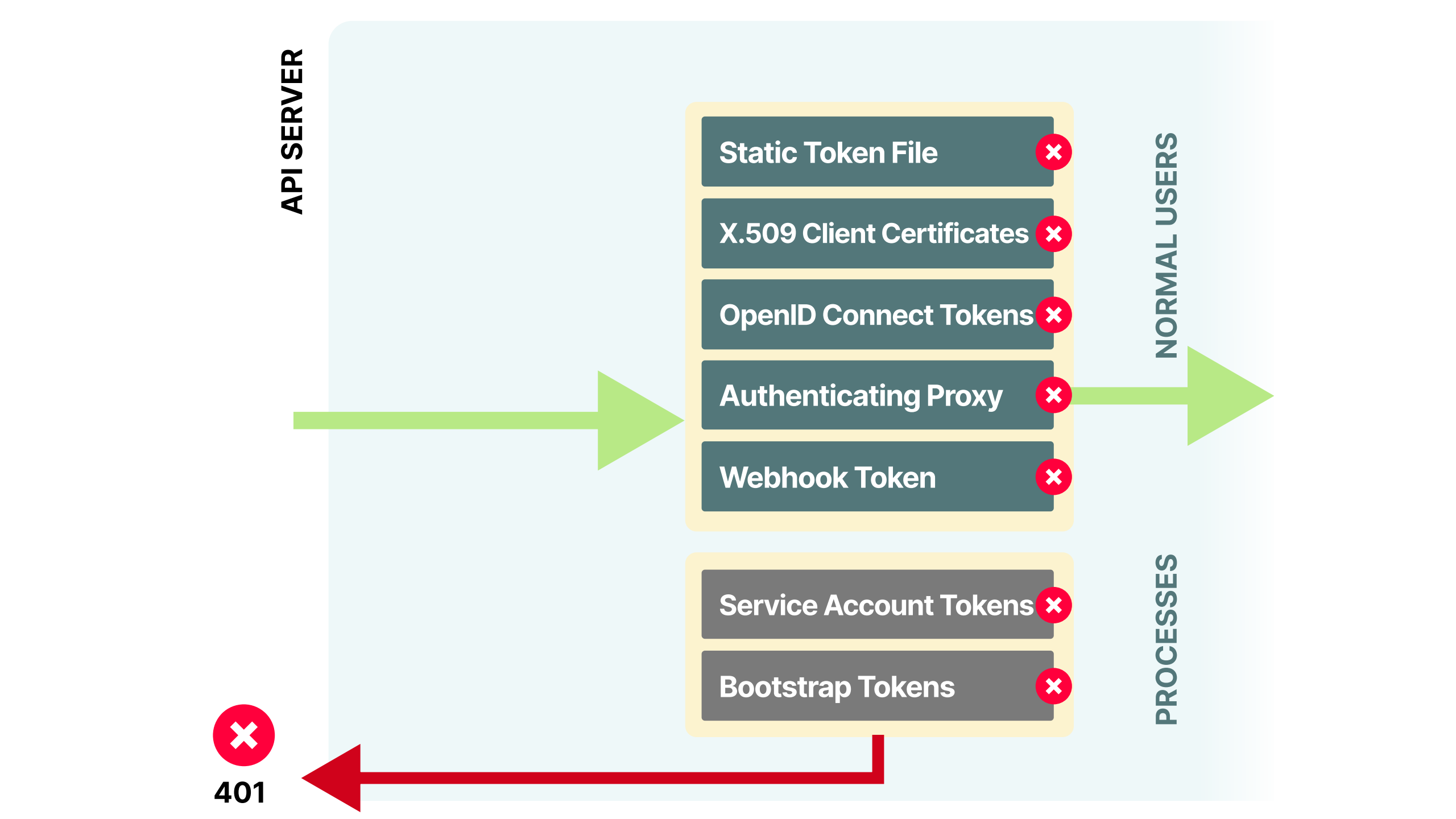

API SERVER
OpenID Connect Tokens
X.509 Client Certificates
Static Token File
PROCESSES
NORMAL USERS

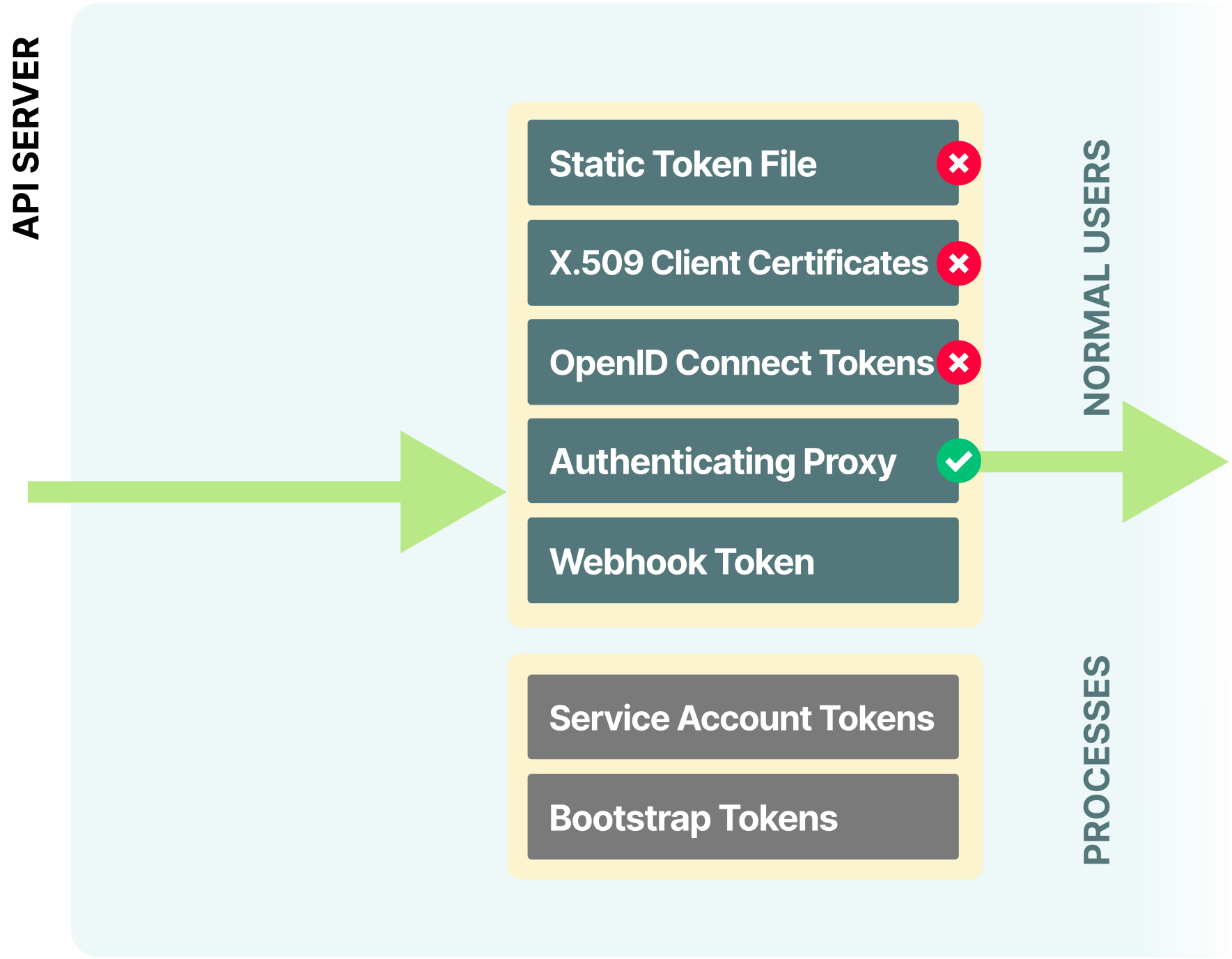

API SERVER
Authenticating Proxy
OpenID Connect Tokens
X.509 Client Certificates
Static Token File
PROCESSES
NORMAL USERS


API SERVER
Webhook Token
Authenticating Proxy
OpenID Connect Tokens
X.509 Client Certificates
Static Token File
PROCESSES
NORMAL USERS


API SERVER
Service Account Tokens
Webhook Token
Authenticating Proxy
OpenID Connect Tokens
X.509 Client Certificates
Static Token File
PROCESSES
NORMAL USERS

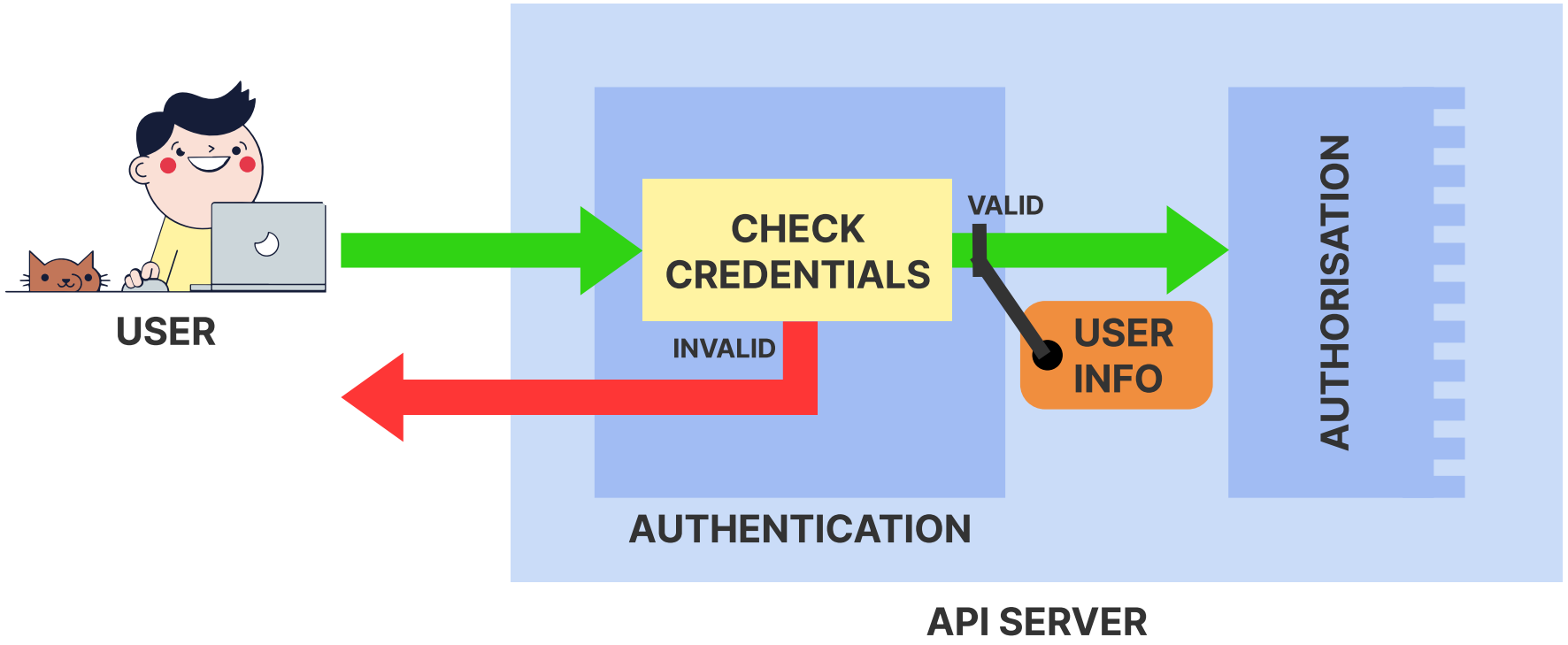

API SERVER
Bootstrap Tokens
Service Account Tokens
Webhook Token
Authenticating Proxy
OpenID Connect Tokens
X.509 Client Certificates
Static Token File
PROCESSES
NORMAL USERS

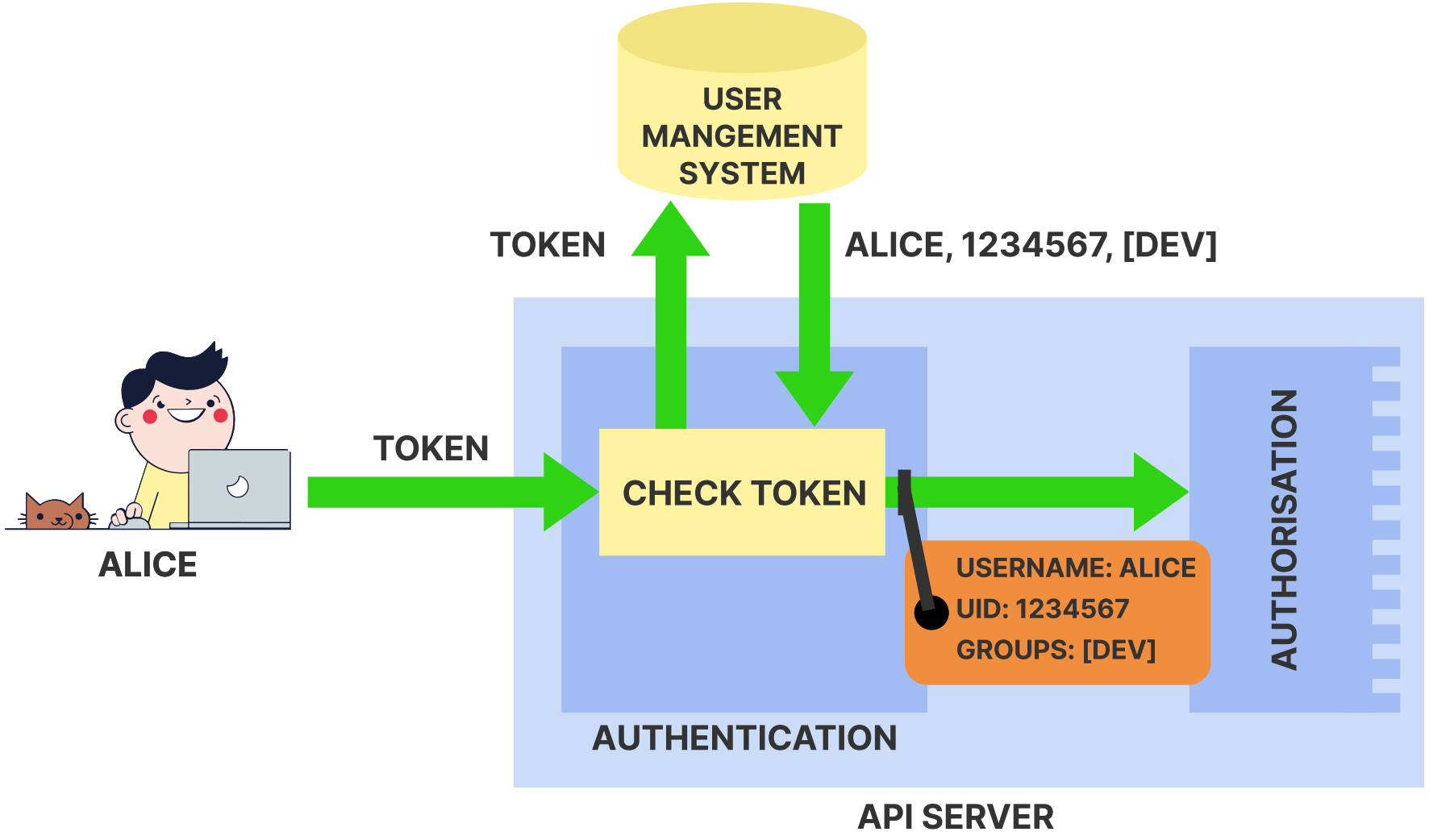

API SERVER
Static Token File
X.509 Client Certificates
OpenID Connect Tokens
Authenticating Proxy
Webhook Token
NORMAL USERS
Service Account Tokens
Bootstrap Tokens
PROCESSES
401


API SERVER
Static Token File
X.509 Client Certificates
OpenID Connect Tokens
Authenticating Proxy
Webhook Token
NORMAL USERS
Service Account Tokens
Bootstrap Tokens
PROCESSES

User info

Authentication


USER
CHECK CREDENTIALS
VALID
INVALID
USER INFO
AUTHORISATION
AUTHENTICATION
API SERVER


USER
MANGEMENT
SYSTEM
TOKEN
ALICE, 1234567, [DEV]
TOKEN
CHECK TOKEN
USERNAME: ALICE
UID: 1234567
GROUPS: [DEV]
AUTHORISATION
AUTHENTICATION
ALICE
API SERVER

User info

Username

UID

Groups (optional)

Extras (optional)

User info

Username

UID

Groups (optional)

Extras (optional)

User info

Username

UID

Groups (optional)

Extras (optional)

User info

Username

UID

Groups (optional)

Extras (optional)

Recap

kubectl and authetication

Recap — kubectl and authetication

1. Kubectl
2. API discovery and negotiation
3. API groups, versions and resources
4. Authentication plugins
5. User types
6. User info

Recap — kubectl and authetication

1. Kubectl
2. API discovery and negotiation
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6. User info

Recap — kubectl and authetication

1. Kubectl
2. API discovery and negotiation
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6. User info

Recap — kubectl and authetication

1. Kubectl
2. API discovery and negotiation
3. API groups, versions and resources
4. **Authentication plugins**
5. User types
6. User info

Recap — kubectl and authetication

1. Kubectl
2. API discovery and negotiation
3. API groups, versions and resources
4. Authentication plugins
5. User types
6. User info

Recap — kubectl and authetication

1. Kubectl
2. API discovery and negotiation
3. API groups, versions and resources
4. Authentication plugins
5. User types
6. User info

Authorization

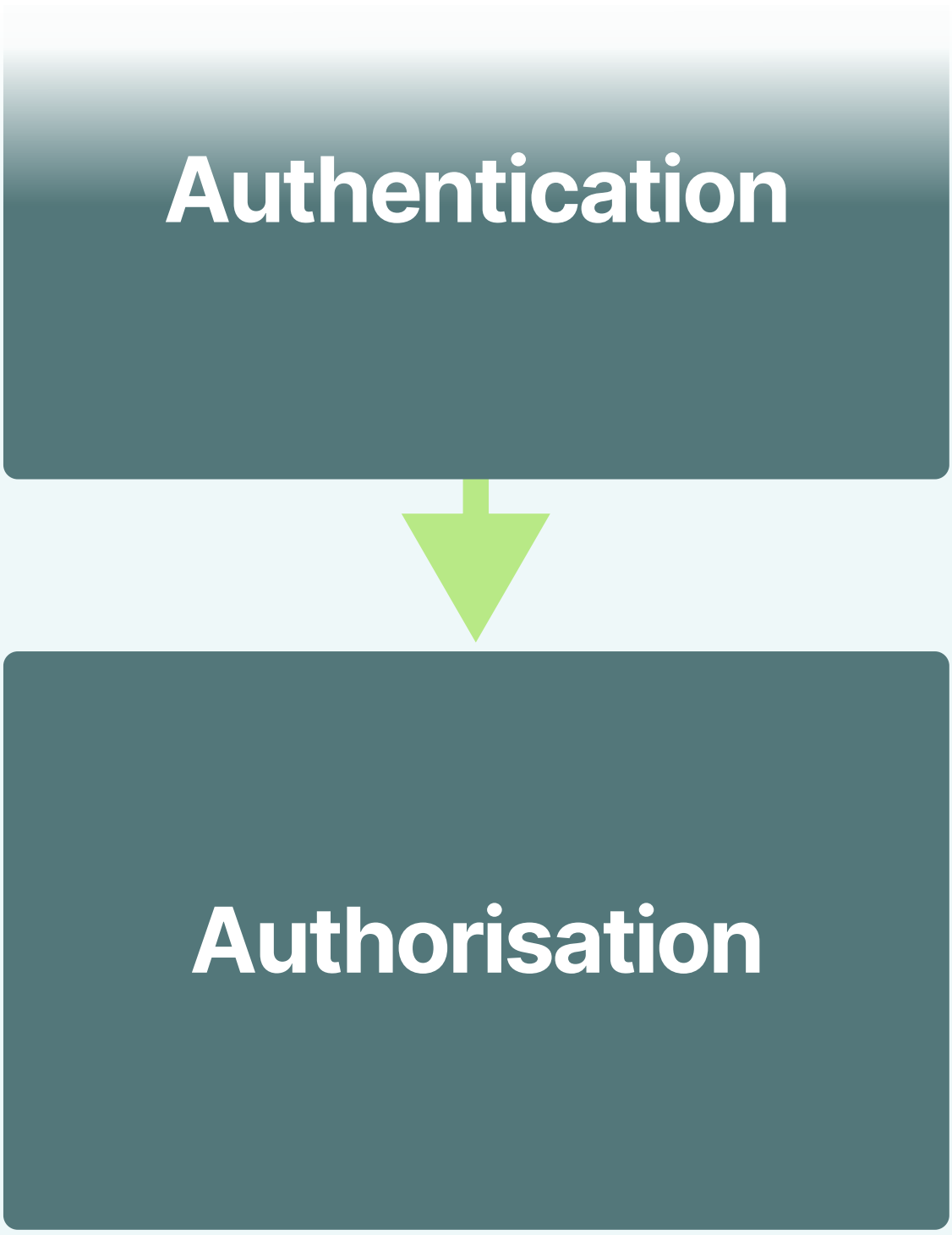
API SERVER

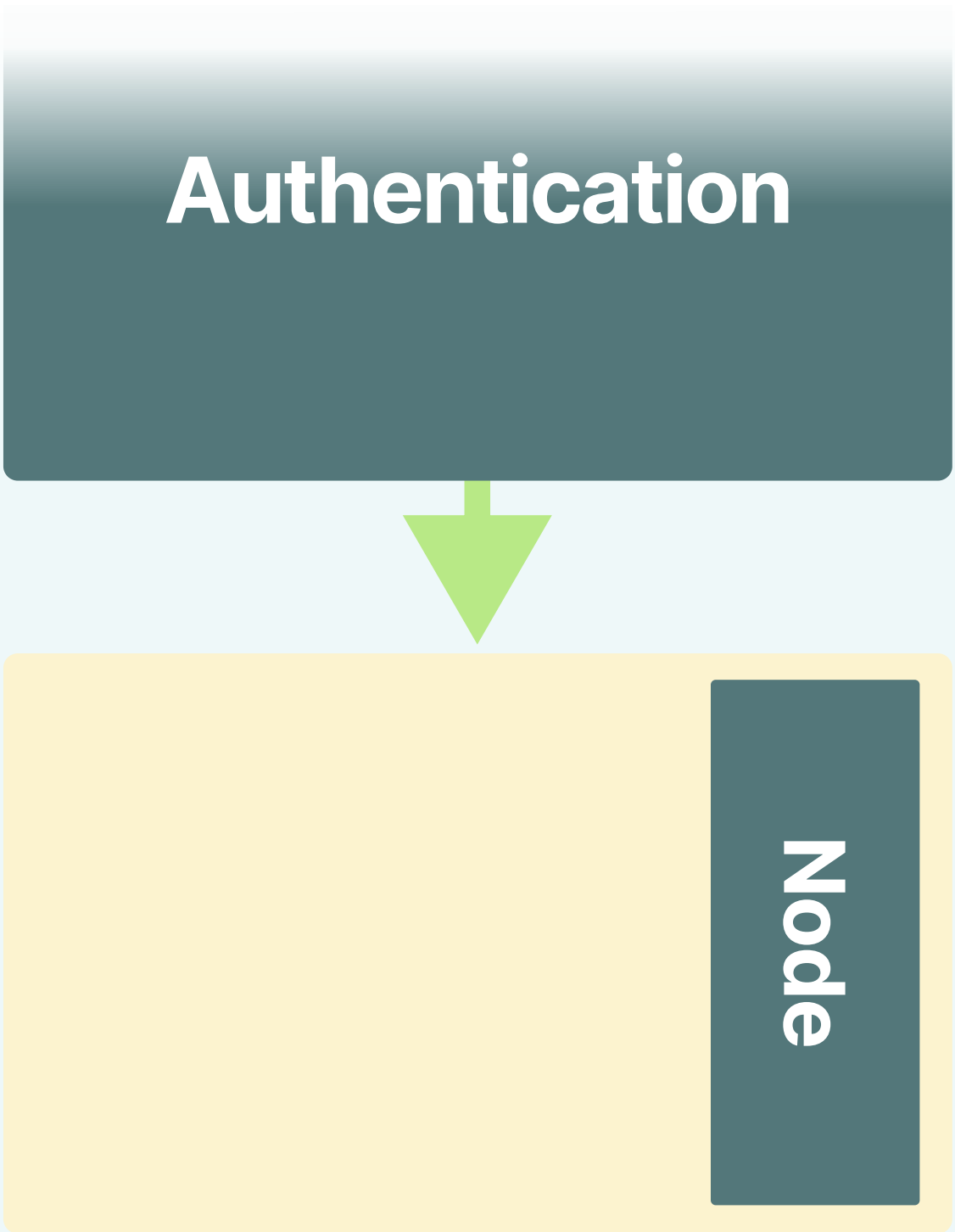


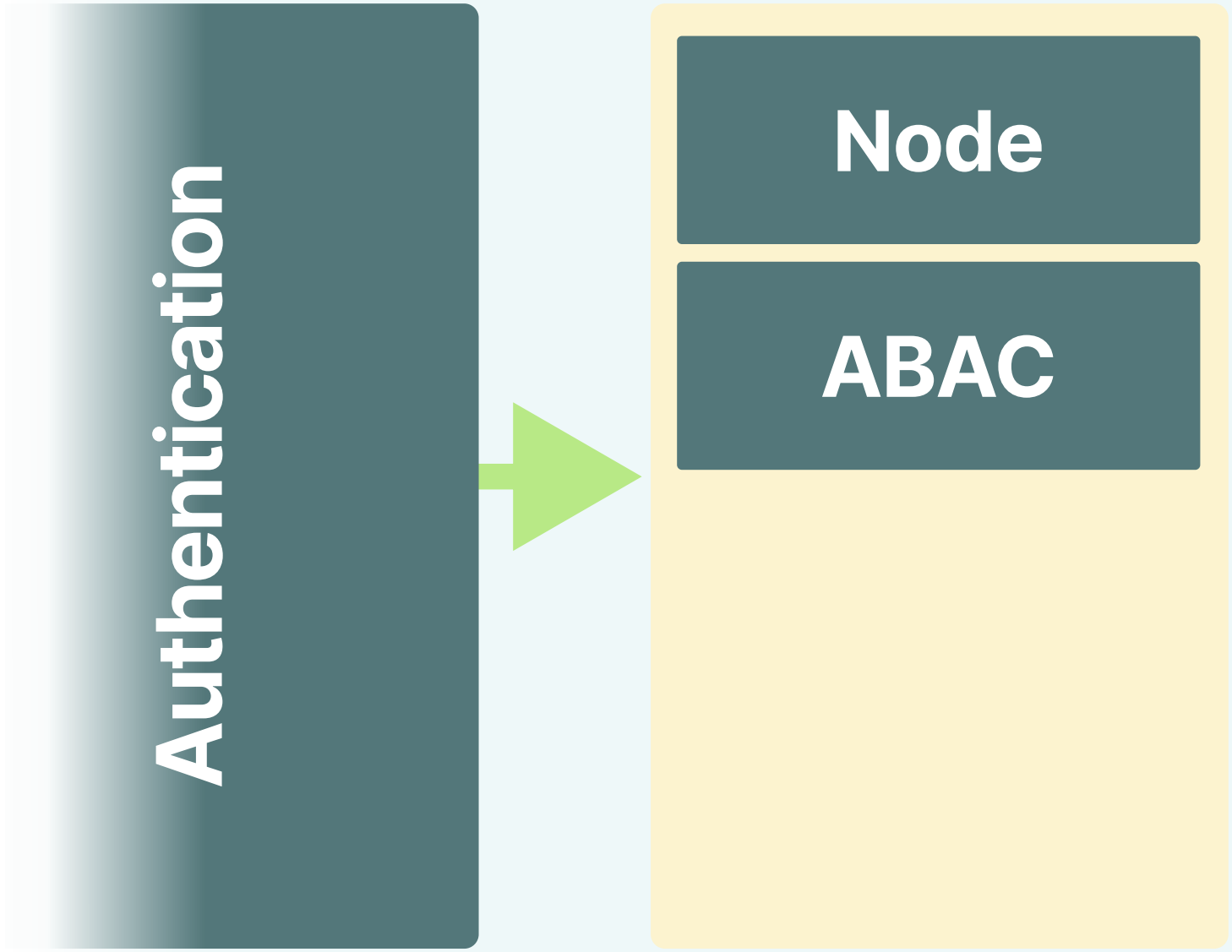


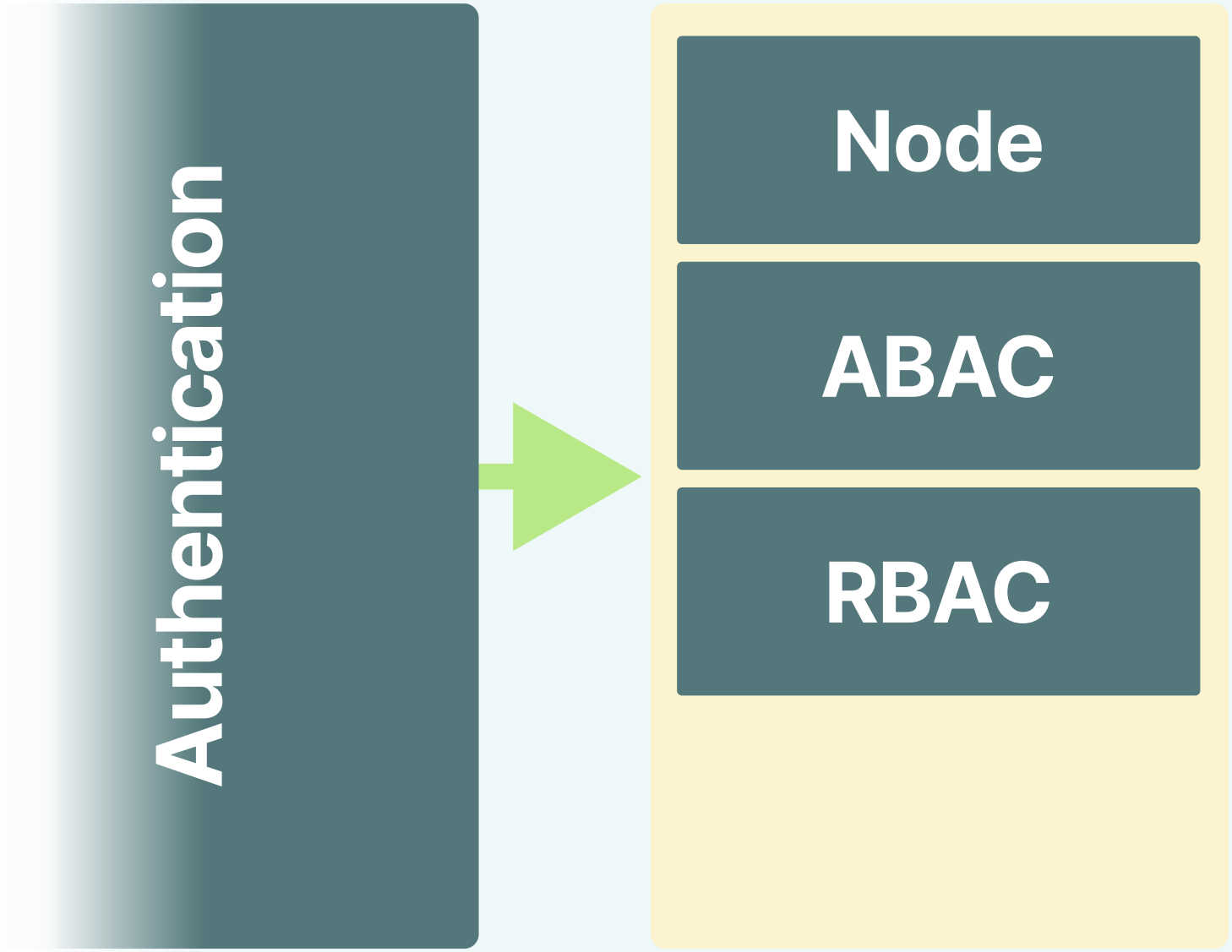
User info

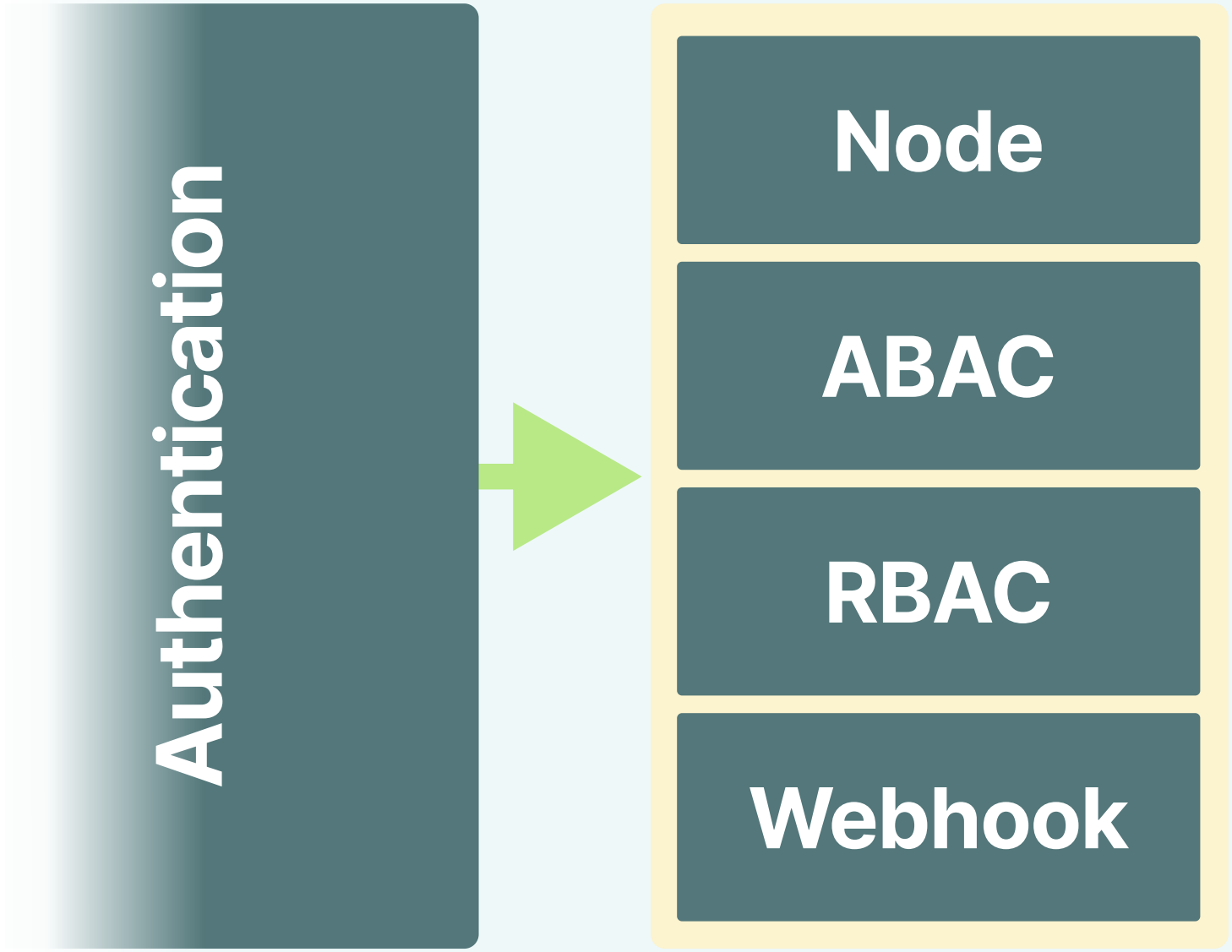
Username
UID
Groups (optional)
Extras (optional)

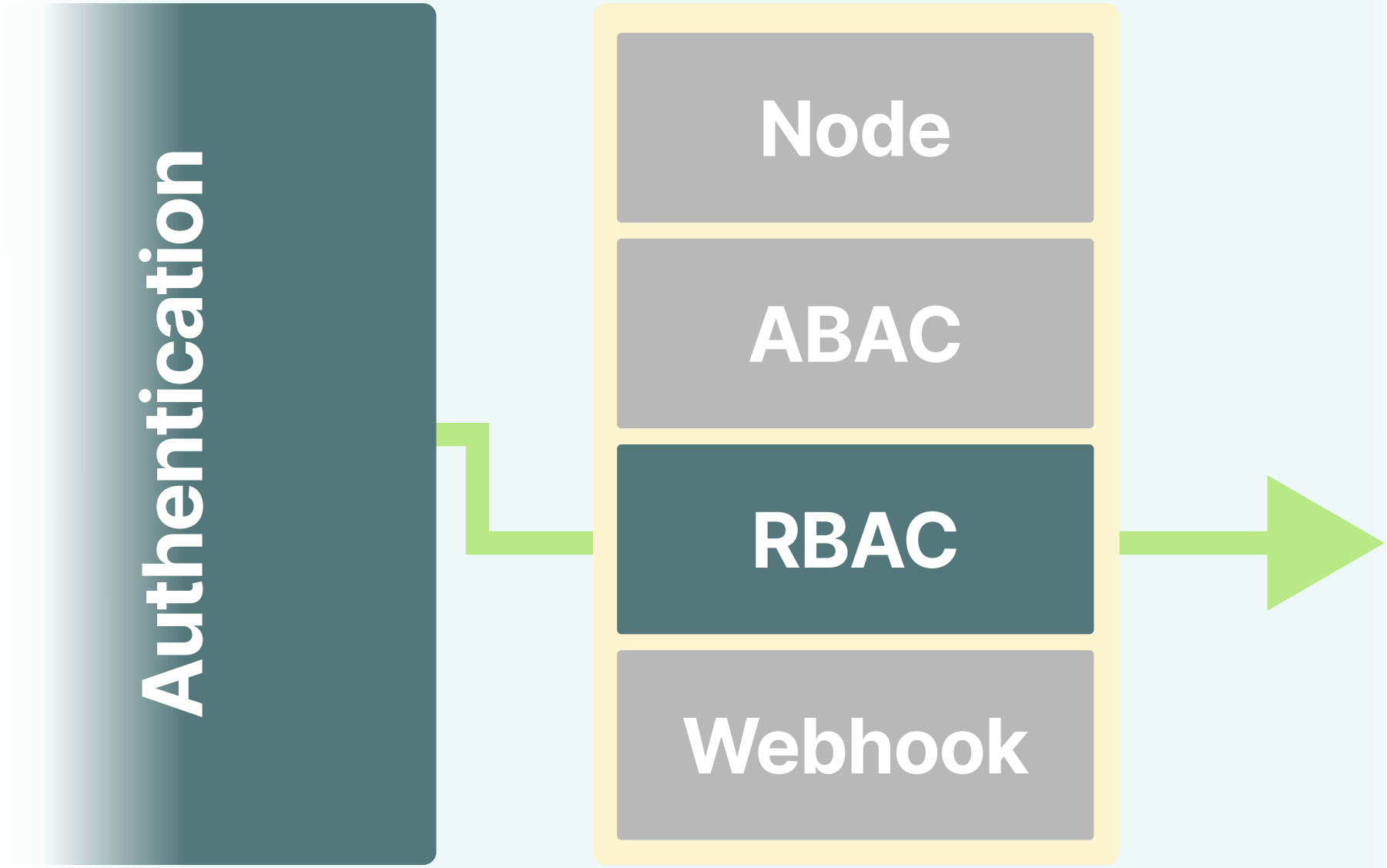

Authentication
Authorisation


Authentication
Node


Authentication
Node
ABAC


Authentication
Node
ABAC
RBAC


Authentication
Node
ABAC
RBAC
Webhook


Authentication
Node
ABAC
RBAC
Webhook

Role-based access control

UID	CLUSTER ROLE	PODS		DEPLOYMENT	
		read	write	read	write
1	admin1	✓	✓	✓	✓
2	debug	✓	✓	✗	✗
3	reviewer	✓	✗	✓	✗



Identity1

Normal user



Identity2

Service Account

UID	CLUSTER ROLE	PODS read	PODS write	DEPLOYMENT read	DEPLOYMENT write
1	admin1	✓	✓	✓	✓
2	debug	✓	✓	✗	✗
3	reviewer	✓	✗	✓	✗

UID	ROLE	PODS read	PODS write	DEPLOYMENT read	DEPLOYMENT write
1	teamA	✓	✓	✓	✓
2	QA	✓	✗	✓	✗

DEV namespace



Identity1

Normal user



Identity2

Service Account

UID	CLUSTER ROLE	PODS read	PODS write	DEPLOYMENT read	DEPLOYMENT write
1	admin1	✓	✓	✓	✓
2	debug	✓	✓	✗	✗
3	reviewer	✓	✗	✓	✗

UID	ROLE	PODS read	PODS write	DEPLOYMENT read	DEPLOYMENT write
1	teamA	✓	✓	✓	✓
2	QA	✓	✗	✓	✗

DEV namespace

ClusterRoleBinding



Identity1

Normal user



Identity2

Service Account

UID	CLUSTER ROLE	PODS read	PODS write	DEPLOYMENT read	DEPLOYMENT write
1	admin1	✓	✓	✓	✓
2	debug	✓	✓	✗	✗
3	reviewer	✓	✗	✓	✗

UID	ROLE	PODS read	PODS write	DEPLOYMENT read	DEPLOYMENT write
1	teamA	✓	✓	✓	✓
2	QA	✓	✗	✓	✗

DEV namespace

ClusterRoleBinding

RoleBinding





Identity1

Normal user

Identity2

Service Account

ClusterRole & Role

Role-based access control

```
~$ cat role.yaml
apiVersion: rbac.authorization.k8s.io/v1
kind: Role
metadata:
  name: pod-reader
  namespace: dev
rules:
- apiGroups: [""]
  resources: ["pods"]
  verbs: ["get", "watch", "list"]
- apiGroups: ["apps"]
  resources: ["deployments"]
  verbs: ["update", "patch"]
```

```
~$ cat role.yaml
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
  name: pod-reader
  namespace: dev
rules:
- apiGroups: [""]
  resources: ["pods"]
  verbs: ["get", "watch", "list"]
- apiGroups: ["apps"]
  resources: ["deployments"]
  verbs: ["update", "patch"]
```

```
~$ cat role.yaml
apiVersion: rbac.authorization.k8s.io/v1
kind: Role
metadata:
  name: pod-reader
  namespace: dev
rules:
- apiGroups: [""]
  resources: ["pods"]
  verbs: ["get", "watch", "list"]
- apiGroups: ["apps"]
  resources: ["deployments"]
  verbs: ["update", "patch"]
```

```
~$ cat role.yaml
apiVersion: rbac.authorization.k8s.io/v1
kind: Role
metadata:
  name: pod-reader
  namespace: dev
rules:
- apiGroups: [""]
  resources: ["pods"]
  verbs: ["get", "watch", "list"]
- apiGroups: ["apps"]
  resources: ["deployments"]
  verbs: ["update", "patch"]
~$ _
```

ClusterRoleBinding & RoleBinding

Role-based access control

```
~$ cat role-binding.yaml
apiVersion: rbac.authorization.k8s.io/v1
kind: RoleBinding
metadata:
  name: team
  namespace: dev
subjects:
- kind: User
  name: jane
  apiGroup: rbac.authorization.k8s.io
roleRef:
  kind: Role
  name: pod-reader
  apiGroup: rbac.authorization.k8s.io
```

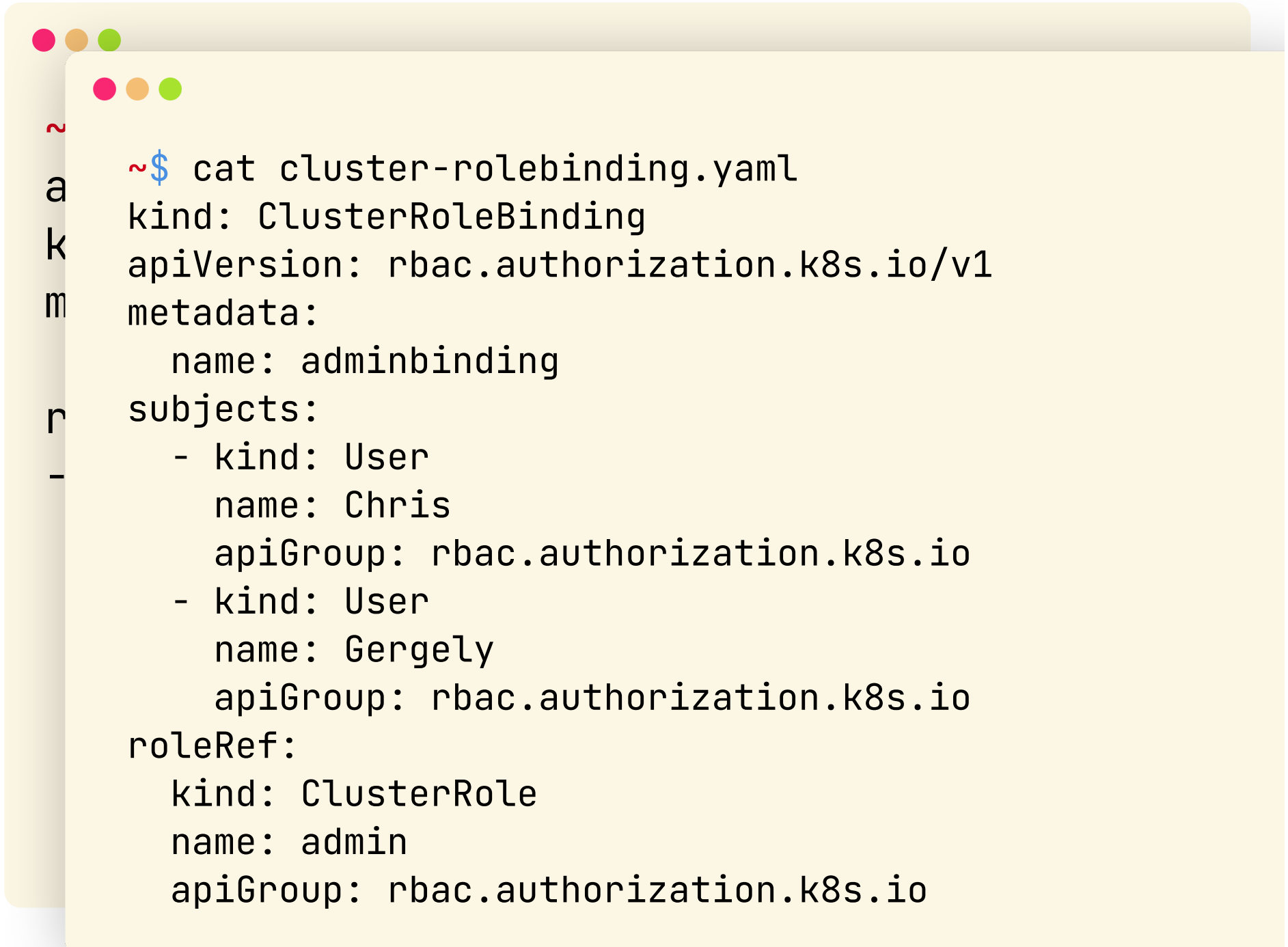
```
~$ cat role-binding.yaml
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRoleBinding
metadata:
  name: read-pods
  namespace: dev
subjects:
- kind: User
  name: jane
  apiGroup: rbac.authorization.k8s.io
roleRef:
  kind: Role
  name: pod-reader
  apiGroup: rbac.authorization.k8s.io
```

```
~$ cat role-binding.yaml
apiVersion: rbac.authorization.k8s.io/v1
kind: RoleBinding # or ClusterRoleBinding
metadata:
  name: read-pods
  namespace: dev
subjects:
- kind: User
  name: jane
  apiGroup: rbac.authorization.k8s.io
roleRef:
  kind: Role
  name: pod-reader
  apiGroup: rbac.authorization.k8s.io
```

```
~$ cat role-binding.yaml
apiVersion: rbac.authorization.k8s.io/v1
kind: RoleBinding # or ClusterRoleBinding
metadata:
  name: read-pods
  namespace: dev
subjects:
- kind: User
  name: jane
  apiGroup: rbac.authorization.k8s.io
roleRef:
  kind: Role
  name: pod-reader
  apiGroup: rbac.authorization.k8s.io
```

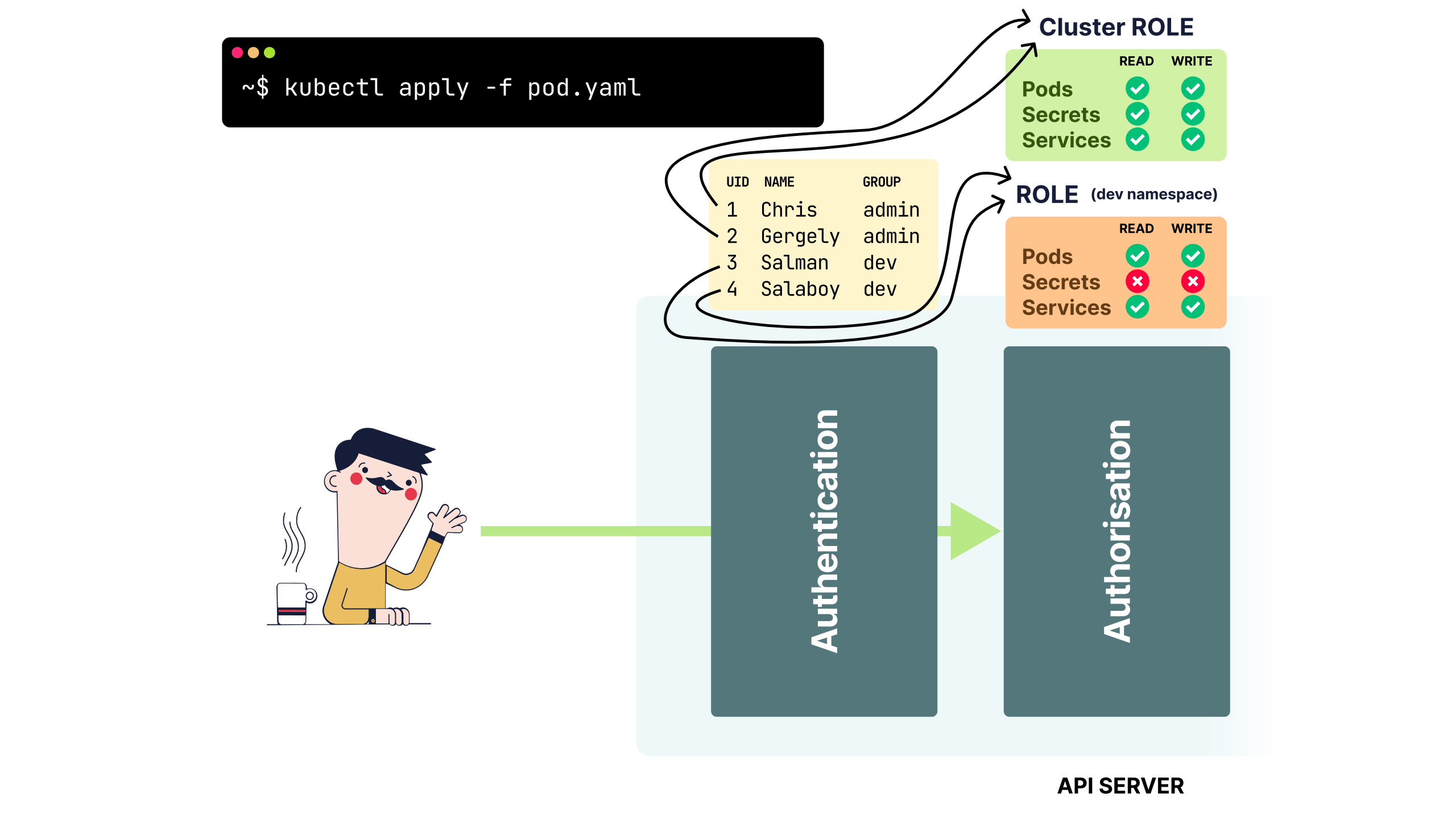
Authentication + authorization

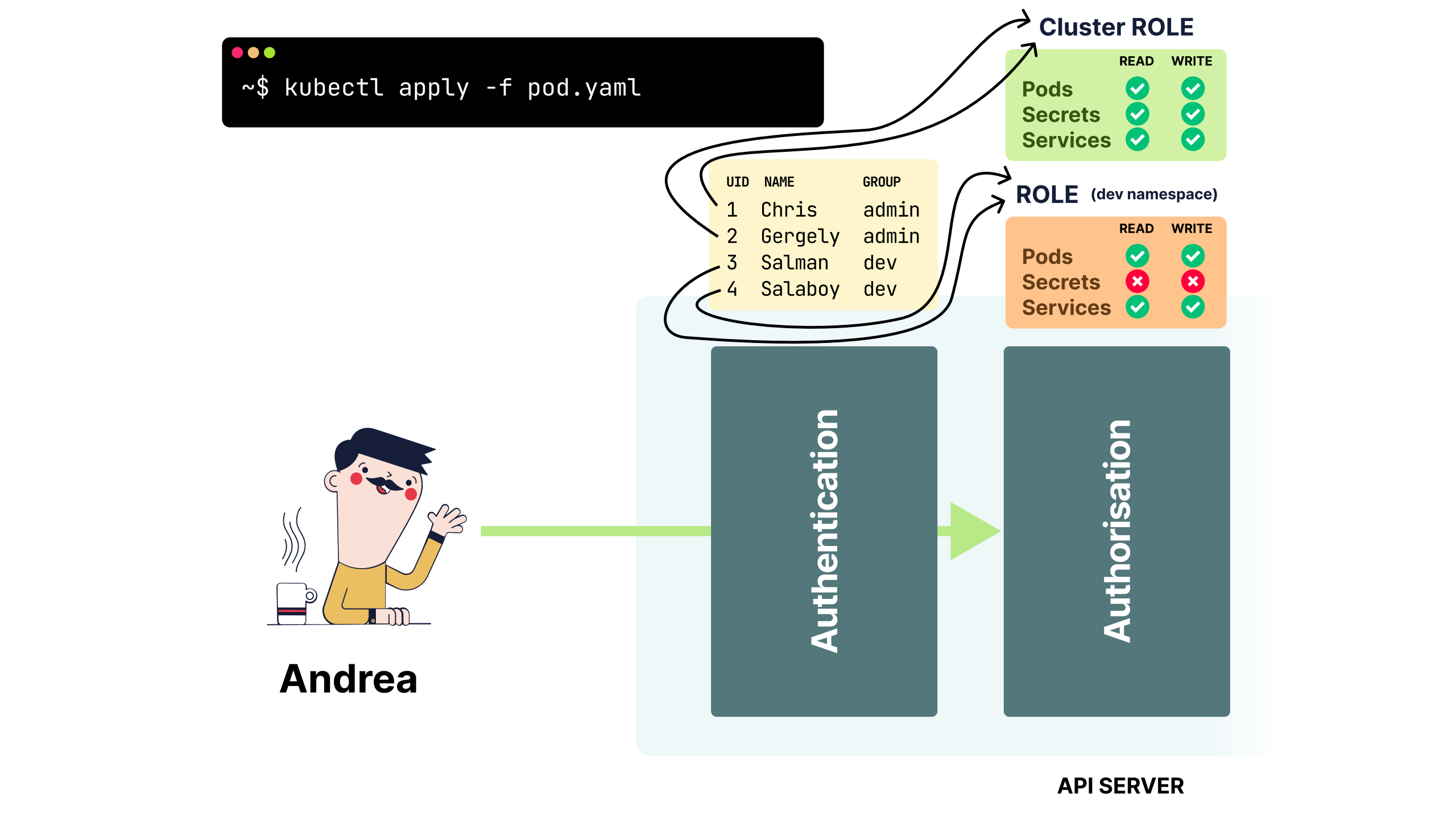
```
~$ cat cluster-role.yaml
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
  name: pod-reader
rules:
- apiGroups: [""]
  resources: ["pods", "secrets", "services"]
  verbs: ["get", "list", "update"]
```

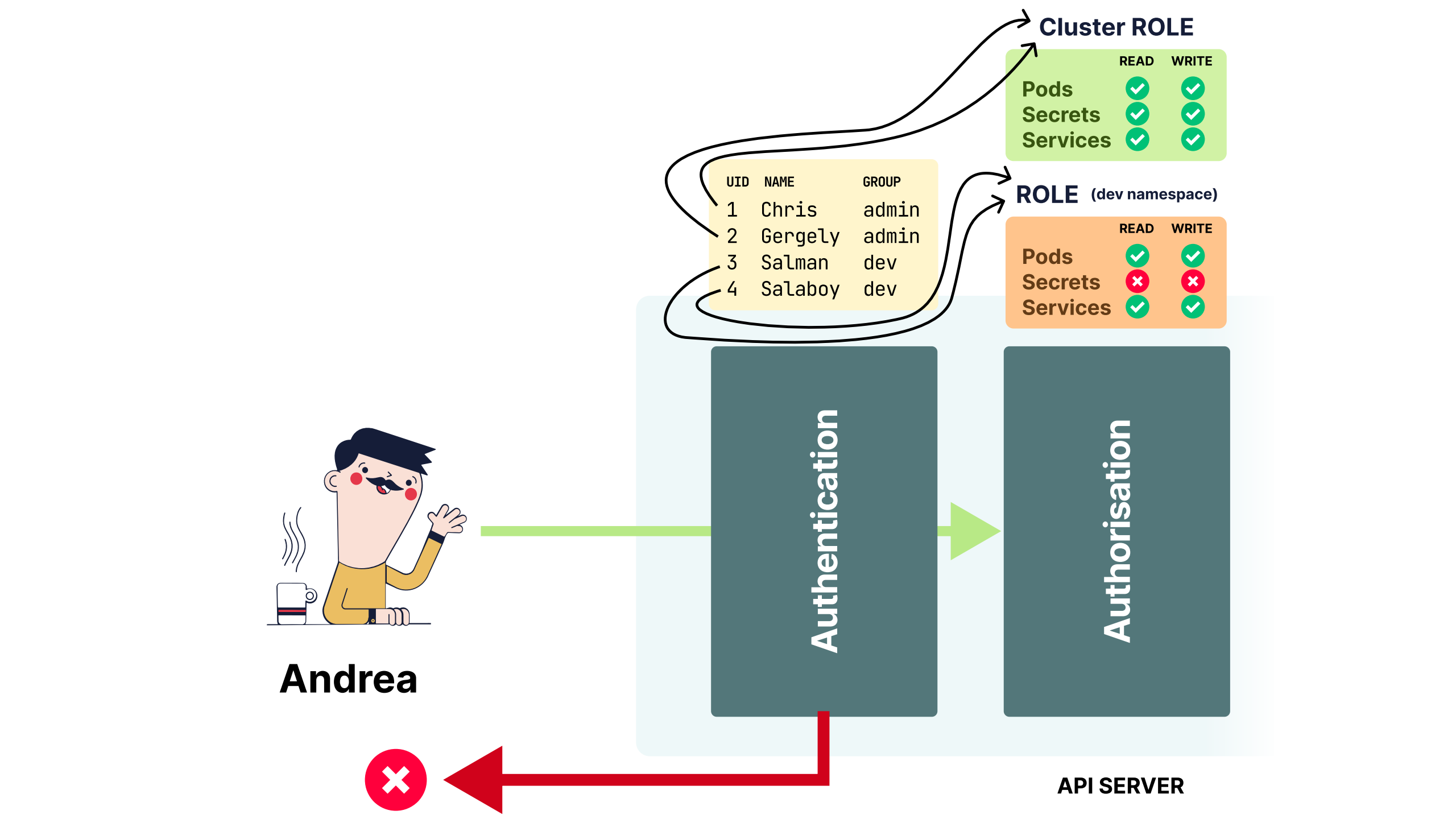

~$ cat cluster-rolebinding.yaml
kind: ClusterRoleBinding
apiVersion: rbac.authorization.k8s.io/v1
metadata:
name: adminbinding
subjects:
- kind: User
name: Chris
apiGroup: rbac.authorization.k8s.io
- kind: User
name: Gergely
apiGroup: rbac.authorization.k8s.io
roleRef:
kind: ClusterRole
name: admin
apiGroup: rbac.authorization.k8s.io

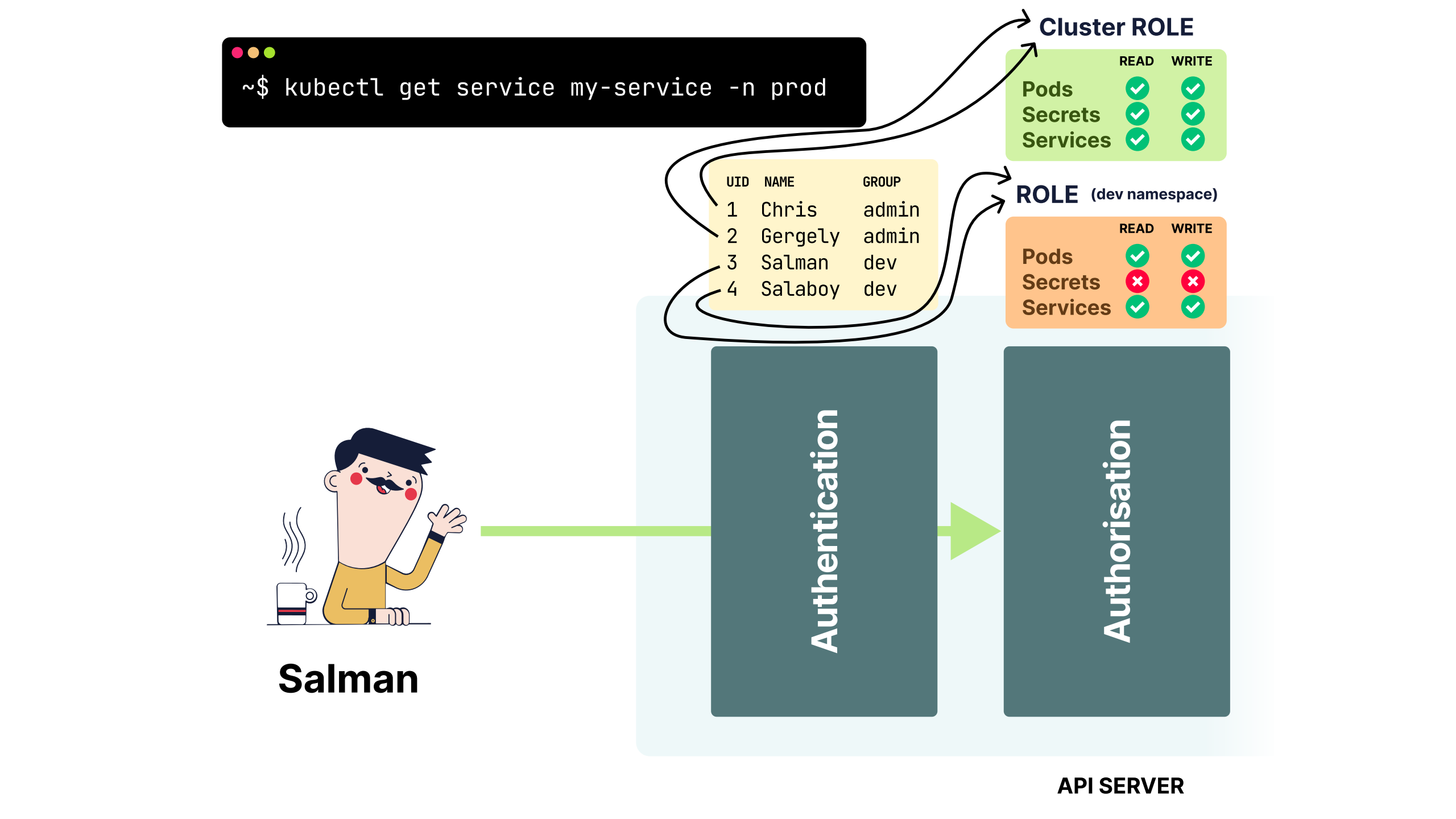
```
~$ cat role.yaml
apiVersion: rbac.authorization.k8s.io/v1
kind: Role
metadata:
  name: pod-reader
  namespace: dev
rules:
- apiGroups: [""]
  resources: ["pods", "services"]
  verbs: ["get", "list", "update"]
```

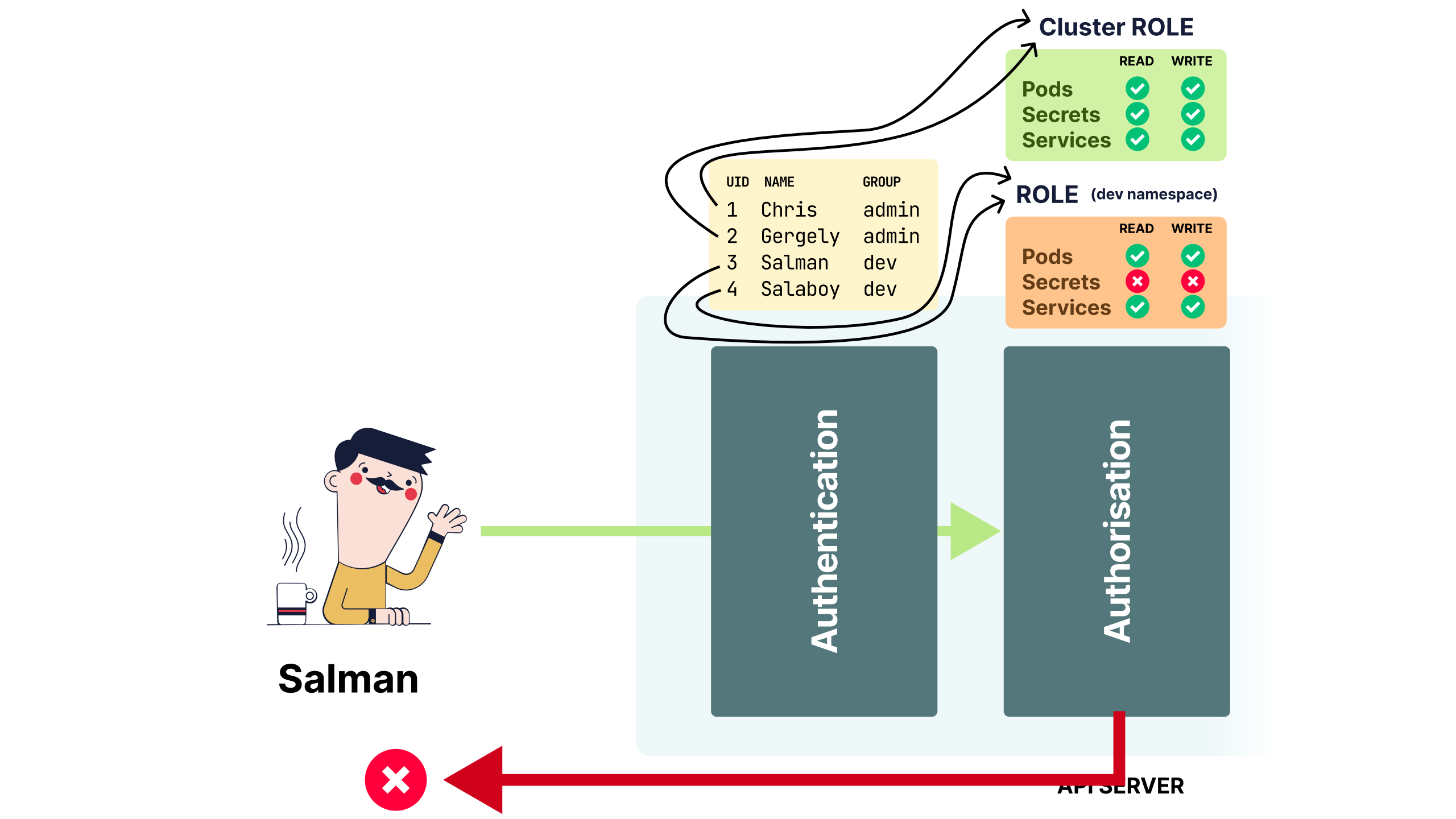
```
~$ cat rolebinding.yaml
kind: RoleBinding
apiVersion: rbac.authorization.k8s.io/v1
metadata:
  name: binding
subjects:
  - kind: User
    name: Salman
    apiGroup: rbac.authorization.k8s.io
  - kind: User
    name: Salaboy
    apiGroup: rbac.authorization.k8s.io
roleRef:
  kind: Role
  name: dev
  apiGroup: rbac.authorization.k8s.io
```

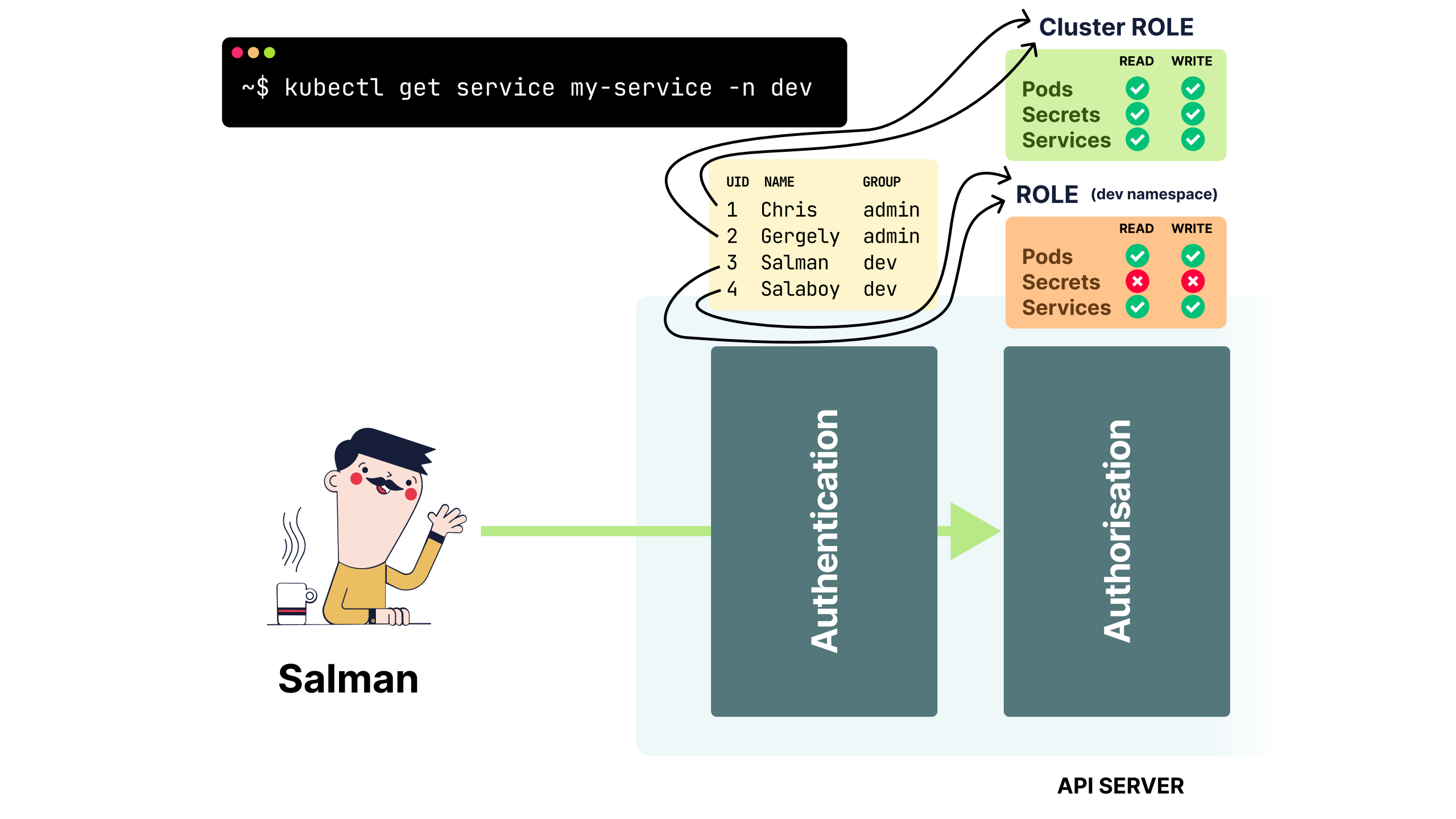

~$ kubectl apply -f pod.yaml
Cluster ROLE
READ
WRITE
Pods
Secrets
Services
UID NAME GROUP
1 Chris admin
2 Gergely admin
3 Salman dev
4 Salaboy dev
ROLE (dev namespace)
READ
WRITE
Pods
Secrets
Services
Authentication
Authorisation
API SERVER

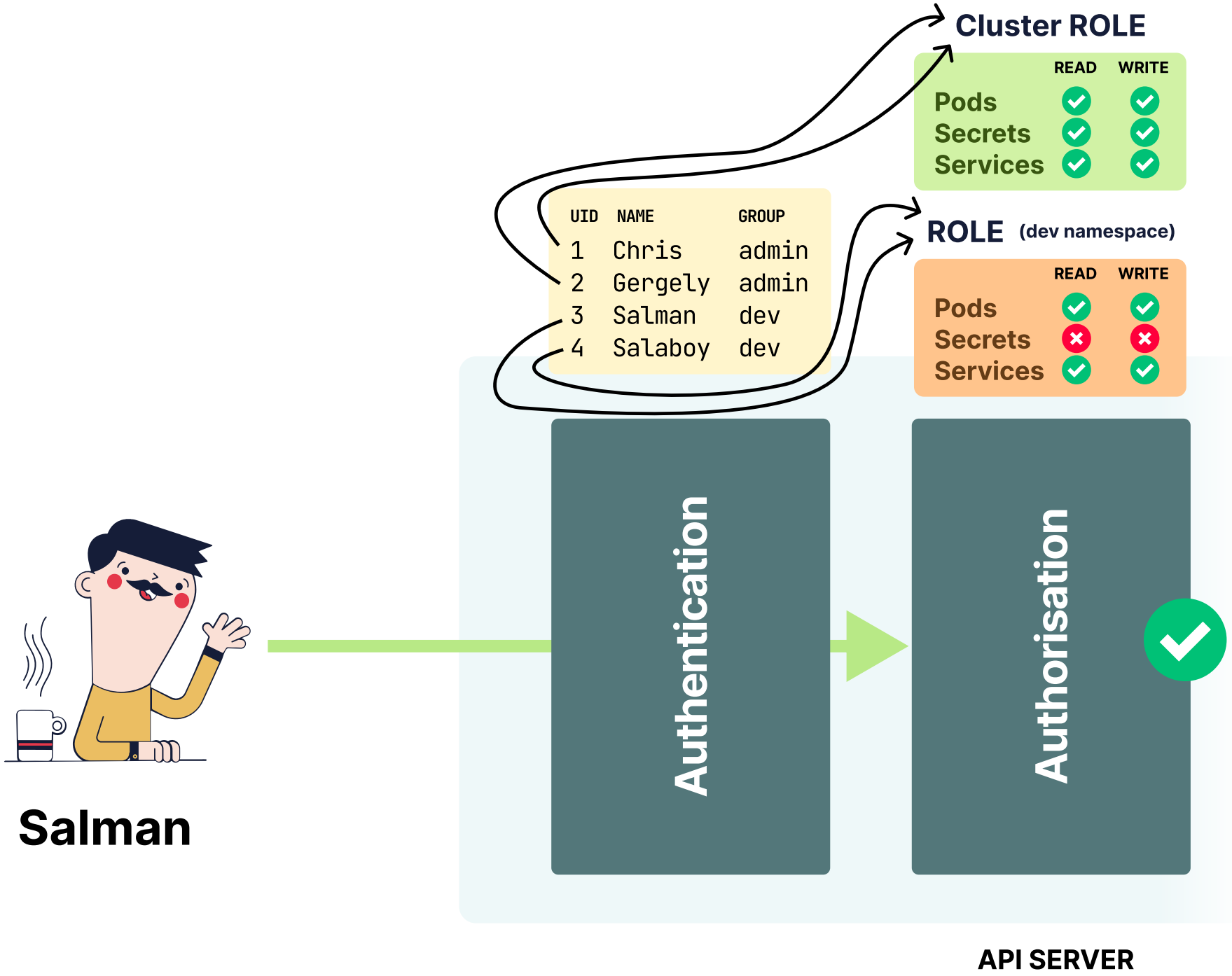

~$ kubectl apply -f pod.yaml
Cluster ROLE
READ
WRITE
Pods
Secrets
Services
UID NAME GROUP
1 Chris admin
2 Gergely admin
3 Salman dev
4 Salaboy dev
ROLE (dev namespace)
READ
WRITE
Pods
Secrets
Services
Authentication
Authorisation
Andrea
API SERVER

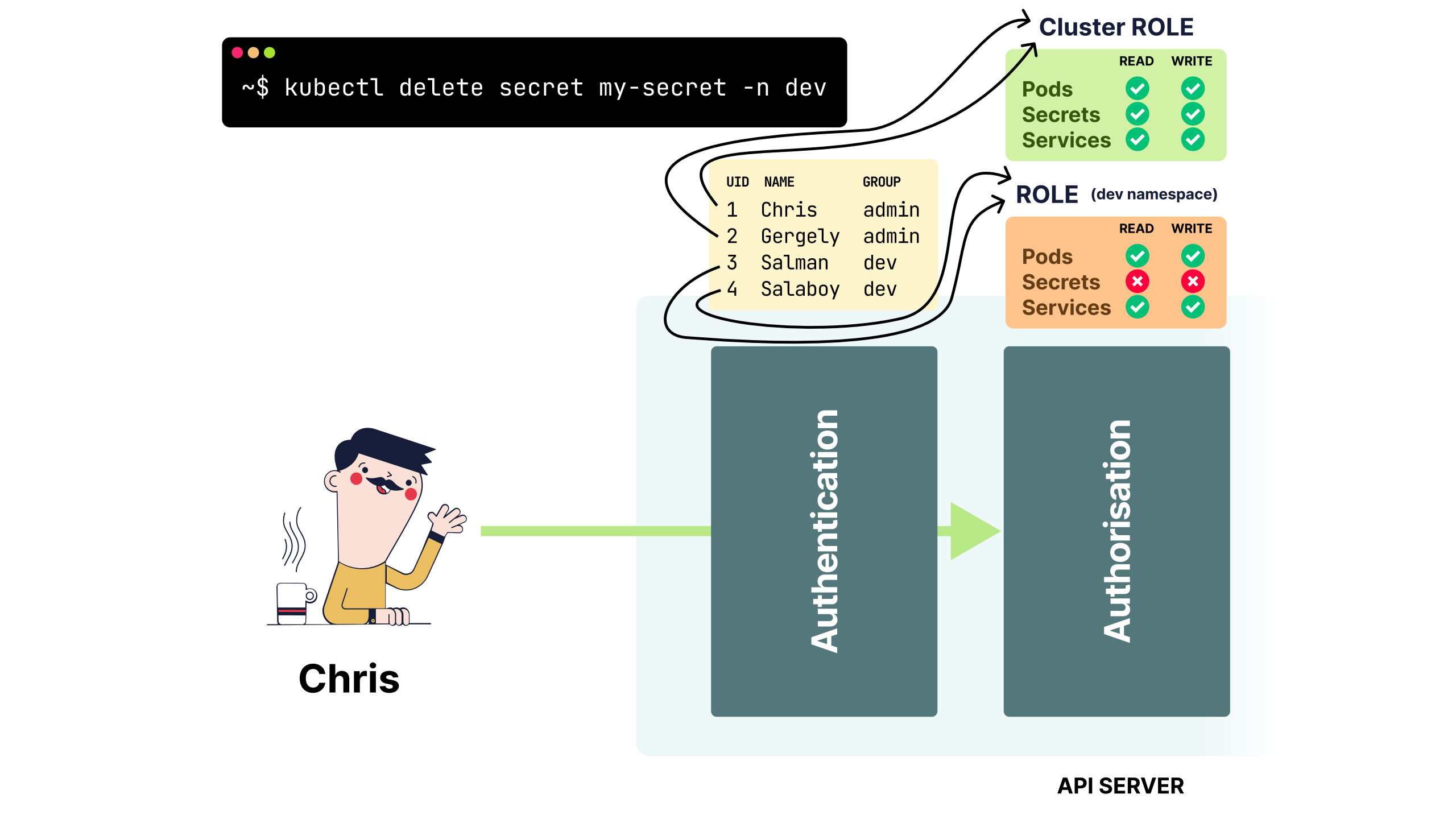

Cluster ROLE
READ
WRITE
Pods
Secrets
Services
UID NAME GROUP
1 Chris admin
2 Gergely admin
3 Salman dev
4 Salaboy dev
ROLE (dev namespace)
READ
WRITE
Pods
Secrets
Services
Authentication
Authorisation
Andrea
API SERVER

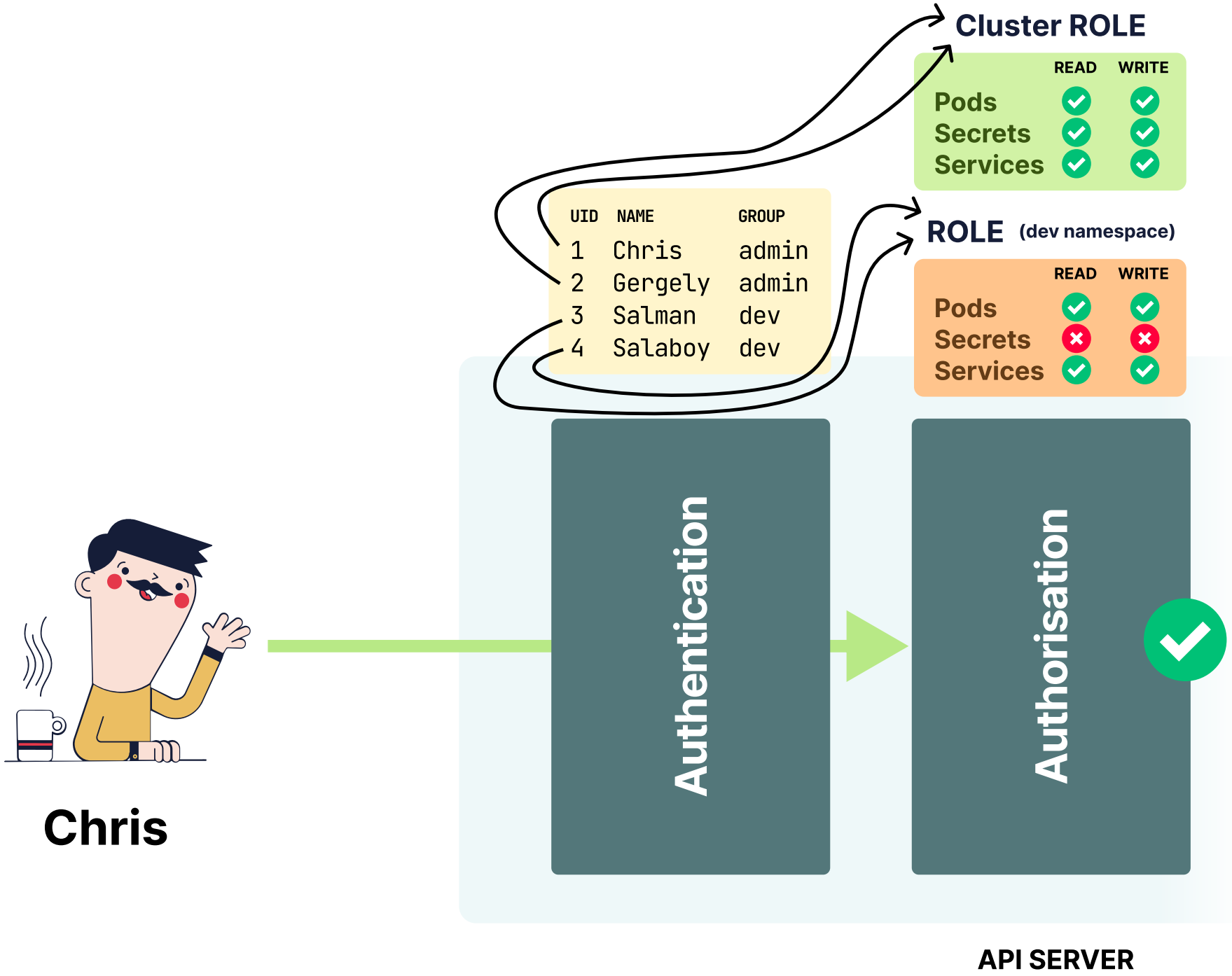

~$ kubectl get service my-service -n prod
Cluster ROLE
READ
WRITE
Pods
Secrets
Services
UID NAME GROUP
1 Chris admin
2 Gergely admin
3 Salman dev
4 Salaboy dev
ROLE (dev namespace)
READ
WRITE
Pods
Secrets
Services
Authentication
Authorisation
Salman
API SERVER


Cluster ROLE
READ
WRITE
Pods
Secrets
Services
UID NAME GROUP
1 Chris admin
2 Gergely admin
3 Salman dev
4 Salaboy dev
ROLE (dev namespace)
READ
WRITE
Pods
Secrets
Services
Authentication
Authorisation
Salman
API SERVER


~$ kubectl get service my-service -n dev
Cluster ROLE
READ
WRITE
Pods
Secrets
Services
UID NAME GROUP
1 Chris admin
2 Gergely admin
3 Salman dev
4 Salaboy dev
ROLE (dev namespace)
READ
WRITE
Pods
Secrets
Services
Authentication
Authorisation
Salman
API SERVER


Cluster ROLE
READ
WRITE
Pods
Secrets
Services
UID NAME GROUP
1 Chris admin
2 Gergely admin
3 Salman dev
4 Salaboy dev
ROLE (dev namespace)
READ
WRITE
Pods
Secrets
Services
Authentication
Authorisation
Salman
API SERVER


~$ kubectl delete secret my-secret -n dev
Cluster ROLE
READ
WRITE
Pods
Secrets
Services
UID NAME GROUP
1 Chris admin
2 Gergely admin
3 Salman dev
4 Salaboy dev
ROLE (dev namespace)
READ
WRITE
Pods
Secrets
Services
Authentication
Authorisation
Chris
API SERVER


Cluster ROLE
READ
WRITE
Pods
Secrets
Services
UID NAME GROUP
1 Chris admin
2 Gergely admin
3 Salman dev
4 Salaboy dev
ROLE (dev namespace)
READ
WRITE
Pods
Secrets
Services
Authentication
Authorisation
Chris
API SERVER



UID	NAME	GROUP
1	Chris	admin
2	Gergely	admin
3	Salman	dev
4	Salaboy	dev

Authentication

Authorisation

Cluster ROLE

	READ	WRITE
Pods	✓	✓
Secrets	✓	✓
Services	✓	✓

ROLE (dev namespace)

	READ	WRITE
Pods	✓	✓
Secrets	✗	✗
Services	✓	✓

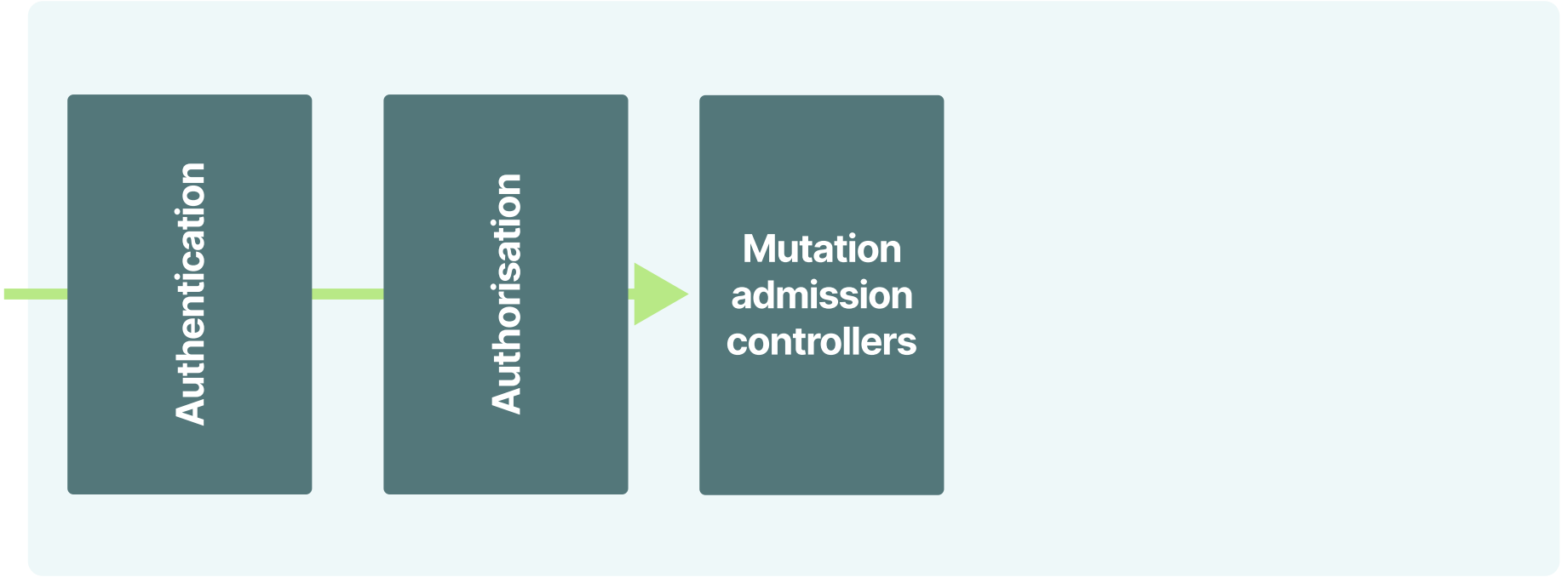
API SERVER

Admission controllers

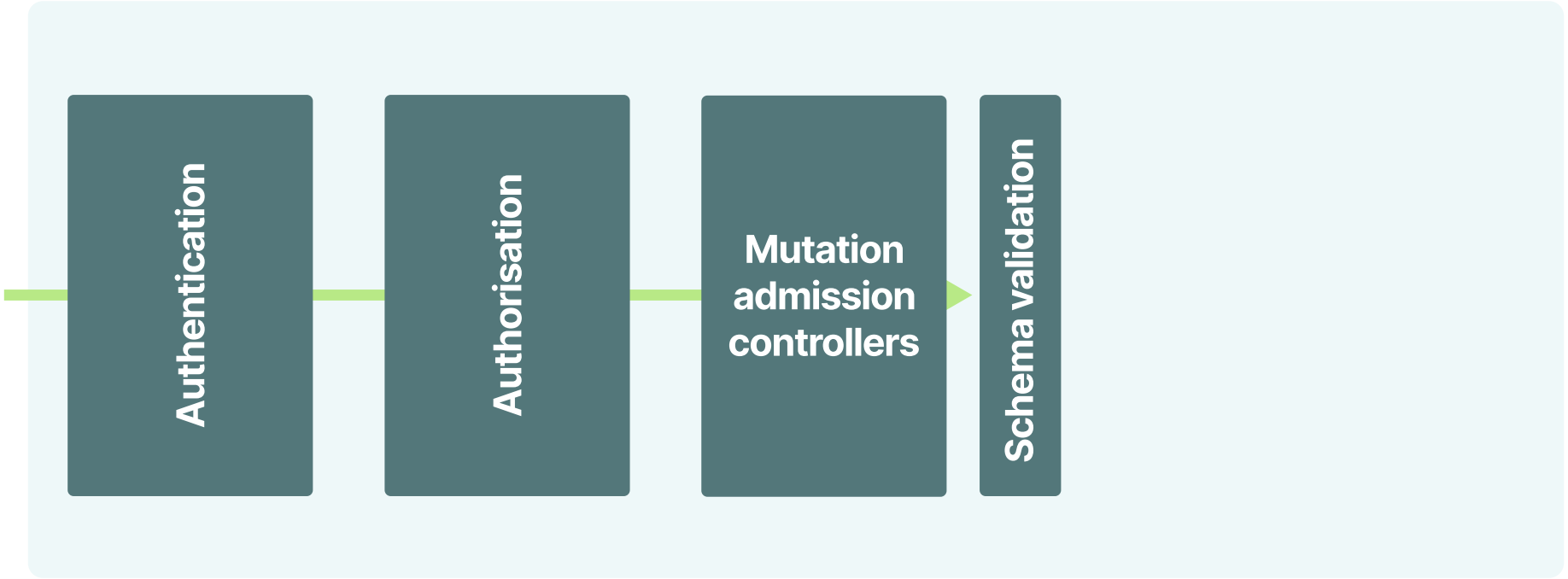
API SERVER



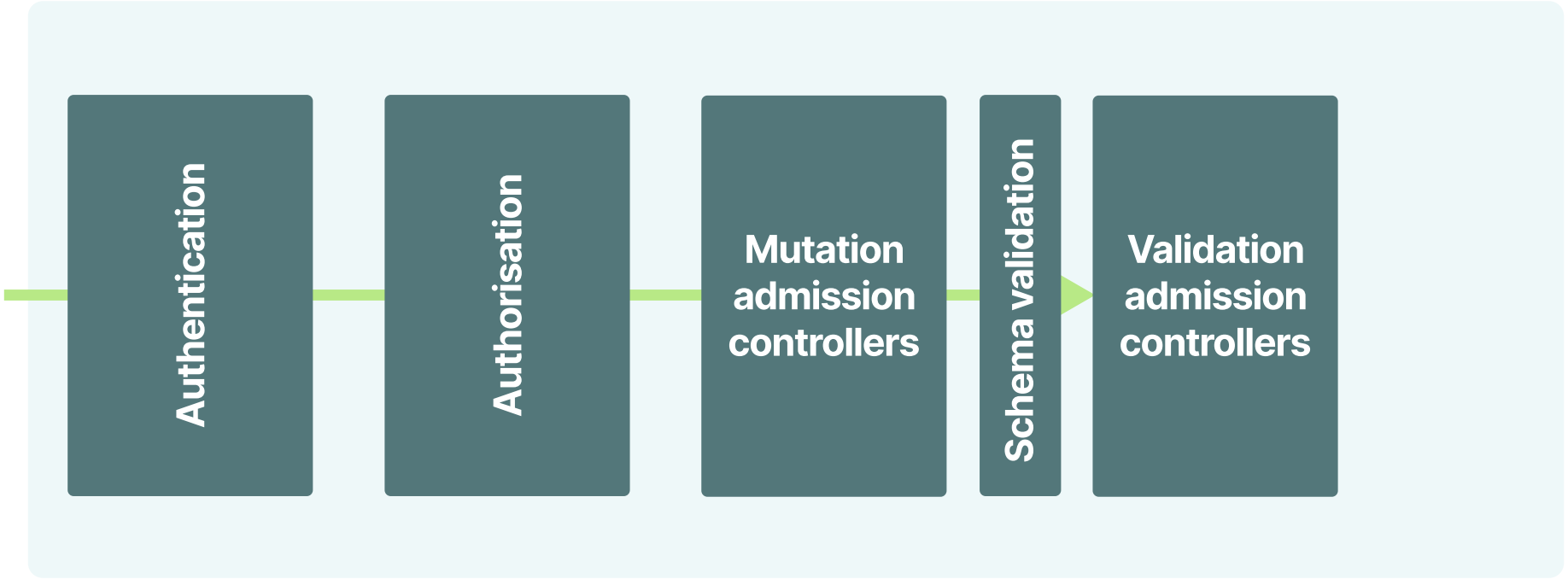
API SERVER


Authentication
Authorisation
Mutation admission controllers

API SERVER


Authentication
Authorisation
Mutation admission controllers
Schema validation

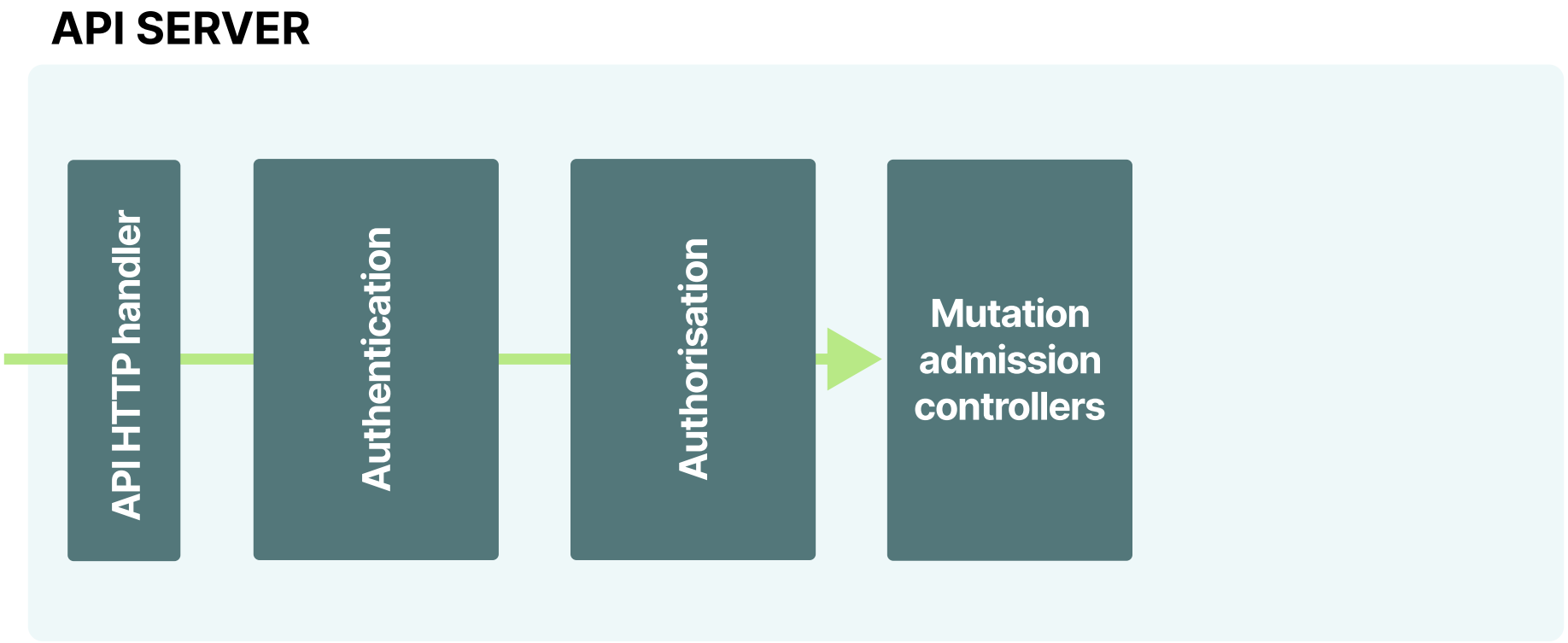
API SERVER


Authentication
Authorisation
Mutation admission controllers
Schema validation
Validation admission controllers

Mutation admission controllers

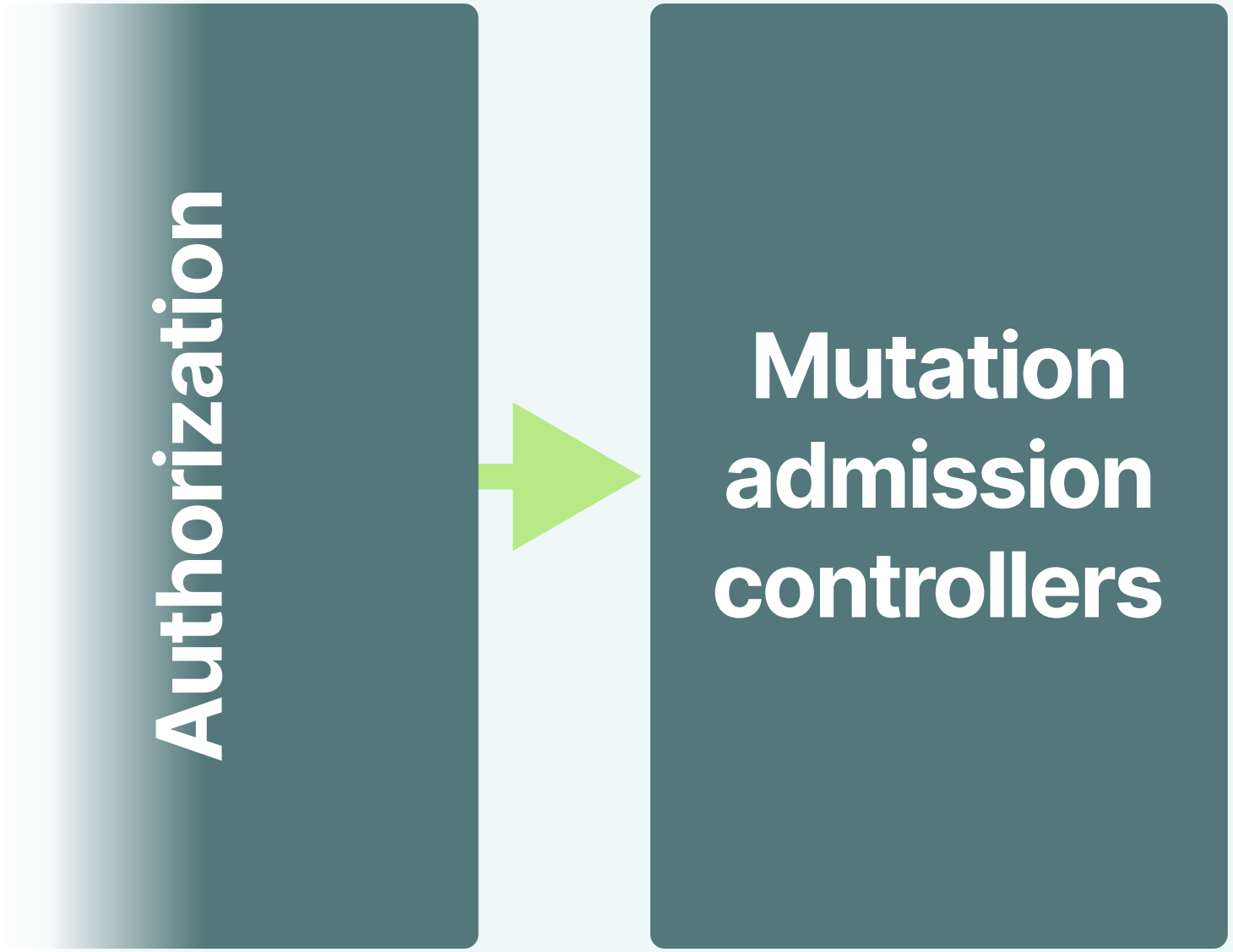
```
~$ cat pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: example-pod
  labels:
    app: web
spec:
  containers:
  - name: app
    image: nginx

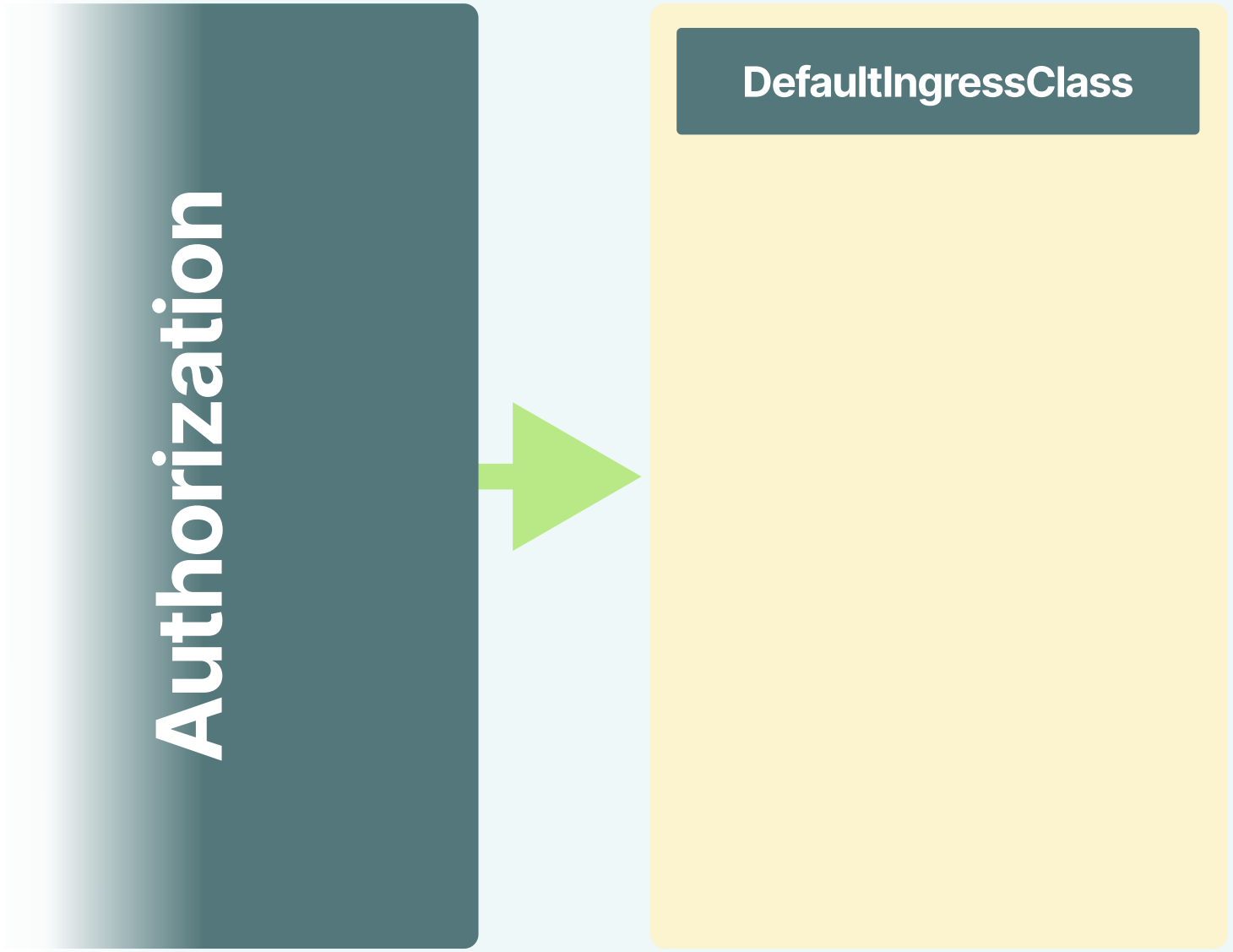
~$ _
```

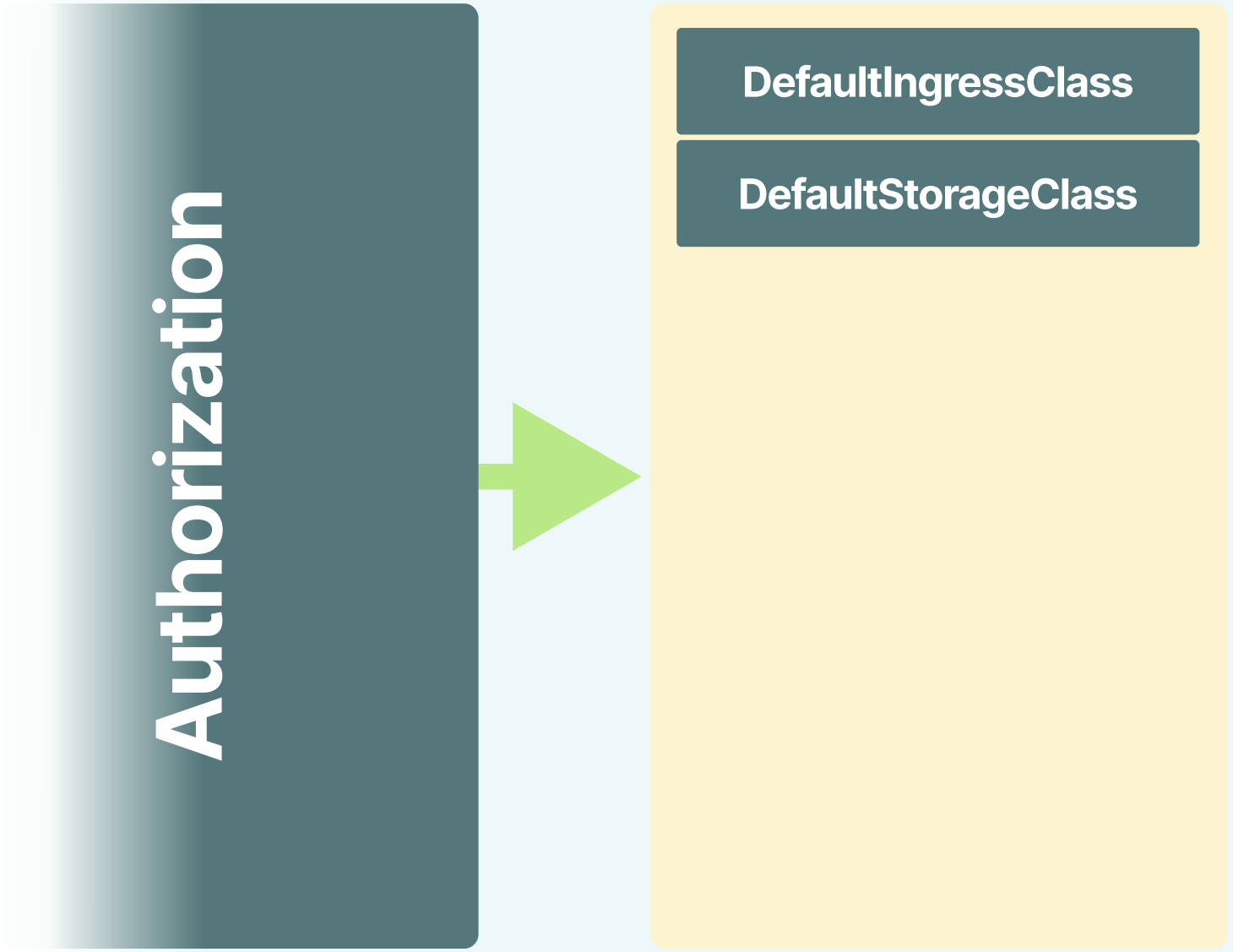

API SERVER
API HTTP handler
Authentication
Authorisation
Mutation admission controllers

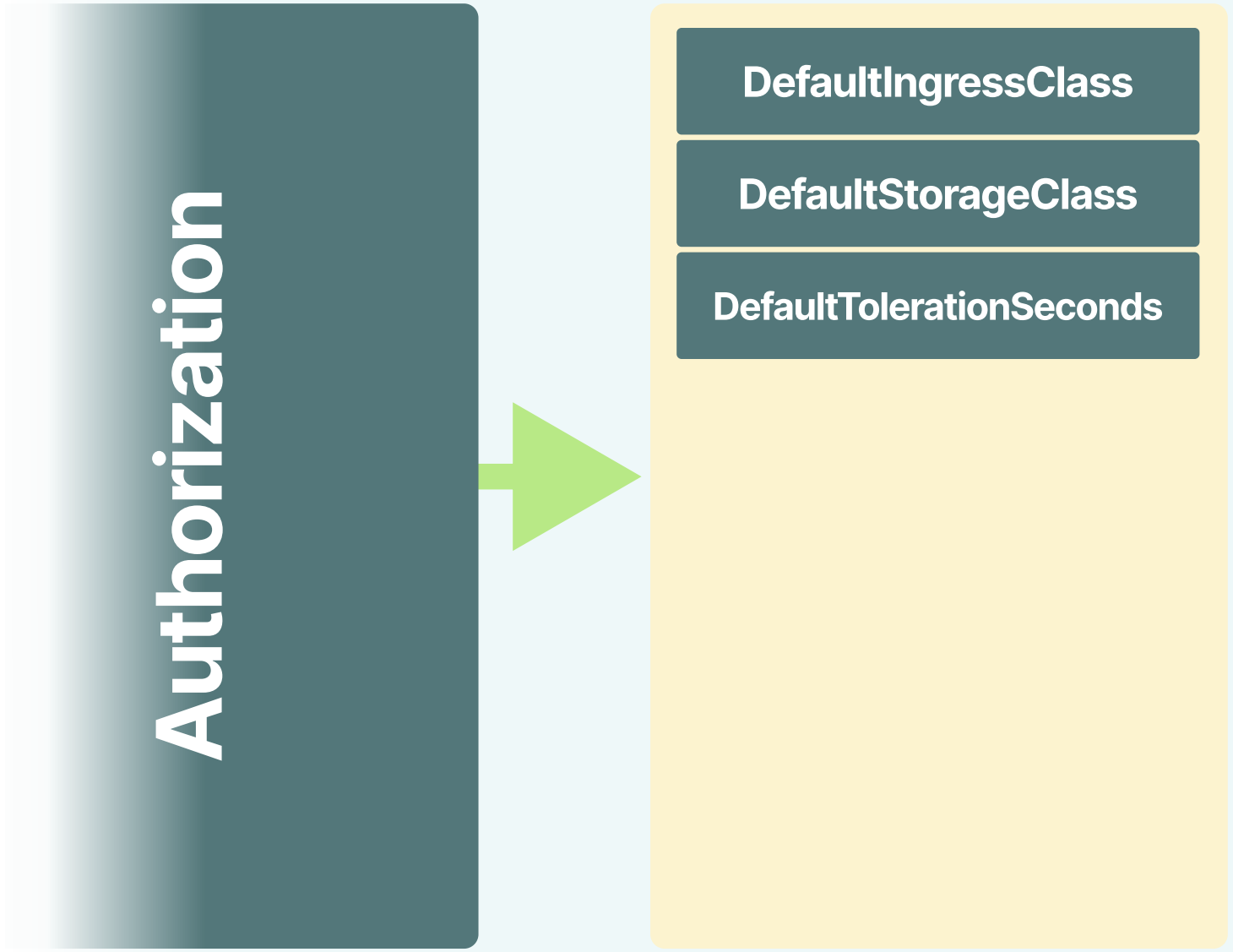
```
~$ cat pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: example-pod
  labels:
    app: web
spec:
  containers:
  - name: app
    image: nginx
    imagePullPolicy: Always

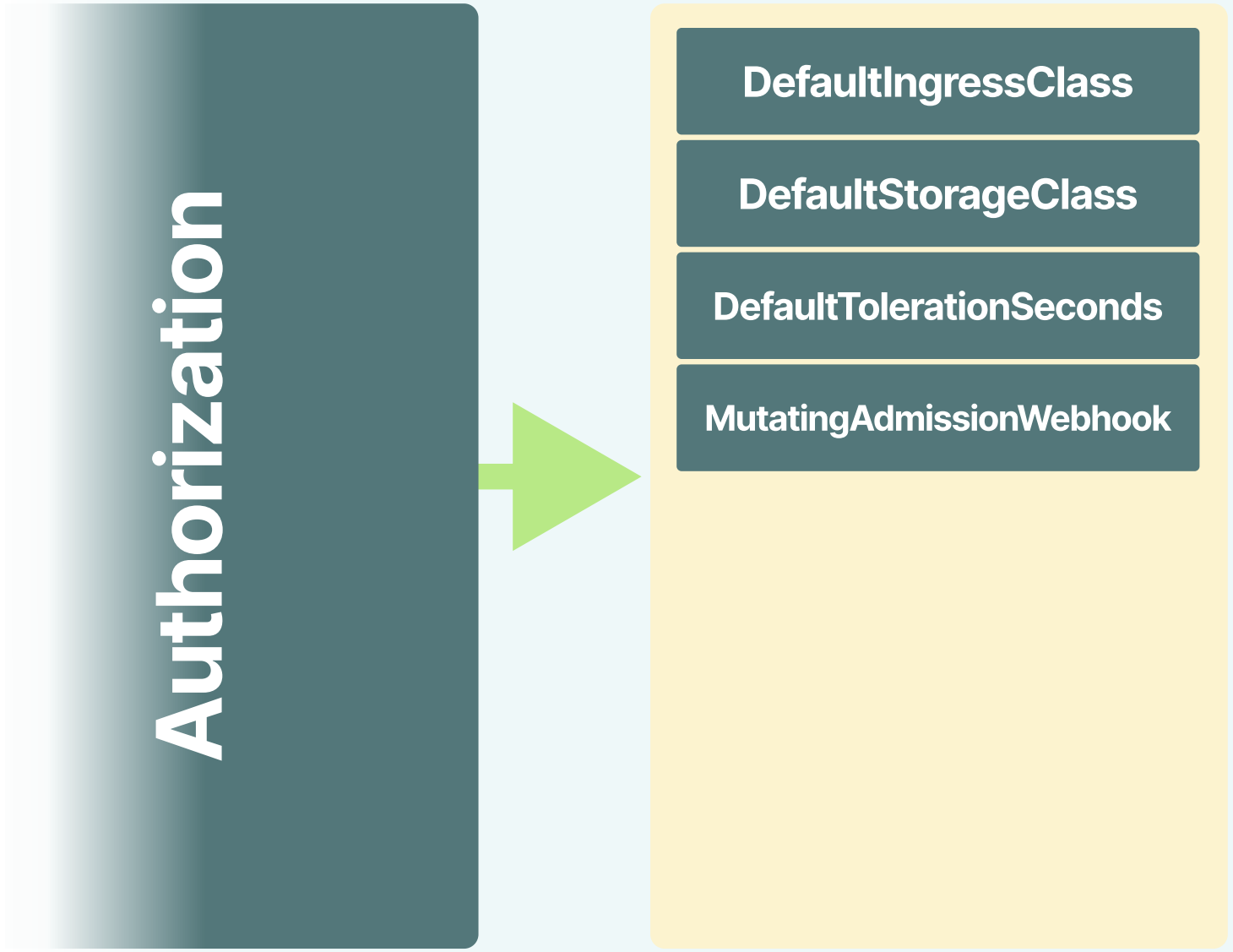
~$ _
```

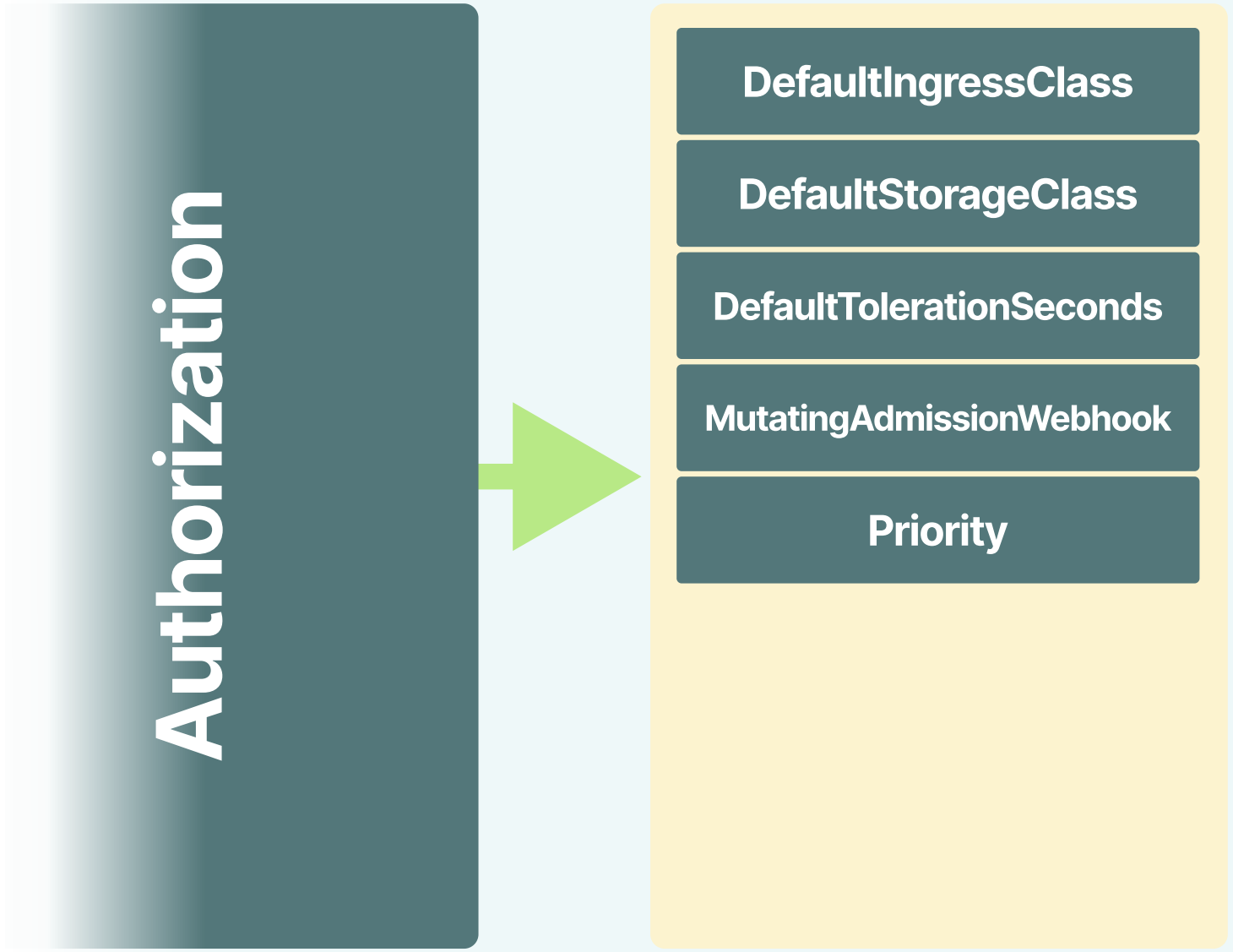

Authorization
Mutation admission controllers

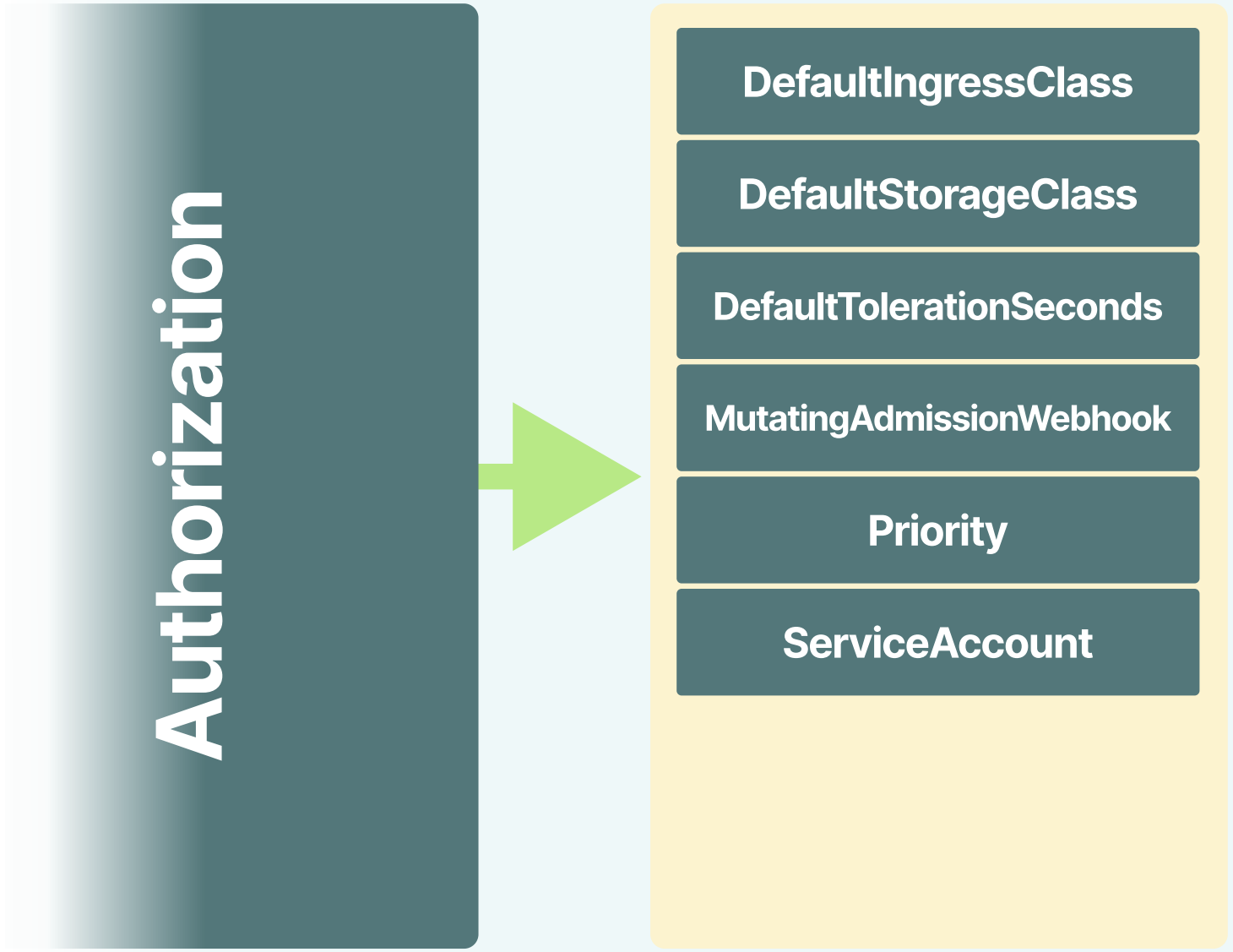

Authorization
DefaultIngressClass

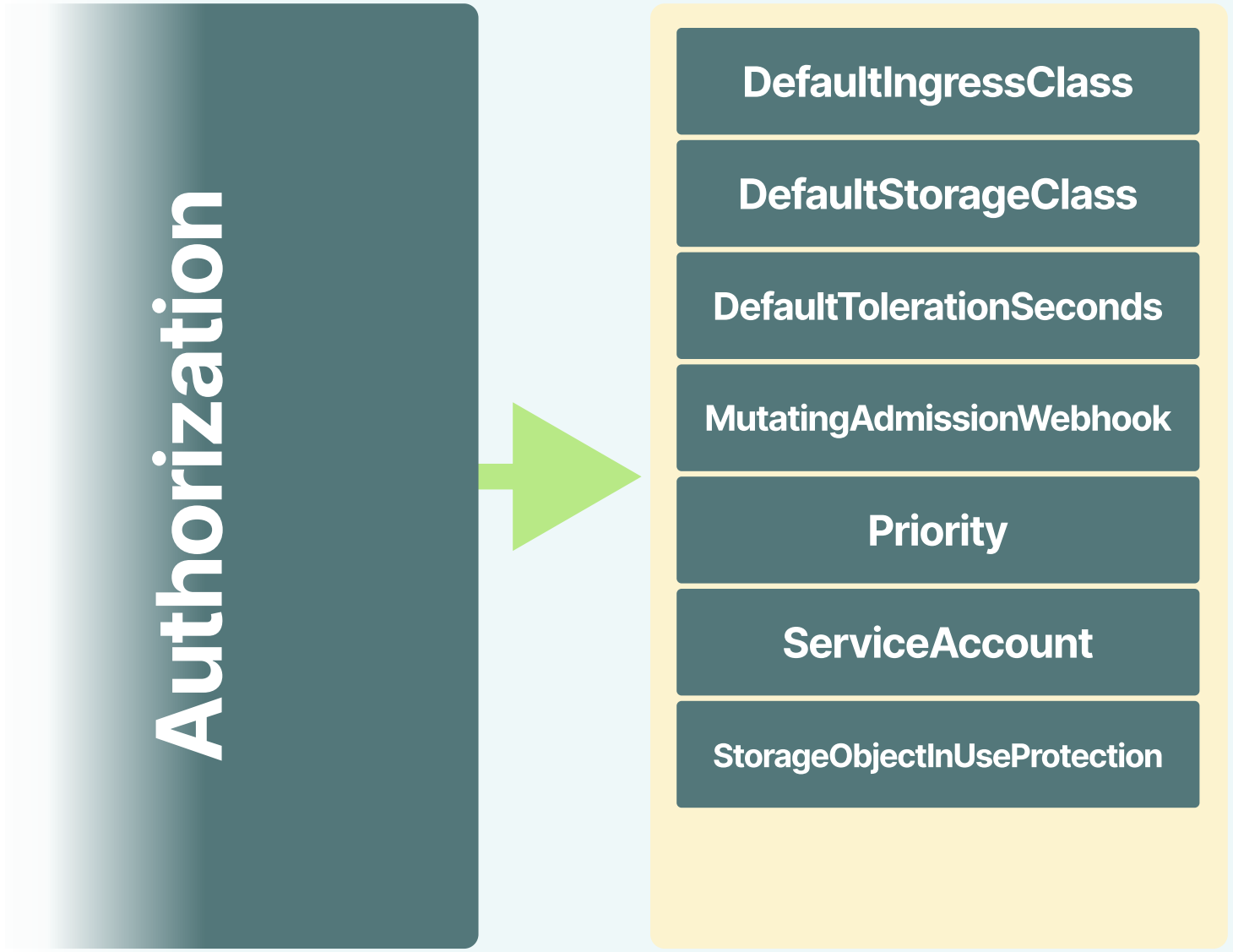

Authorization
DefaultIngressClass
DefaultStorageClass

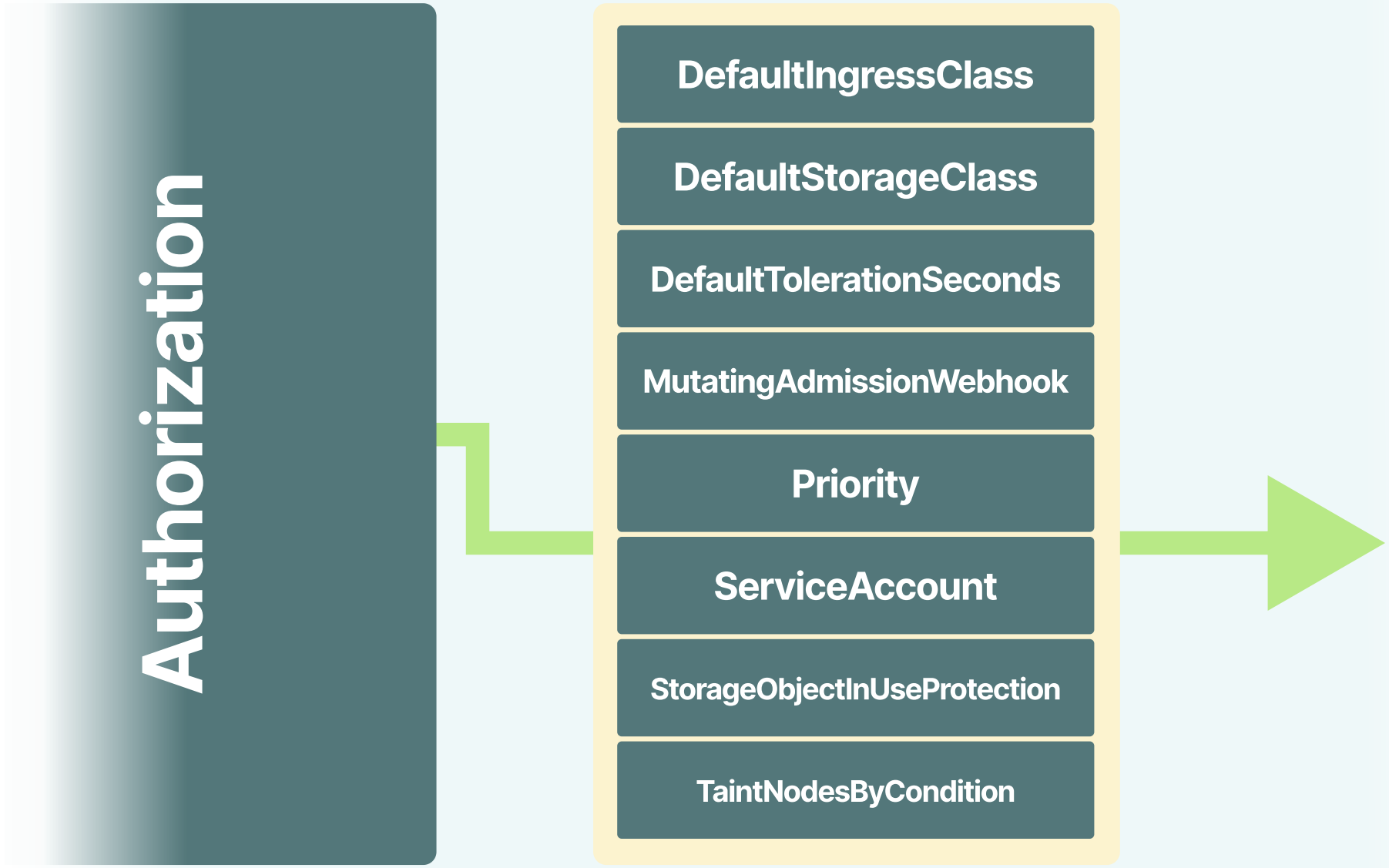

Authorization
DefaultIngressClass
DefaultStorageClass
DefaultTolerationSeconds


Authorization
DefaultIngressClass
DefaultStorageClass
DefaultTolerationSeconds
MutatingAdmissionWebhook


Authorization
DefaultIngressClass
DefaultStorageClass
DefaultTolerationSeconds
MutatingAdmissionWebhook
Priority


Authorization
DefaultIngressClass
DefaultStorageClass
DefaultTolerationSeconds
MutatingAdmissionWebhook
Priority
ServiceAccount


Authorization
DefaultIngressClass
DefaultStorageClass
DefaultTolerationSeconds
MutatingAdmissionWebhook
Priority
ServiceAccount
StorageObjectInUseProtection


Authorization
DefaultIngressClass
DefaultStorageClass
DefaultTolerationSeconds
MutatingAdmissionWebhook
Priority
ServiceAccount
StorageObjectInUseProtection
TaintNodesByCondition

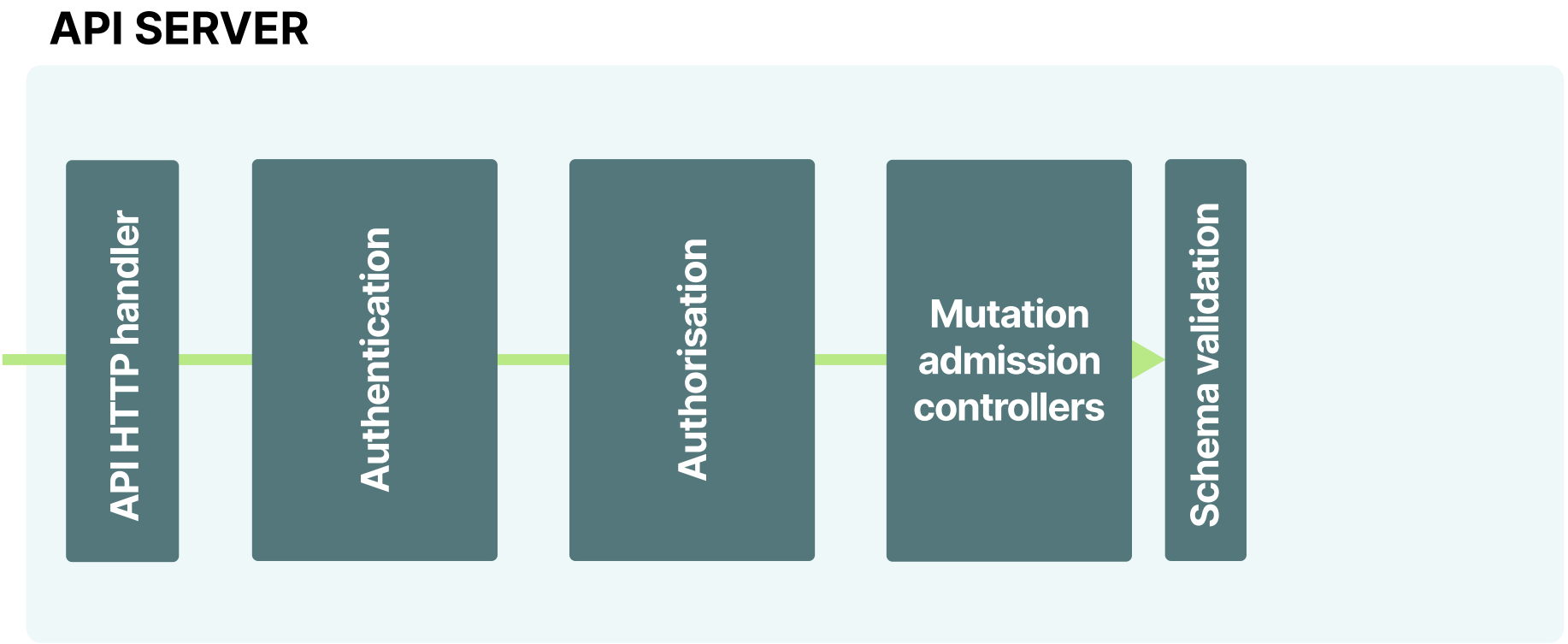
```
~$ cat pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: example-pod
  labels:
    app: web
spec:
  containers:
  - name: app
    image: nginx
```

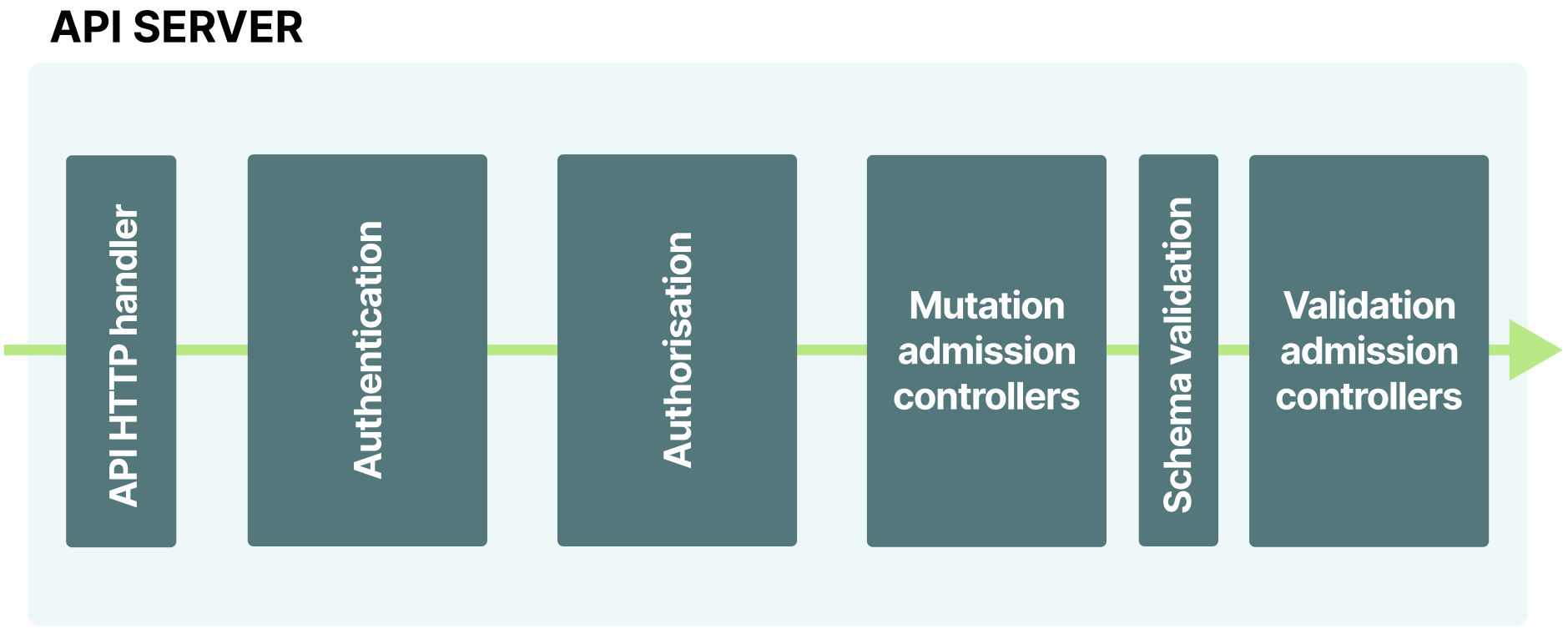
BEFORE

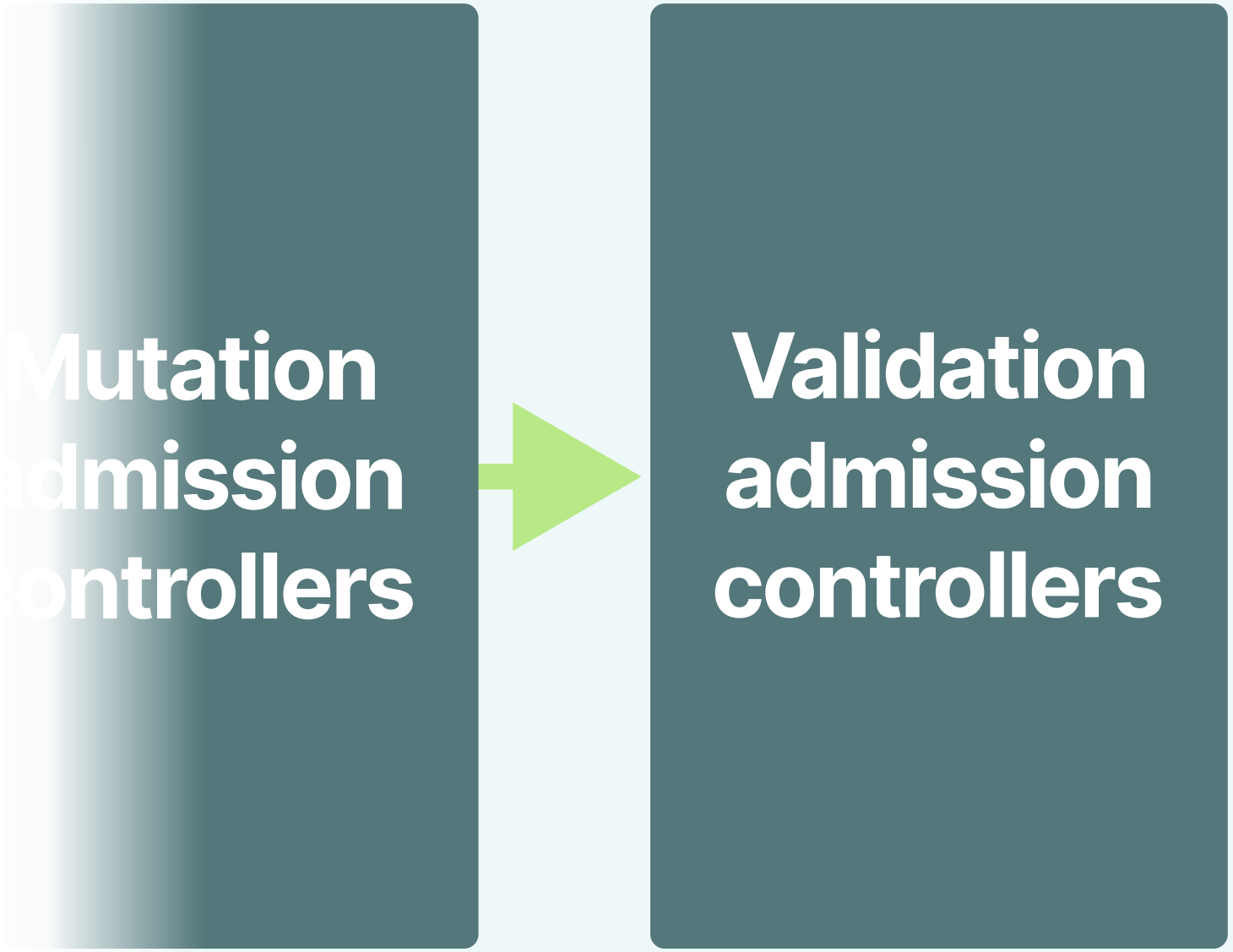
```
~$ cat pod.yaml
apiVersion: v1
kind: Pod
metadata:
  creationTimestamp: "2021-06-24T08:26:21Z"
  generateName: v1-57cbf4bb76-
  labels:
    app:web
  name: exaple-pod
  namespace: default
  ownerReferences:
  - apiVersion: apps/v1
    blockOwnerDeletion: true
    controller: true
    kind: ReplicaSet
    name: v1-57cbf4bb76
    uid: e6b0a616-a50e-4f64-b55e-9b46f30dddf8
  resourceVersion: "49626"
  uid: c2abed7f-7e3d-4559-9508-e3f956c28c37
spec:
  containers:
  - image: learnk8s/deployment-strategies:v2
    imagePullPolicy: IfNotPresent
    livenessProbe:
      failureThreshold: 1
      httpGet:
        path: /health
        port: 8080
        scheme: HTTP
      initialDelaySeconds: 6
      periodSeconds: 5
      successThreshold: 1
      timeoutSeconds: 1
    name: web
    ports:
    - containerPort: 8080
      protocol: TCP
    readinessProbe:
      failureThreshold: 1
      httpGet:
        path: /ready
        port: 8080
        scheme: HTTP
      initialDelaySeconds: 15
      periodSeconds: 5
      successThreshold: 1
      timeoutSeconds: 1
    resources: {}
    terminationMessagePath: /dev/termination-log
    terminationMessagePolicy: File
    volumeMounts:
    - mountPath: /var/run/secrets/kubernetes.io/serviceaccount
      name: default-token-bpjhp
      readOnly: true
  dnsPolicy: ClusterFirst
  enableServiceLinks: true
  nodeName: minikube
  preemptionPolicy: PreemptLowerPriority
  priority: 0
  restartPolicy: Always
  schedulerName: default-scheduler
  securityContext: {}
  serviceAccount: default
  serviceAccountName: default
  terminationGracePeriodSeconds: 30
  tolerations:
  - effect: NoExecute
    key: node.kubernetes.io/not-ready
    operator: Exists
    tolerationSeconds: 300
  - effect: NoExecute
    key: node.kubernetes.io/unreachable
    operator: Exists
    tolerationSeconds: 300
  volumes:
  - name: default-token-bpjhp
    secret:
      defaultMode: 420
      secretName: default-token-bpjhp
```

AFTER

Validation admission controllers


API SERVER
API HTTP handler
Authentication
Authorisation
Mutation admission controllers
Schema validation

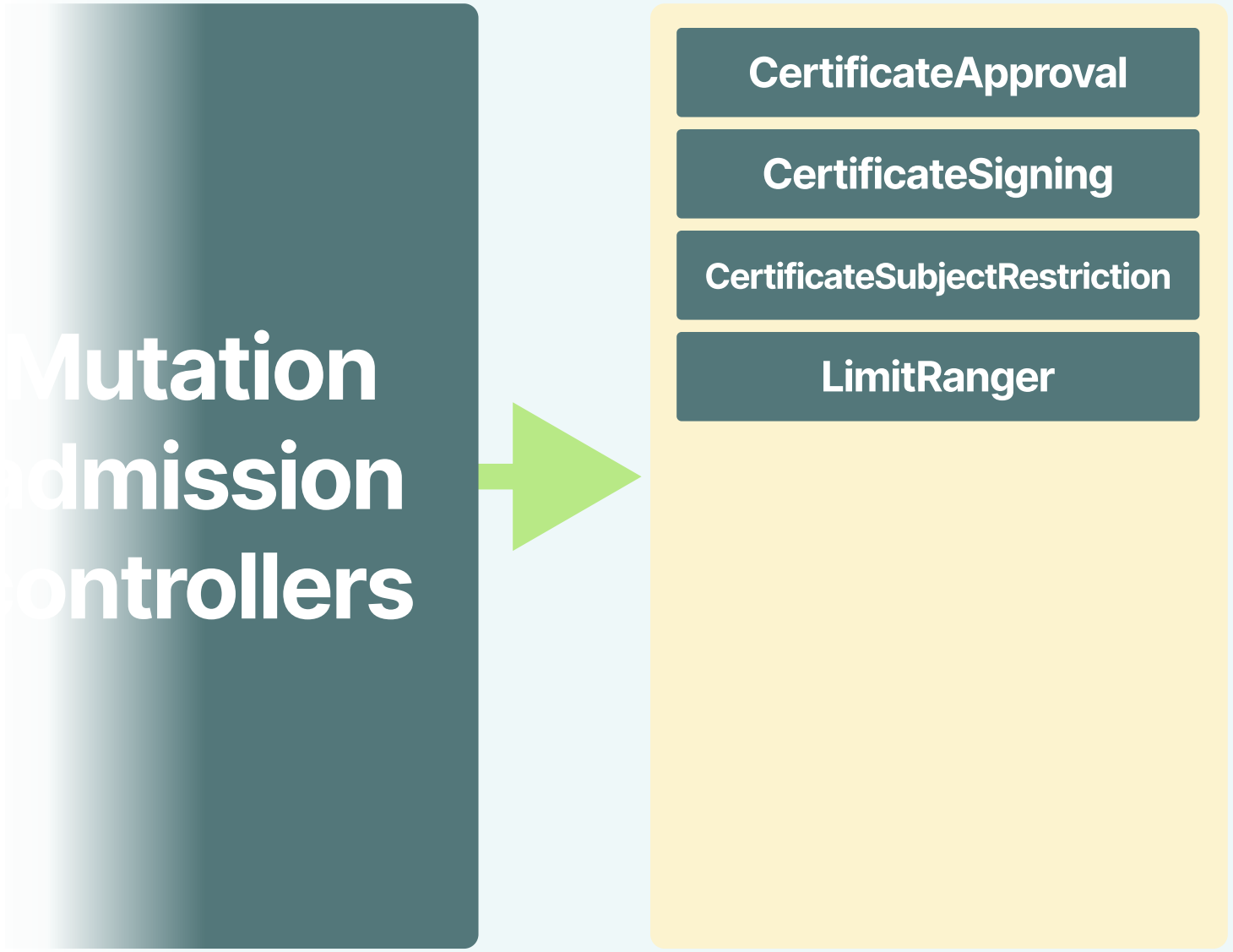

API SERVER
API HTTP handler
Authentication
Authorisation
Mutation admission controllers
Schema validation
Validation admission controllers

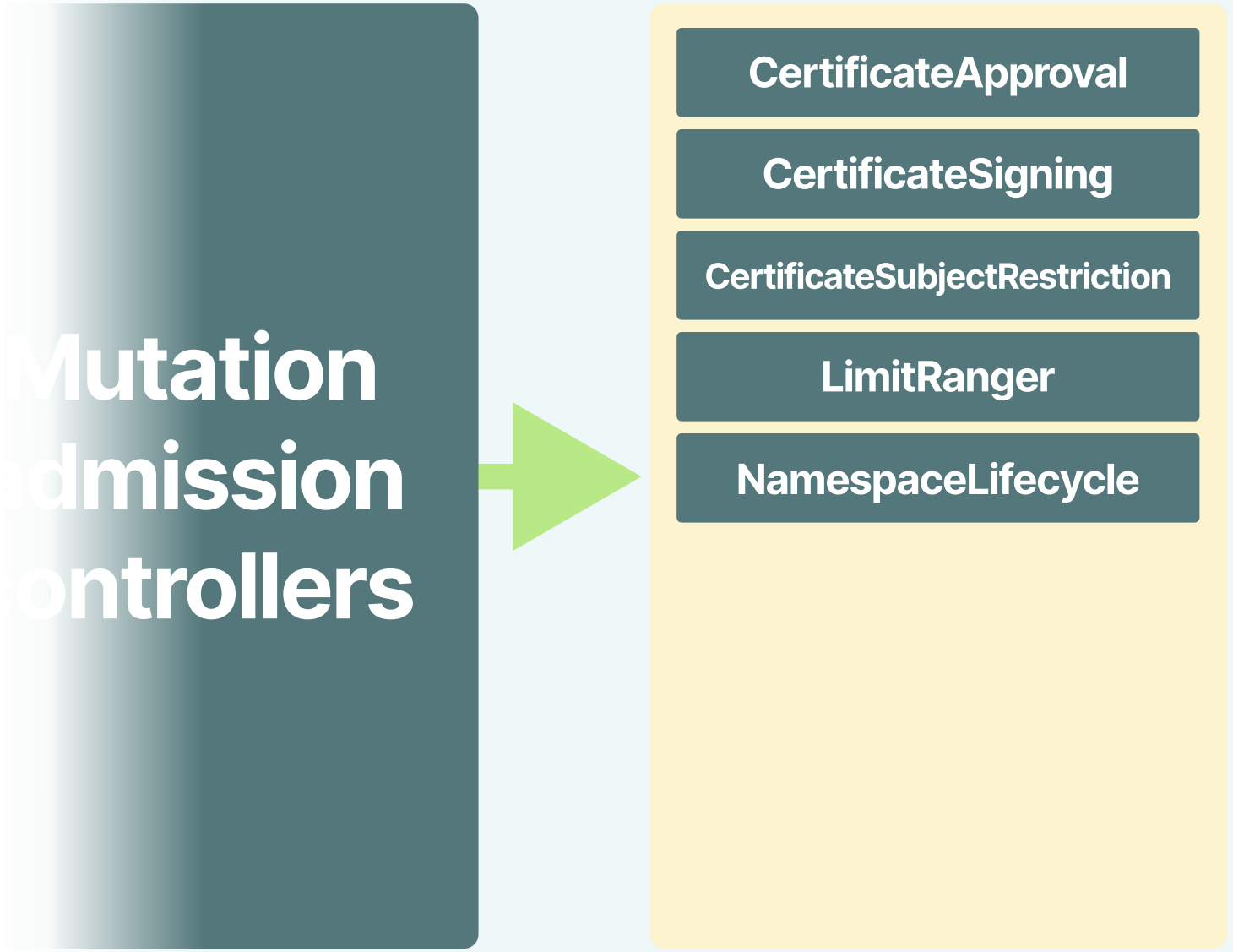

Mutation
admission
controllers
Validation
admission
controllers

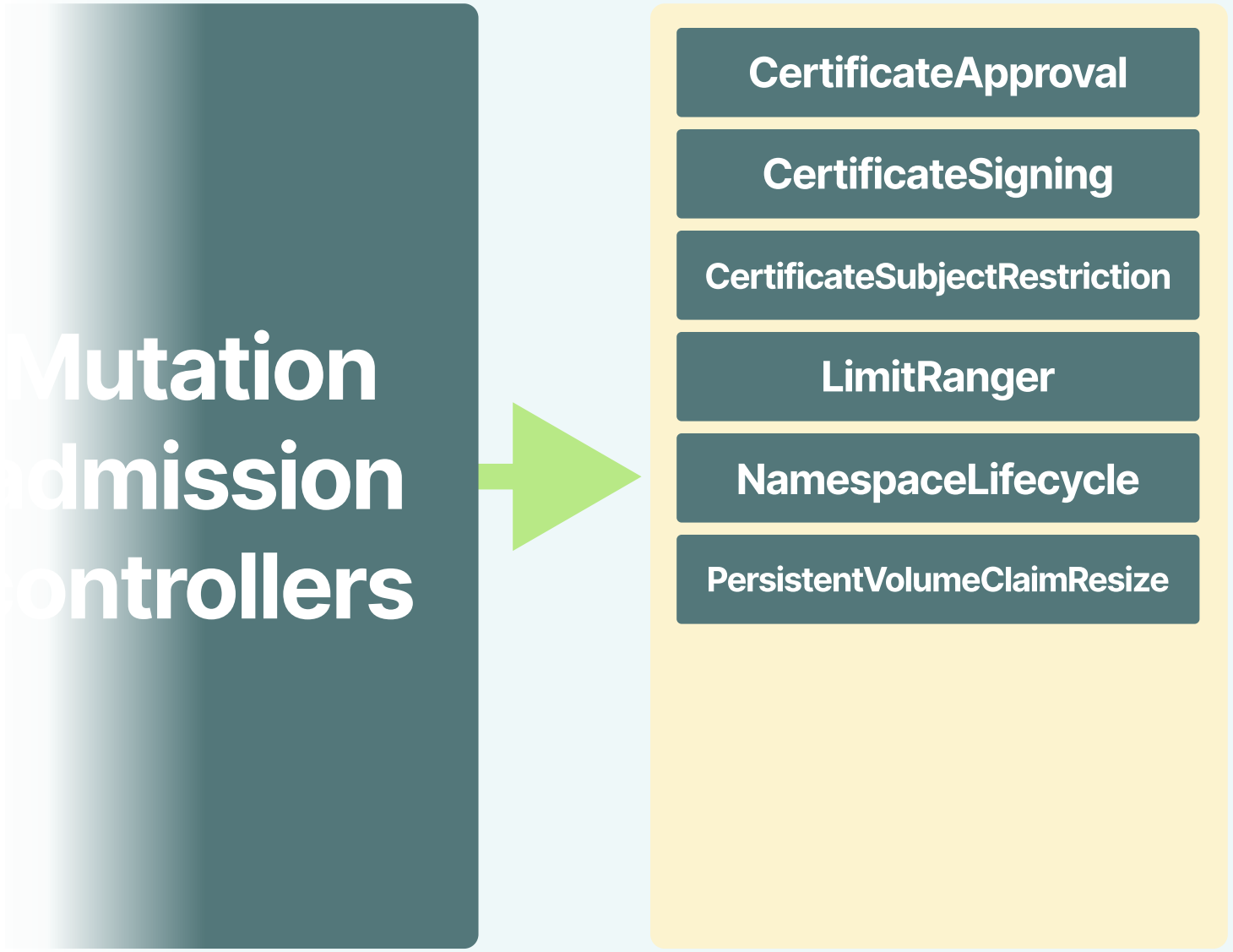

Mutation
dmission
ontrollers
CertificateApproval

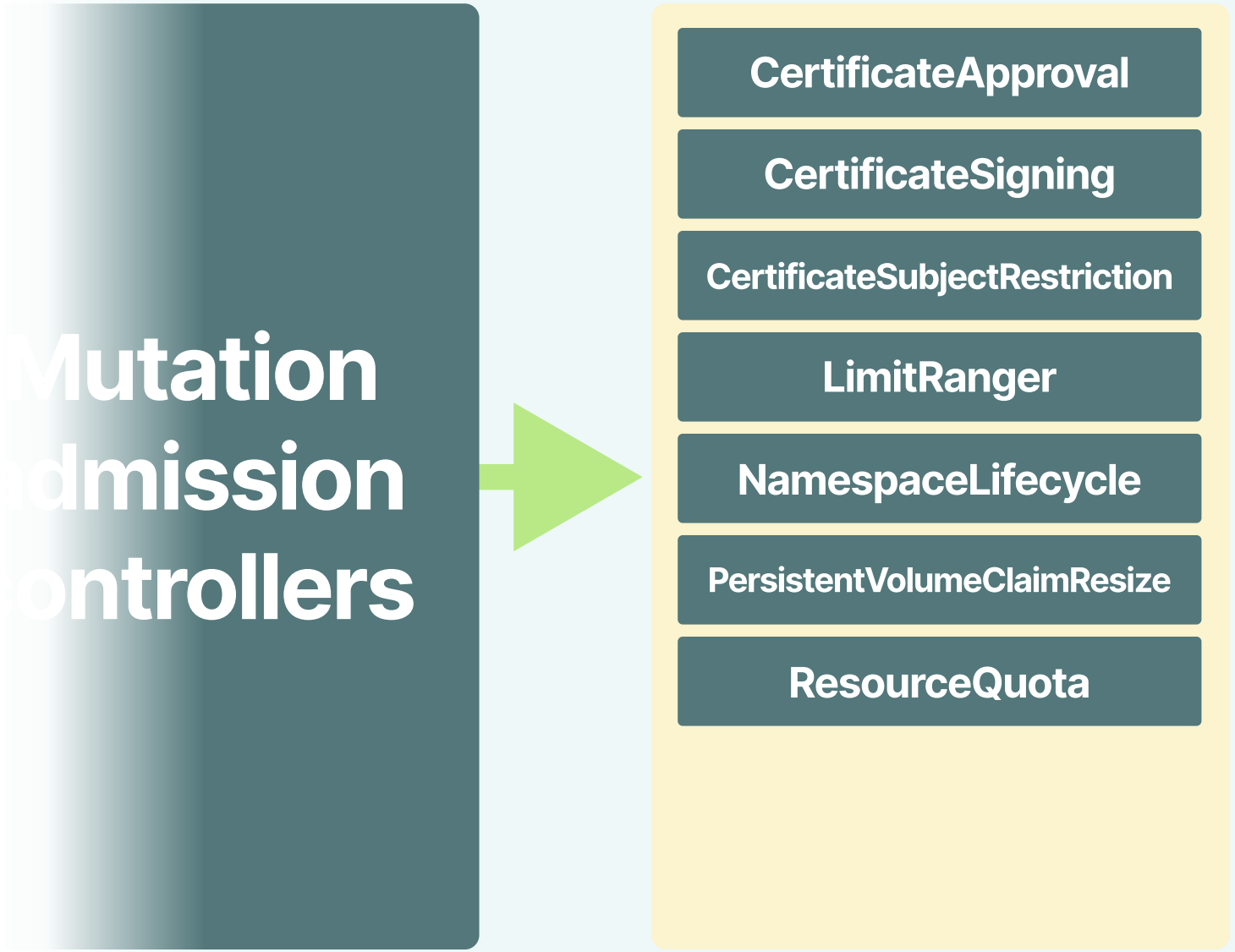

Mutation
dmission
ontrollers
CertificateApproval
CertificateSigning

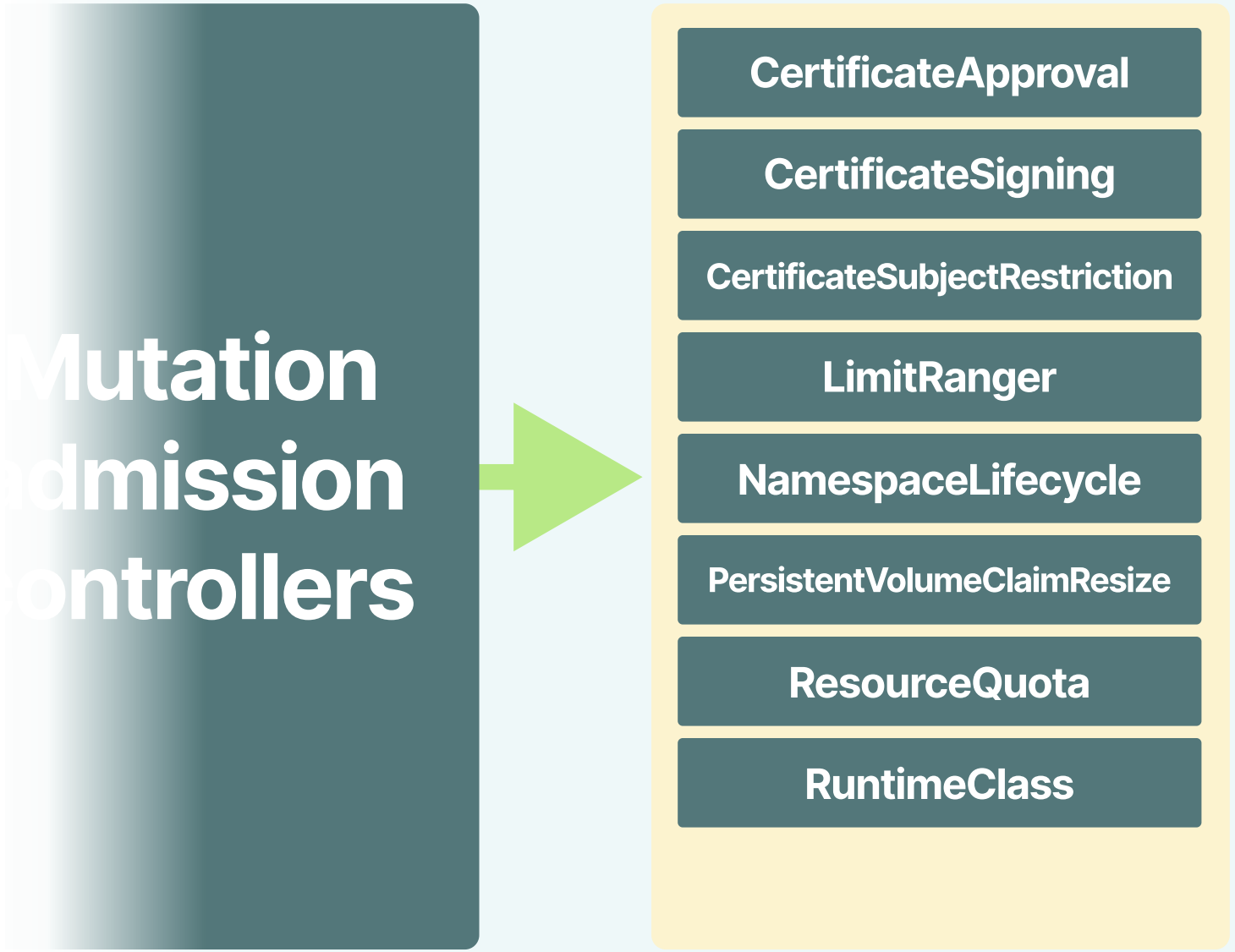

Mutation
dmission
ontrollers
CertificateApproval
CertificateSigning
CertificateSubjectRestriction

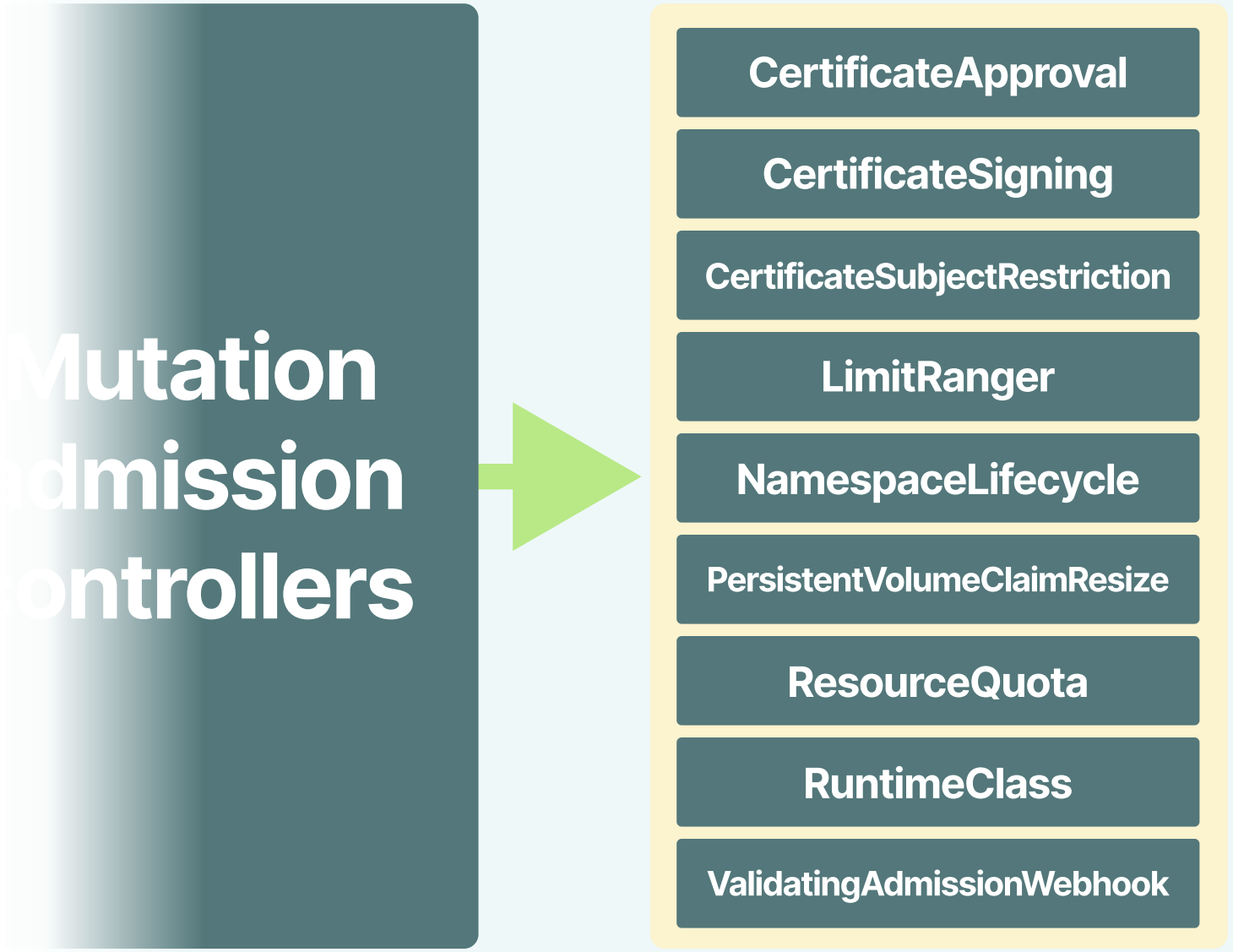

Mutation
dmission
ontrollers
CertificateApproval
CertificateSigning
CertificateSubjectRestriction
LimitRanger

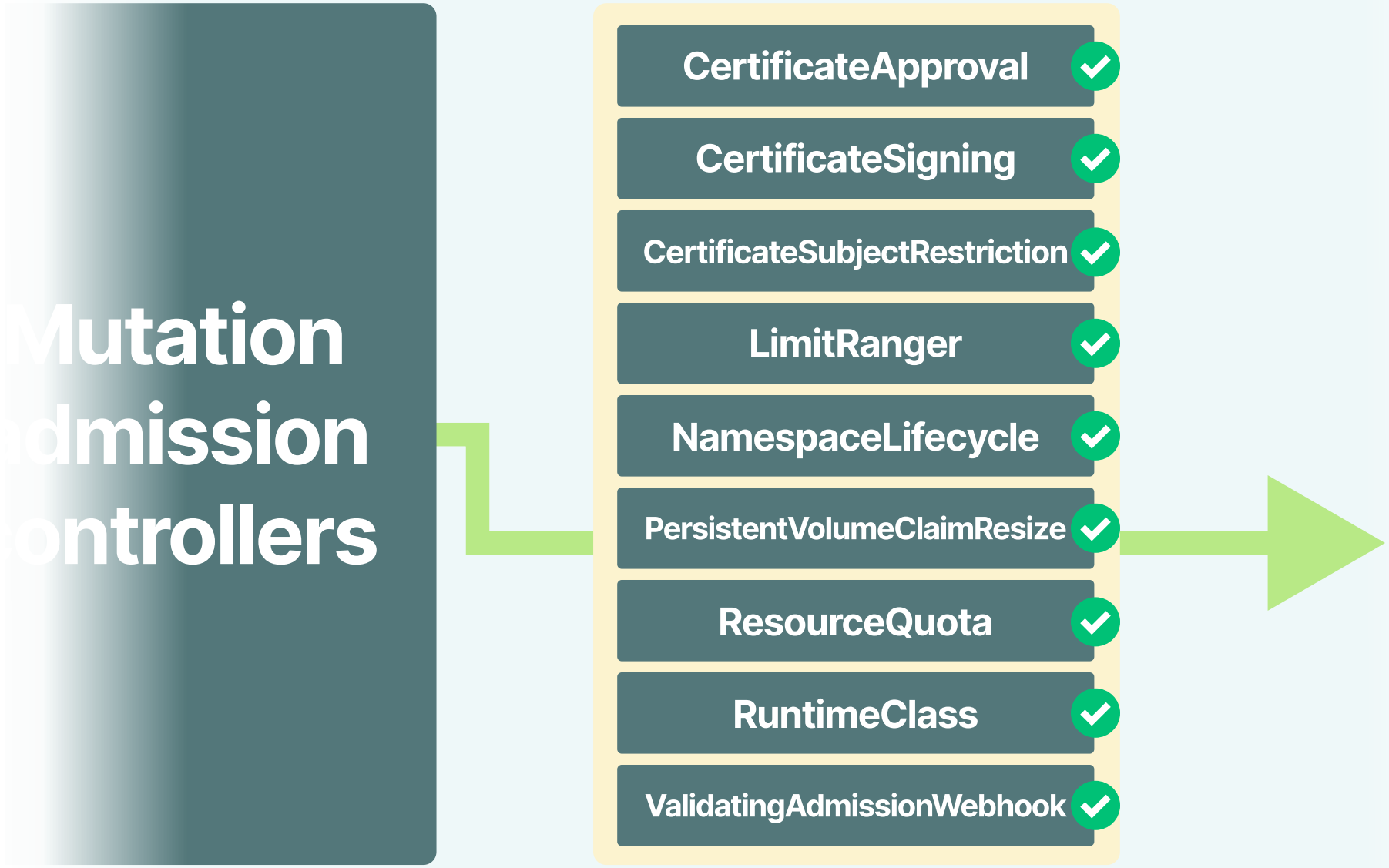

Mutation
dmission
ontrollers
CertificateApproval
CertificateSigning
CertificateSubjectRestriction
LimitRanger
NamespaceLifecycle

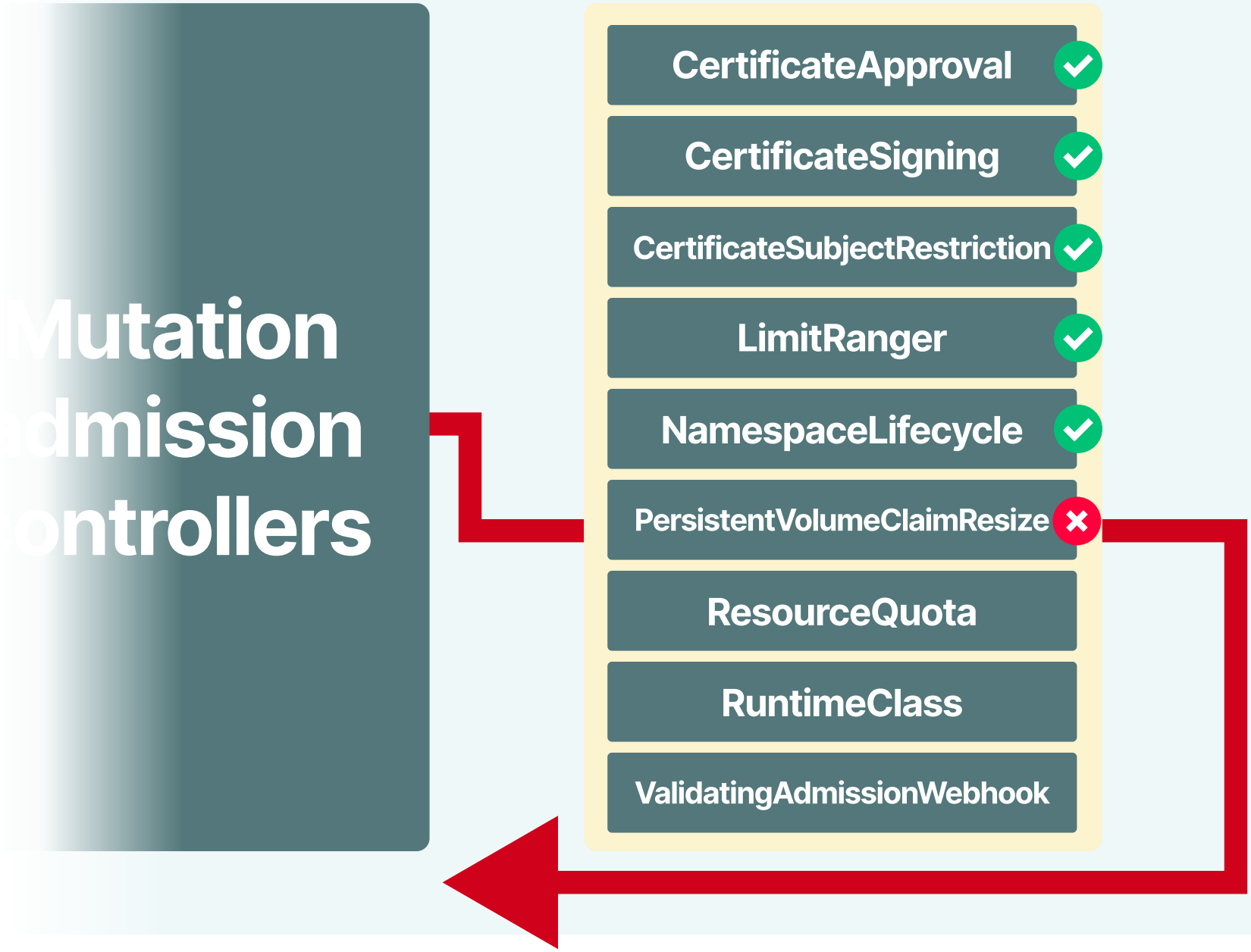

Mutation
dmission
ontrollers
CertificateApproval
CertificateSigning
CertificateSubjectRestriction
LimitRanger
NamespaceLifecycle
PersistentVolumeClaimResize


Mutation
dmission
ontrollers
CertificateApproval
CertificateSigning
CertificateSubjectRestriction
LimitRanger
NamespaceLifecycle
PersistentVolumeClaimResize
ResourceQuota

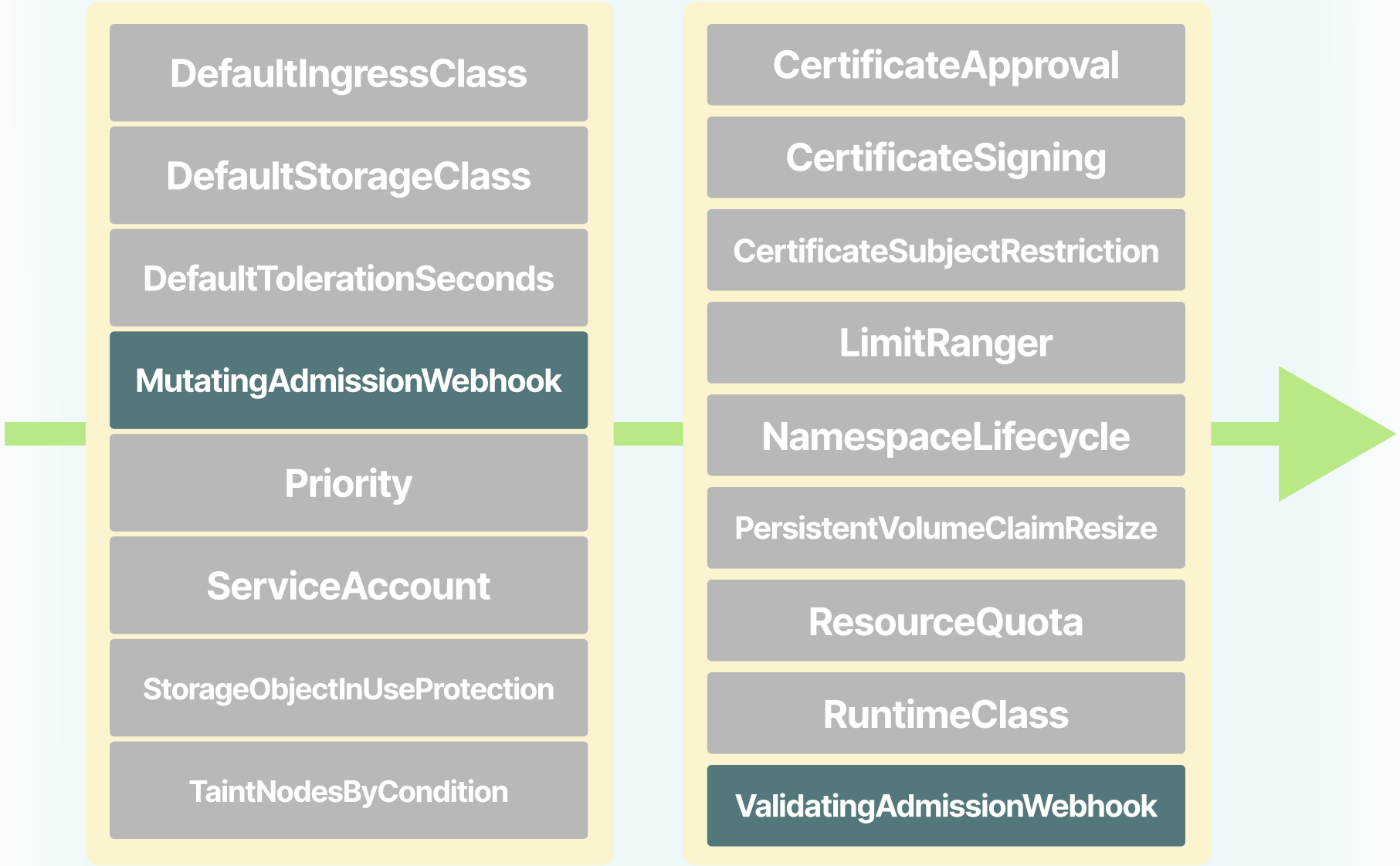

Mutation
dmission
ontrollers
CertificateApproval
CertificateSigning
CertificateSubjectRestriction
LimitRanger
NamespaceLifecycle
PersistentVolumeClaimResize
ResourceQuota
RuntimeClass

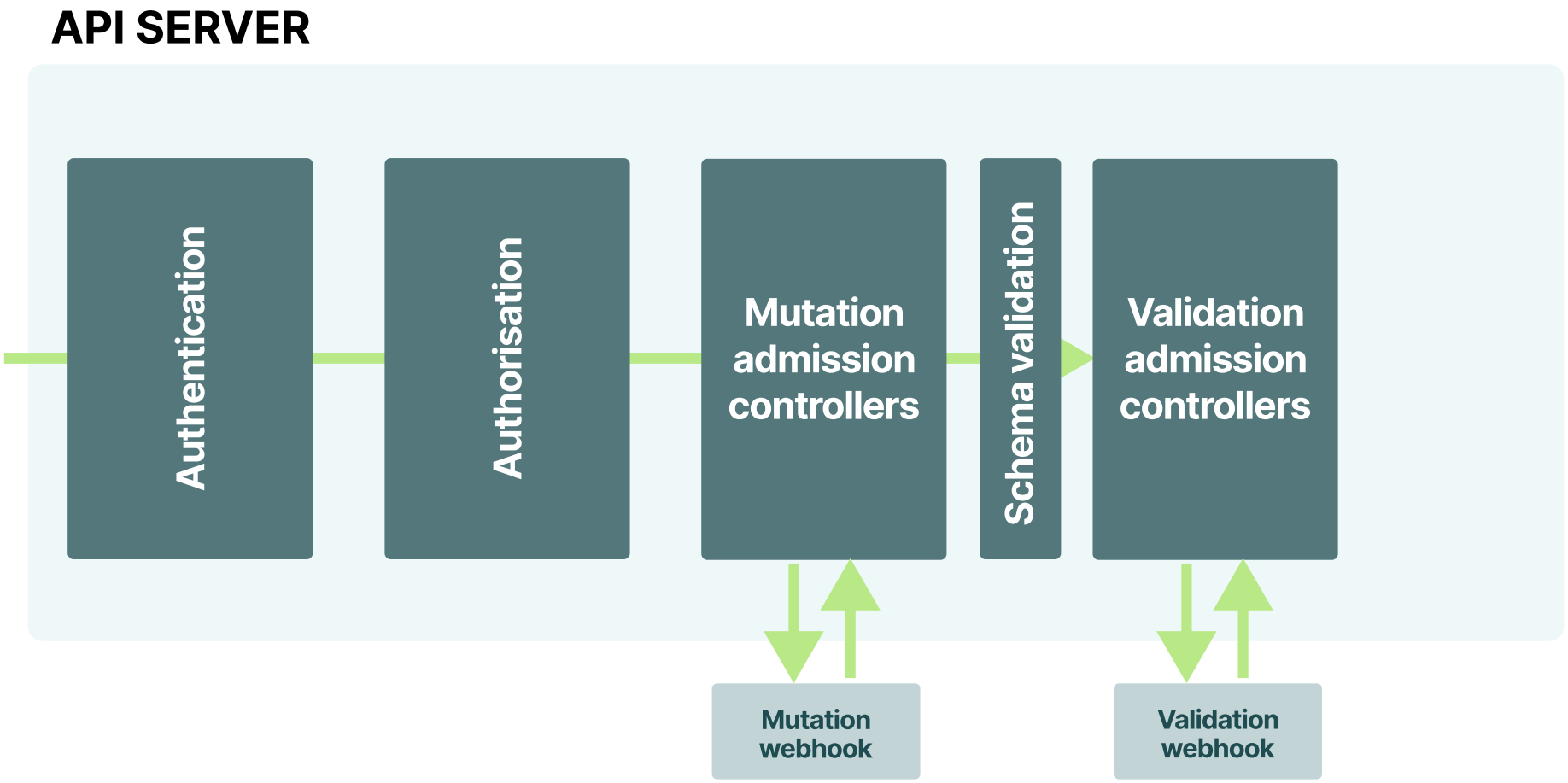

Mutation
dmission
ontrollers
CertificateApproval
CertificateSigning
CertificateSubjectRestriction
LimitRanger
NamespaceLifecycle
PersistentVolumeClaimResize
ResourceQuota
RuntimeClass
ValidatingAdmissionWebhook


Mutation
dmission
ontrollers
CertificateApproval
CertificateSigning
CertificateSubjectRestriction
LimitRanger
NamespaceLifecycle
PersistentVolumeClaimResize
ResourceQuota
RuntimeClass
ValidatingAdmissionWebhook

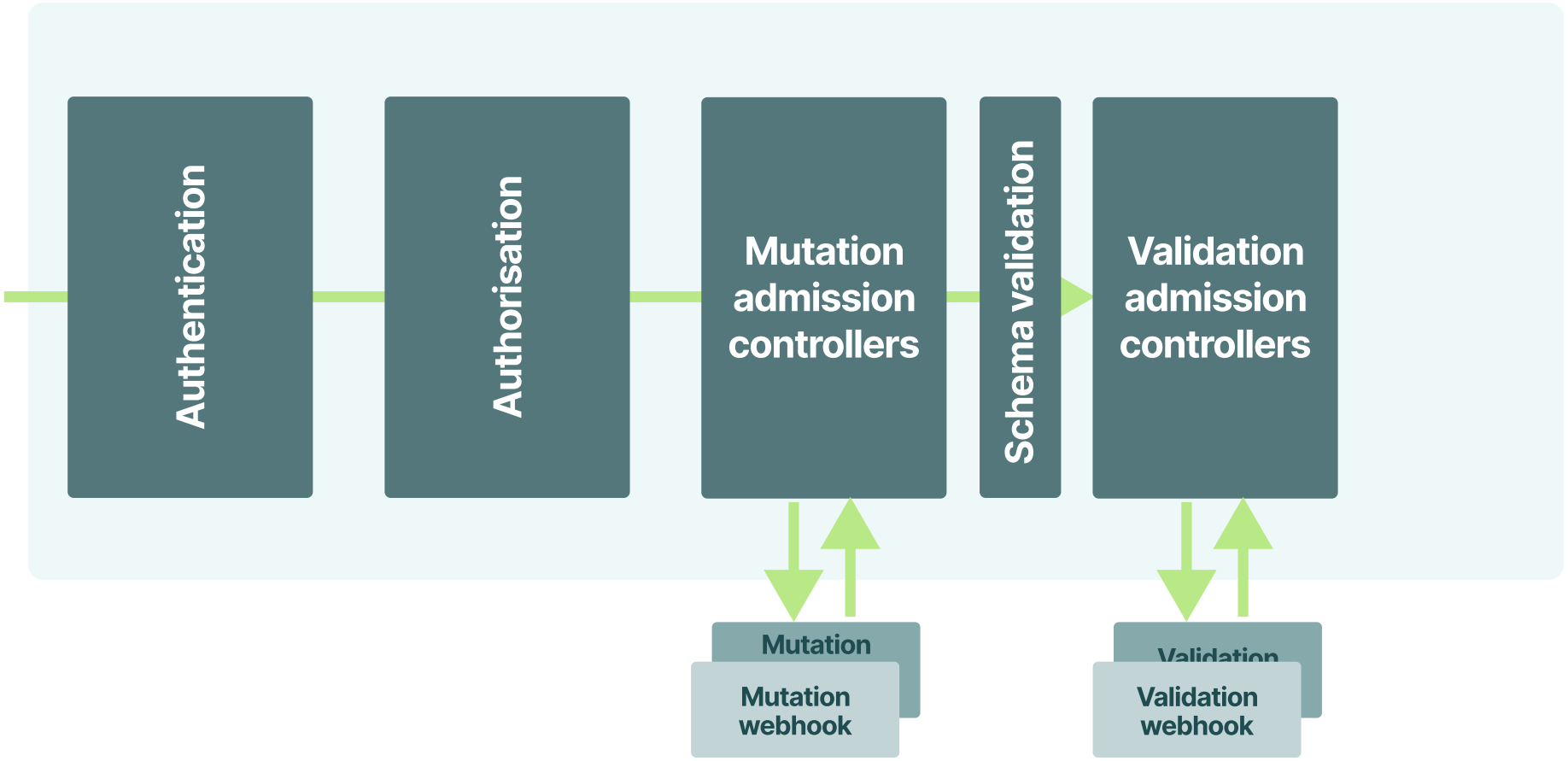

Mutation
dmission
ontrollers
CertificateApproval
CertificateSigning
CertificateSubjectRestriction
LimitRanger
NamespaceLifecycle
PersistentVolumeClaimResize
ResourceQuota
RuntimeClass
ValidatingAdmissionWebhook

Mutation & validation admission webhook

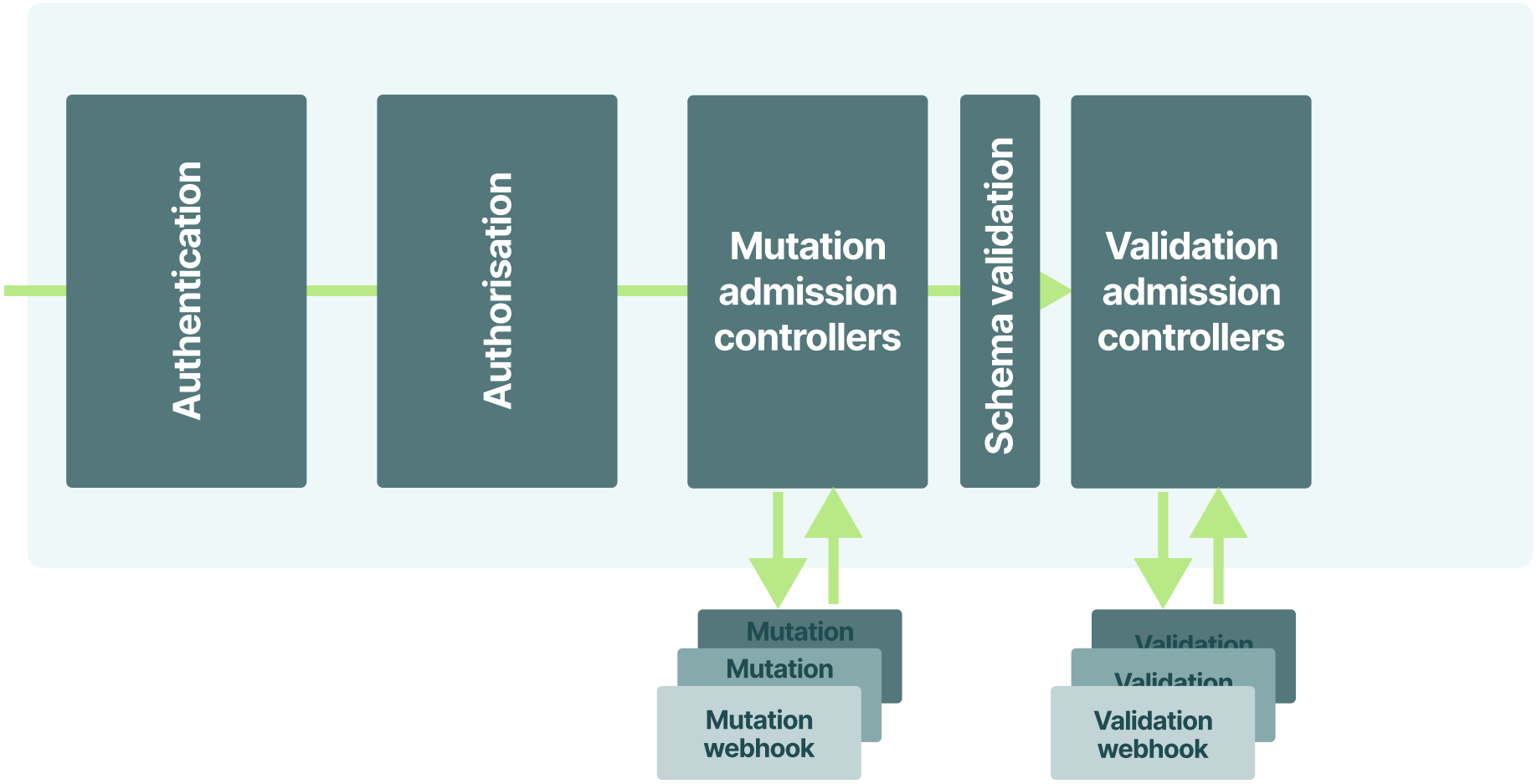

DefaultIngressClass
DefaultStorageClass
DefaultTolerationSeconds
MutatingAdmissionWebhook
Priority
ServiceAccount
StorageObjectInUseProtection
TaintNodesByCondition
CertificateApproval
CertificateSigning
CertificateSubjectRestriction
LimitRanger
NamespaceLifecycle
PersistentVolumeClaimResize
ResourceQuota
RuntimeClass
ValidatingAdmissionWebhook


API SERVER
Authentication
Authorisation
Mutation admission controllers
Schema validation
Validation admission controllers
Mutation webhook
Validation webhook

API SERVER


Authentication
Authorisation
Mutation admission controllers
Schema validation
Validation admission controllers
Mutation
Mutation webhook
Validation
Validation webhook

API SERVER


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Mutation & validation admission webhook

Add labels, annotations

Inject sidecars

Restrict images, ingress

Restrict insecure conf

Mutation & validation admission webhook

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Add labels, annotations

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Restrict insecure conf

Mutation & validation admission webhook

Examples

```
~$ cat pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: example-pod
  labels:
    app: web
spec:
  containers:
  - name: app
    image: my-private-reg/nginx:v1

~$ _
```

```
~$ cat pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: example-pod
  labels:
    app: web
spec:
  containers:
  - name: app
    image: my-private-reg/nginx:v1
  securityContext:
          privileged: true
```

```
~$ cat pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: example-pod
  labels:
    app: web
spec:
  containers:
  - name: app
    image: my-private-reg/nginx:v1
  - name: sidecar
    image: istio/sidecar
```

```
~$ cat ingress.yaml
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: ingress-wildcard-host
spec:
  rules:
  - host: "foo.bar.com"
    http:
      paths:
      - pathType: Prefix
        path: "/bar"
        backend:
          service:
            name: service1
            port:
              number: 80

~$ _
```



Open Policy Agent



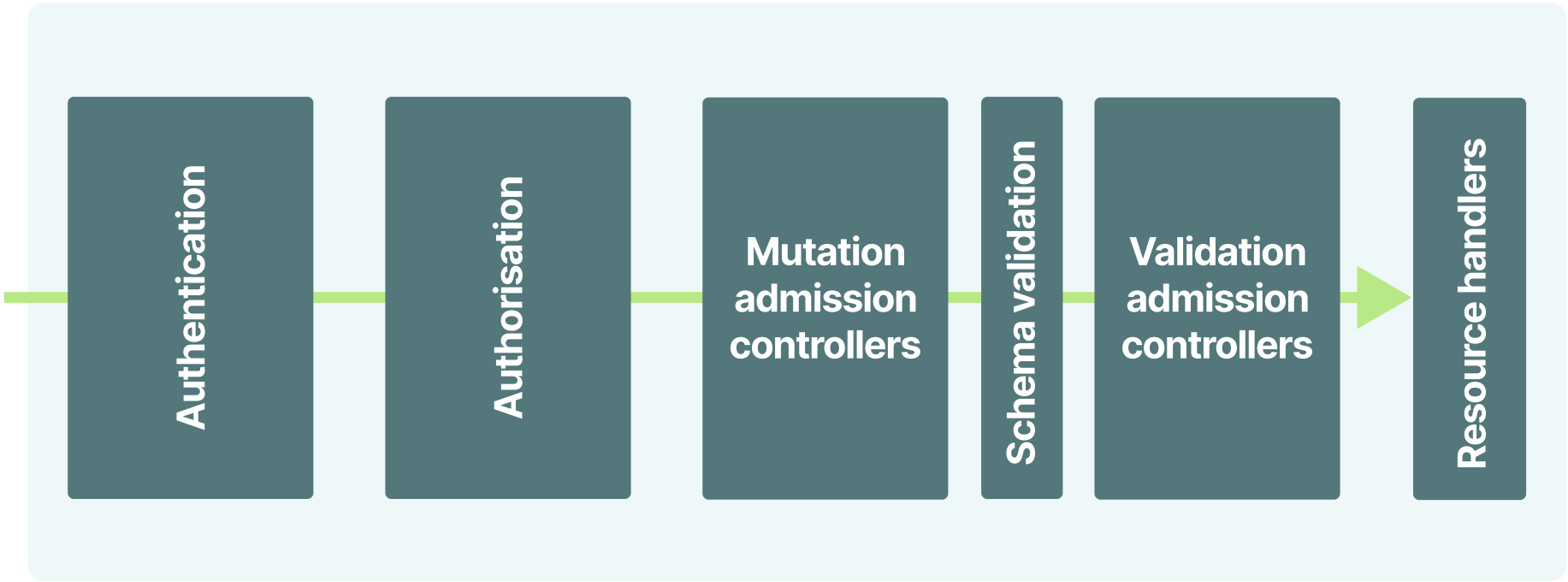
Kyverno



KUBEWARDEN

Storing the object

API SERVER


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Storing the object

Deserialize the request

Construct runtime obj

Persist in ETCD

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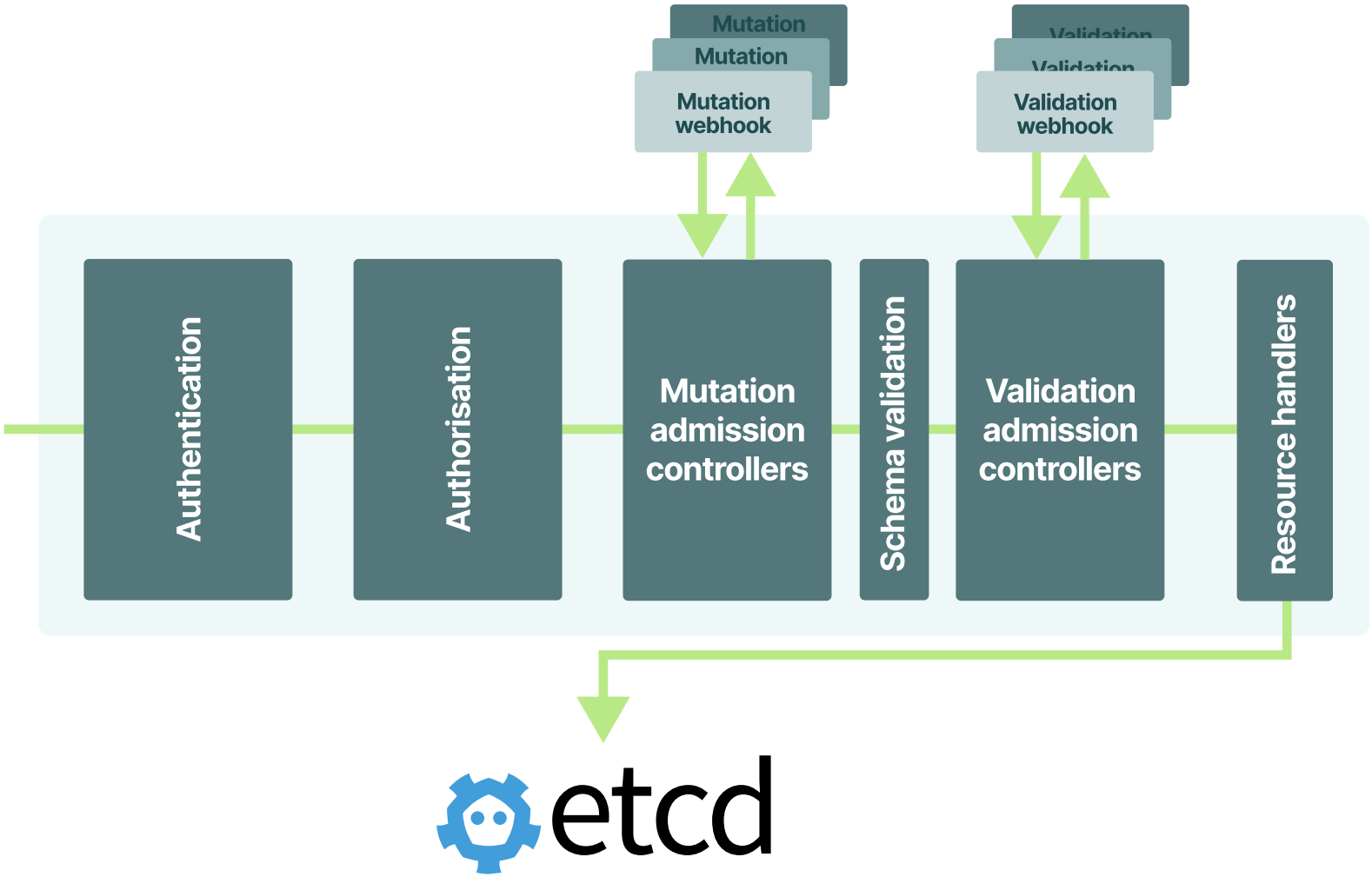
Extending the API

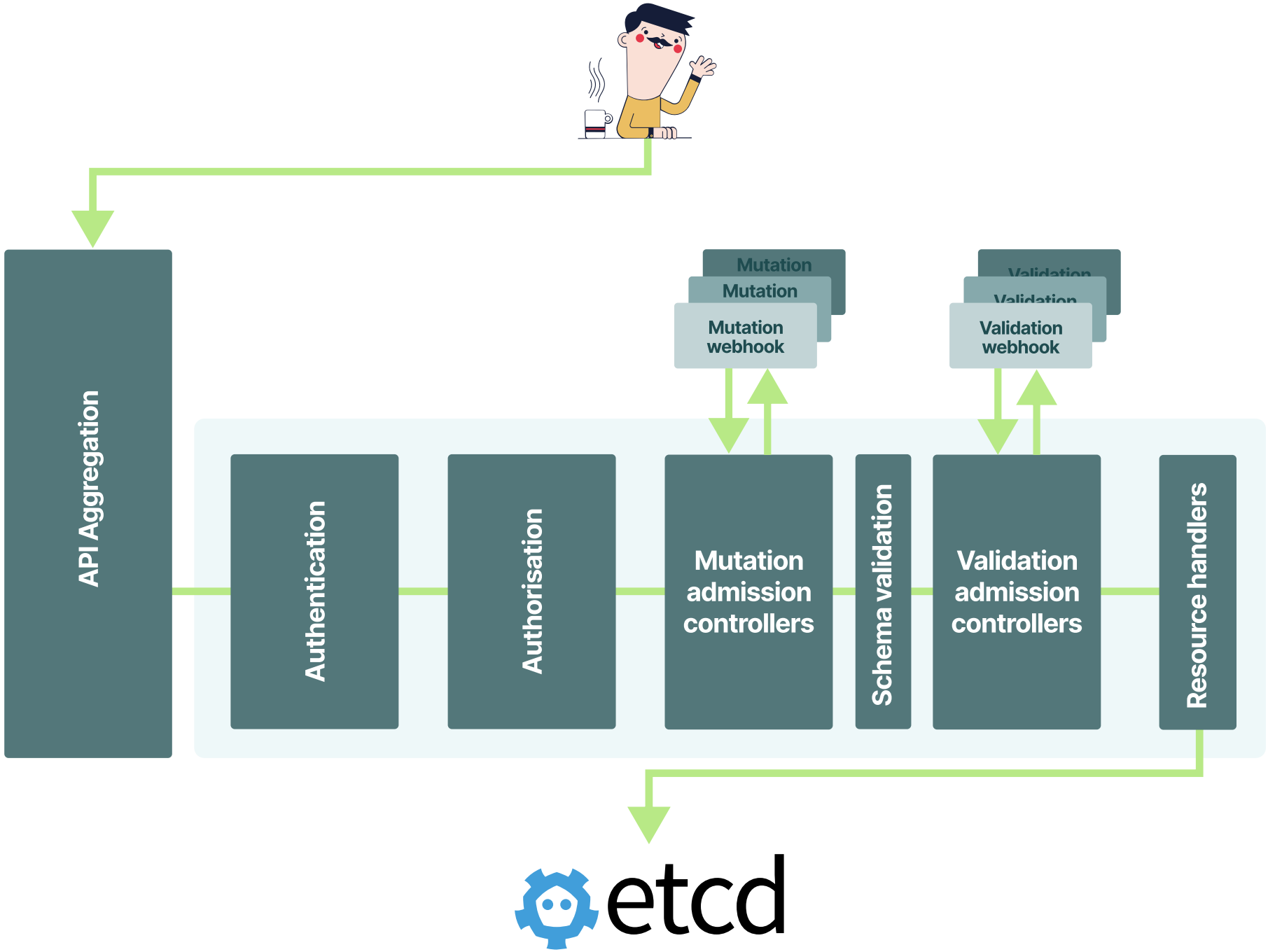


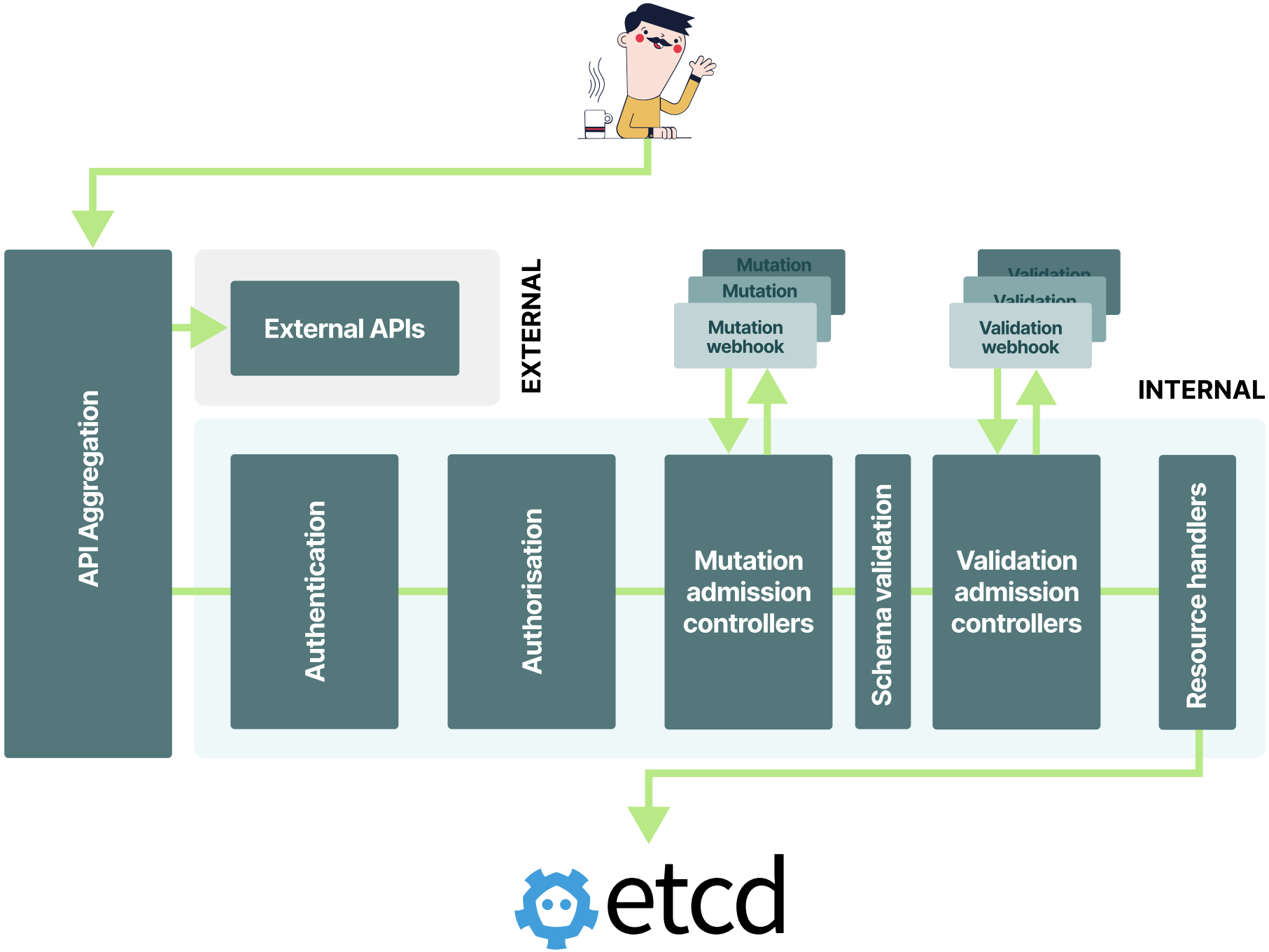
API SERVER

etcd




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API Aggregation
External APIs
EXTERNAL
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Mutation
Mutation webhook
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Validation
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INTERNAL
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Registering APIs

Service discovery

Reverse Proxy

Extending the API

Registering APIs

Service discovery

Reverse Proxy

Extending the API

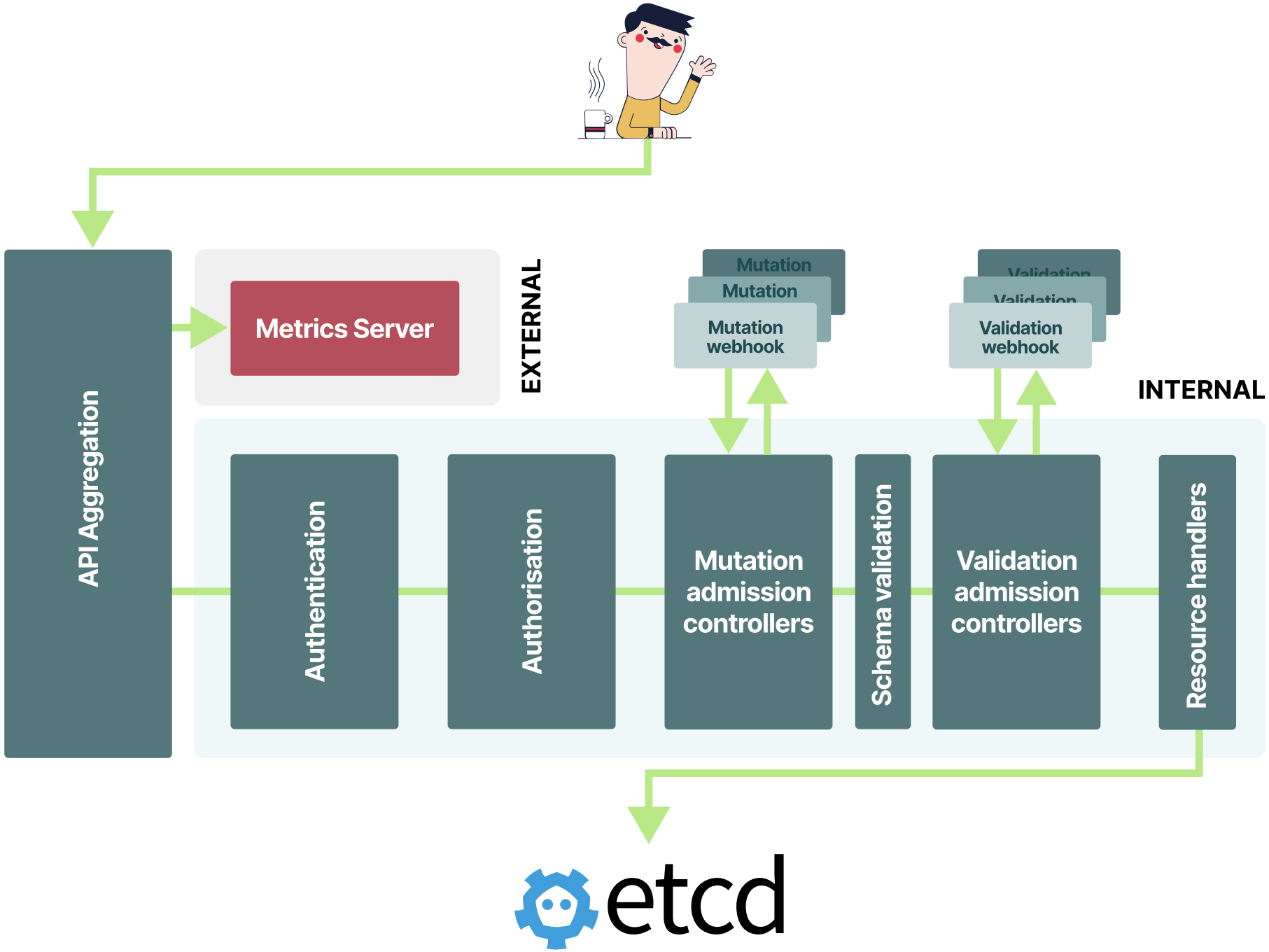
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Metrics server


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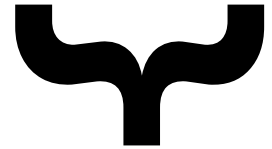

/apis/metrics.k8s.io/v1beta1/pods
Group

/apis/metrics.k8s.io/v1beta1/pods



Version

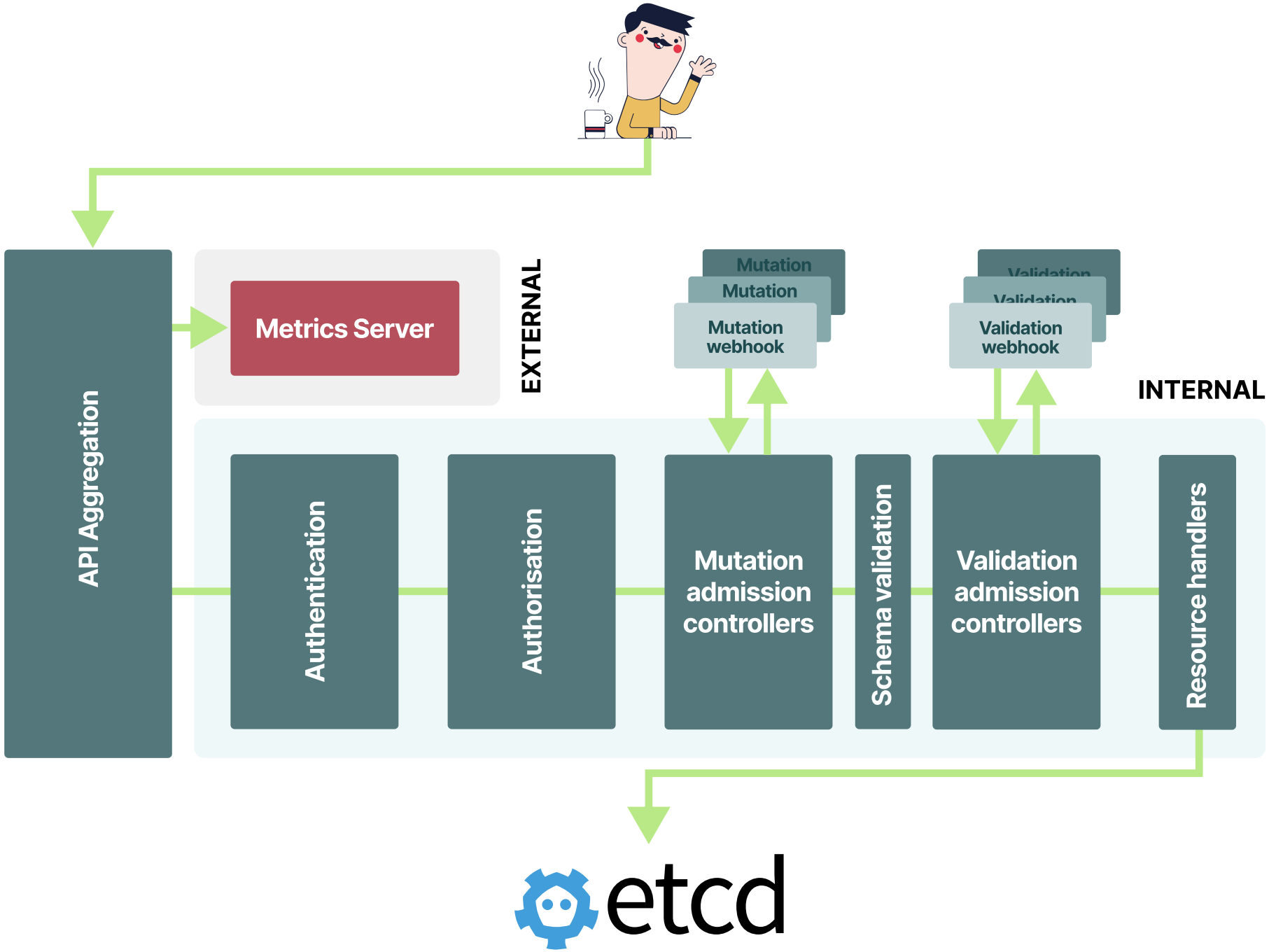
/apis/metrics.k8s.io/v1beta1/pods

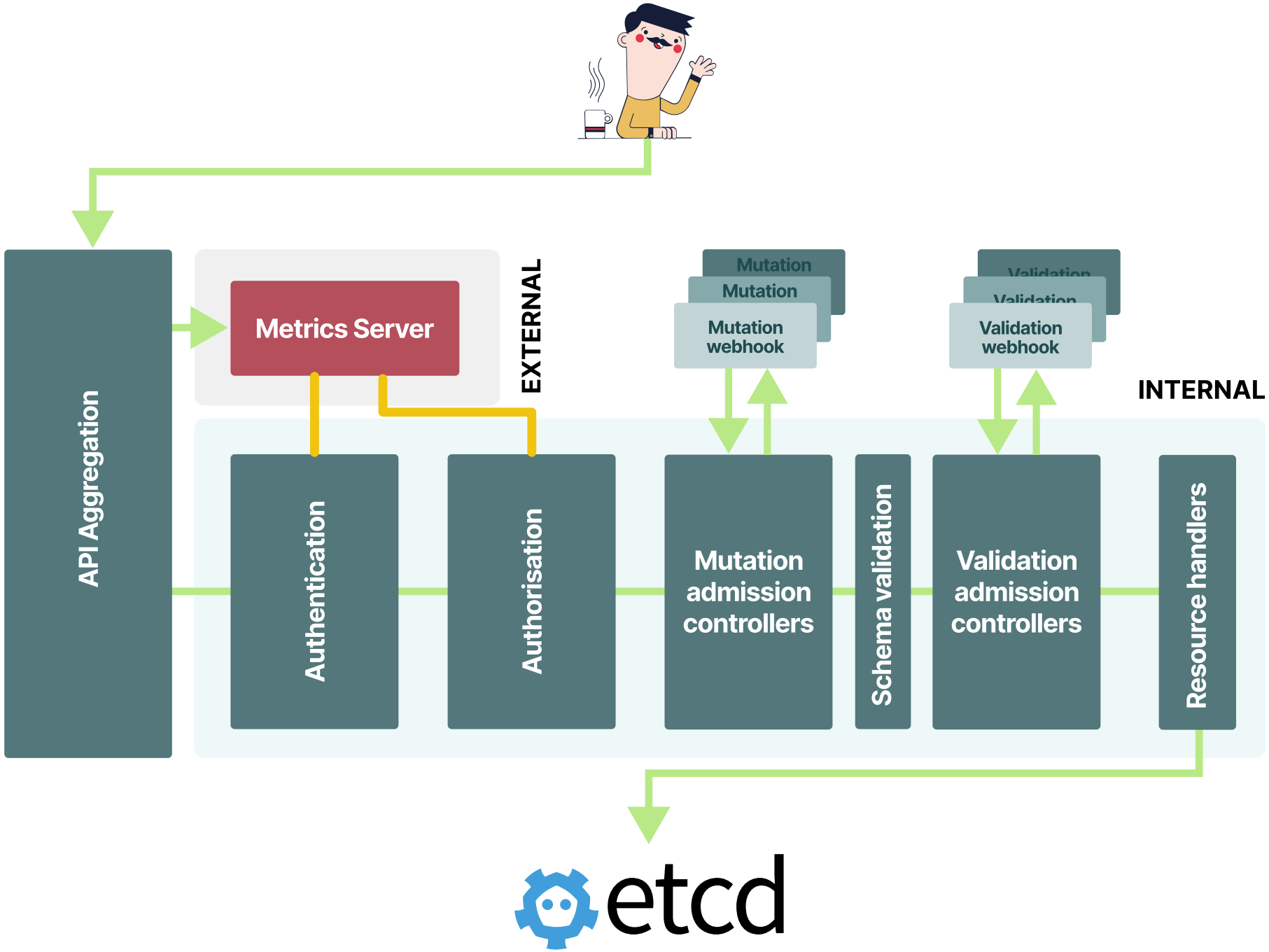


Resource

```
~$ kubectl get --raw "/apis/metrics.k8s.io/v1beta1/pods"
```

```
~$ kubectl get --raw "/apis/metrics.k8s.io/v1beta1/pods"
{
  "kind": "PodMetricsList",
  "apiVersion": "metrics.k8s.io/v1beta1",
  "metadata": {
    "selfLink": "/apis/metrics.k8s.io/v1beta1/pods"
  },
  "items": [
    {
      "metadata": {
        "name": "v1-57cbf4bb76-h7fpw",
        "namespace": "default",
        "creationTimestamp": "2021-06-28T03:09:19Z"
      },
      "timestamp": "2021-06-28T03:09:16Z",
      "window": "30s",
      "containers": [
        {
          "name": "web",
          "usage": {
```


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Recap

API

Recap — API

1. RBAC
2. Admission controllers
3. Resource handlers
4. API aggregation and extensions

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Thanks.